

Supplementary Material

Advancing monitoring of antimicrobial resistance trajectories using flexible spatiotemporal modelling: a change-point analysis

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Glossary

AMU= Antimicrobial usage

AMR= Antimicrobial resistance

CRE= Carbapenem-resistant Enterobacterales

DDD= Daily defined dose

DID= Daily defined doses per 1000 inhabitants

EDF= Effective degrees of freedom

EU= European union

IQR= Interquartile range

GAM= Generalised additive model

SE= Standard error

US= United States of America

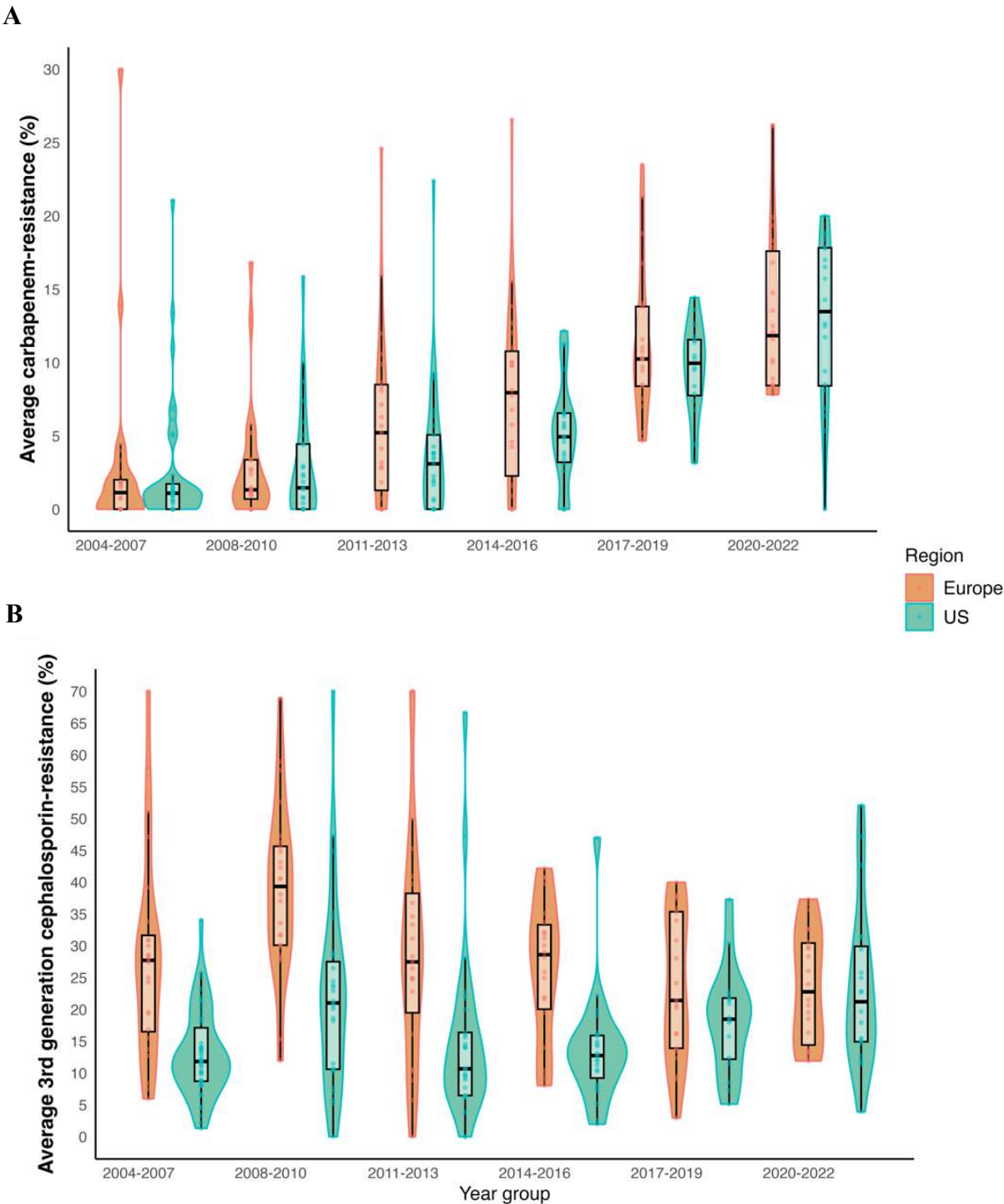
3GCR= Third generation cephalosporin resistance

Table S1. Number of Enterobacterales cultures by US-states and European countries

US states								Europe			
ID	State	N	%	ID	State	N	%	ID	Country	N	%
1	Alabama	262	0.9	25	Nebraska	1227	4.3	1	Austria	570	0.9
2	Arizona	430	1.5	26	New Hampshire	69	0.2	2	Belgium	4492	7.0
3	Arkansas	102	0.4	27	New Jersey	850	3.0	3	Croatia	1883	2.9
4	California	831	2.9	28	New Mexico	86	0.3	4	Czech Republic	1326	2.1
5	Colorado	789	2.7	29	New York	3713	12.9	5	Denmark	1991	3.1
6	Connecticut	114	0.4	30	North Carolina	351	1.2	6	Finland	511	0.8
7	Delaware	63	0.2	31	North Dakota	523	1.8	7	France	9708	15.0
8	D. Columbia	28	0.1	32	Ohio	1133	3.9	8	Germany	4680	7.3
9	Florida	3473	12.1	33	Oklahoma	71	0.2	9	Greece	2334	3.6
10	Georgia	877	3.0	34	Oregon	161	0.6	10	Hungary	1411	2.2
11	Illinois	935	3.2	35	Pennsylvania	1734	6.0	11	Ireland	1542	2.4
12	Indiana	996	3.5	36	South Carolina	34	0.1	12	Italy	9209	14.3
13	Iowa	236	0.8	37	South Dakota	113	0.4	13	Latvia	532	0.8
14	Kansas	392	1.4	38	Tennessee	462	1.6	14	Lithuania	660	1.0
15	Kentucky	873	3.0	39	Texas	1217	4.2	15	Netherlands	1498	2.3
16	Louisiana	278	1.0	40	Utah	194	0.7	16	Poland	1681	2.6
17	Maine	43	0.1	41	Vermont	128	0.4	17	Portugal	3567	5.5
18	Maryland	390	1.4	42	Virginia	304	1.1	18	Romania	1575	2.4
19	Massachusetts	755	2.6	43	Washington	692	2.4	19	Slovak Republic	101	0.2
20	Michigan	902	3.1	44	West Virginia	78	0.3	20	Slovenia	370	0.6
21	Minnesota	131	0.5	45	Wisconsin	2141	7.4	21	Spain	9726	15.1
22	Mississippi	152	0.5					22	Sweden	1450	2.2
23	Missouri	419	1.5					23	Switzerland	1212	1.9
24	Montana	41	0.1					24	United Kingdom	2511	3.9
Total				28793				64540			

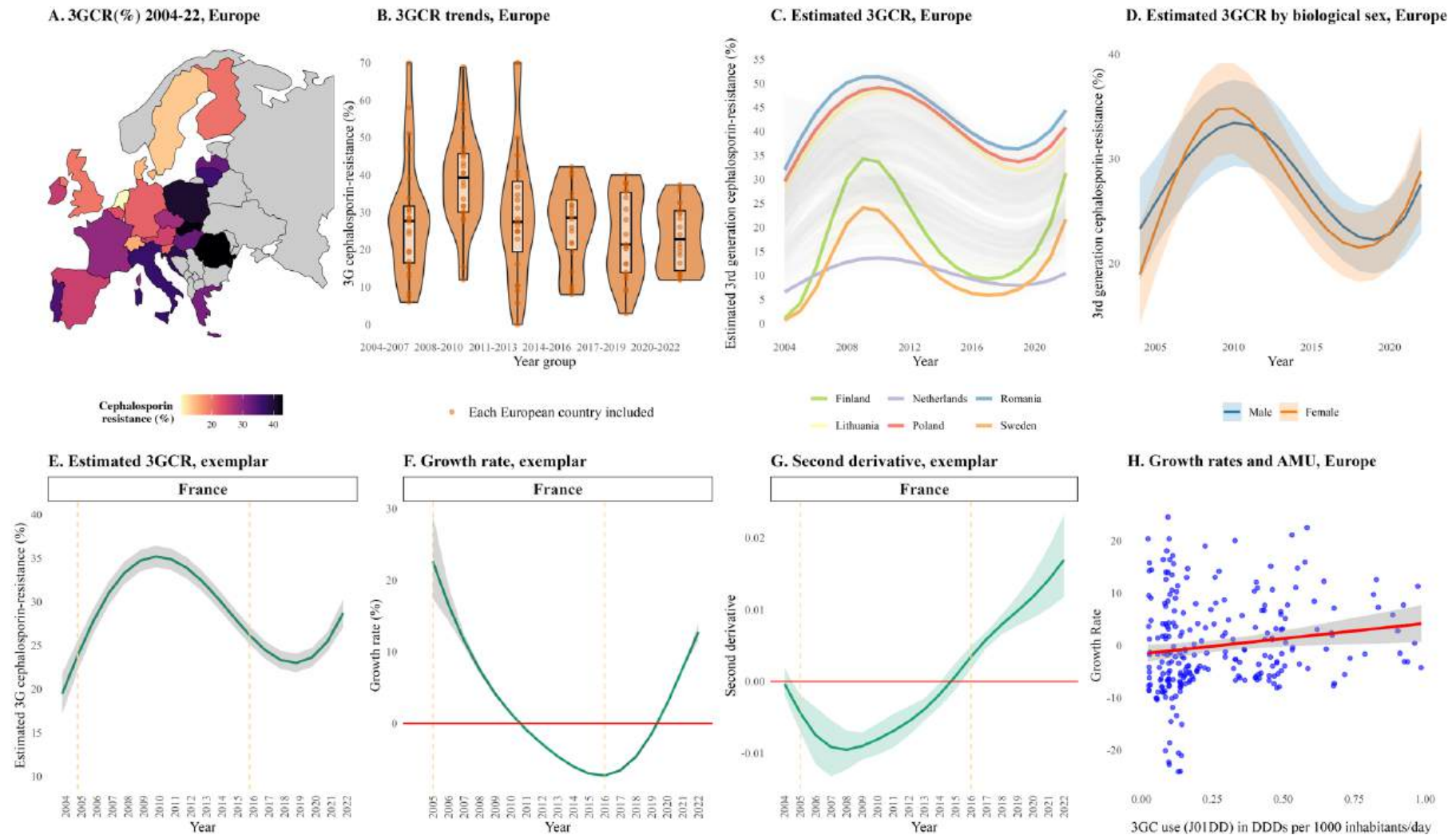
Notes: US= United States. D.= District. Lighter red indicates fewer cultures relative to the respective region (US or Europe). The percentage (%) means the percentage regarding the total for each respective region (US or Europe).

Figure S1. Overall antibiotic resistance combinations among Enterobacterales isolates over time and across US-states and European countries, 2004-2022.



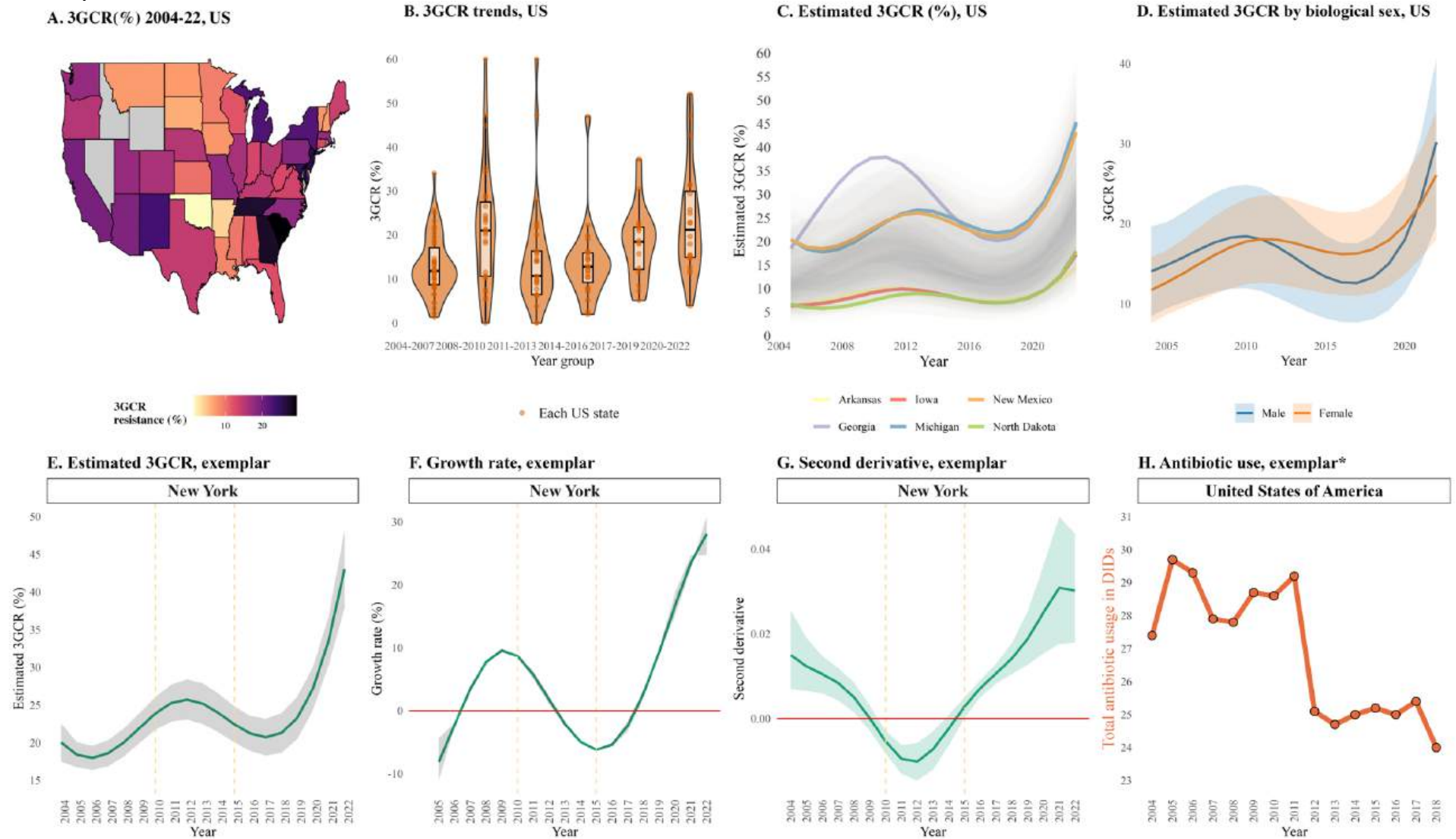
Notes: US= United States of America

Figure S2. Main findings from spatio-temporal GAM analyses of 3GCRE growth rates and change points across Europe and France as an exemplar, 2004-2022.



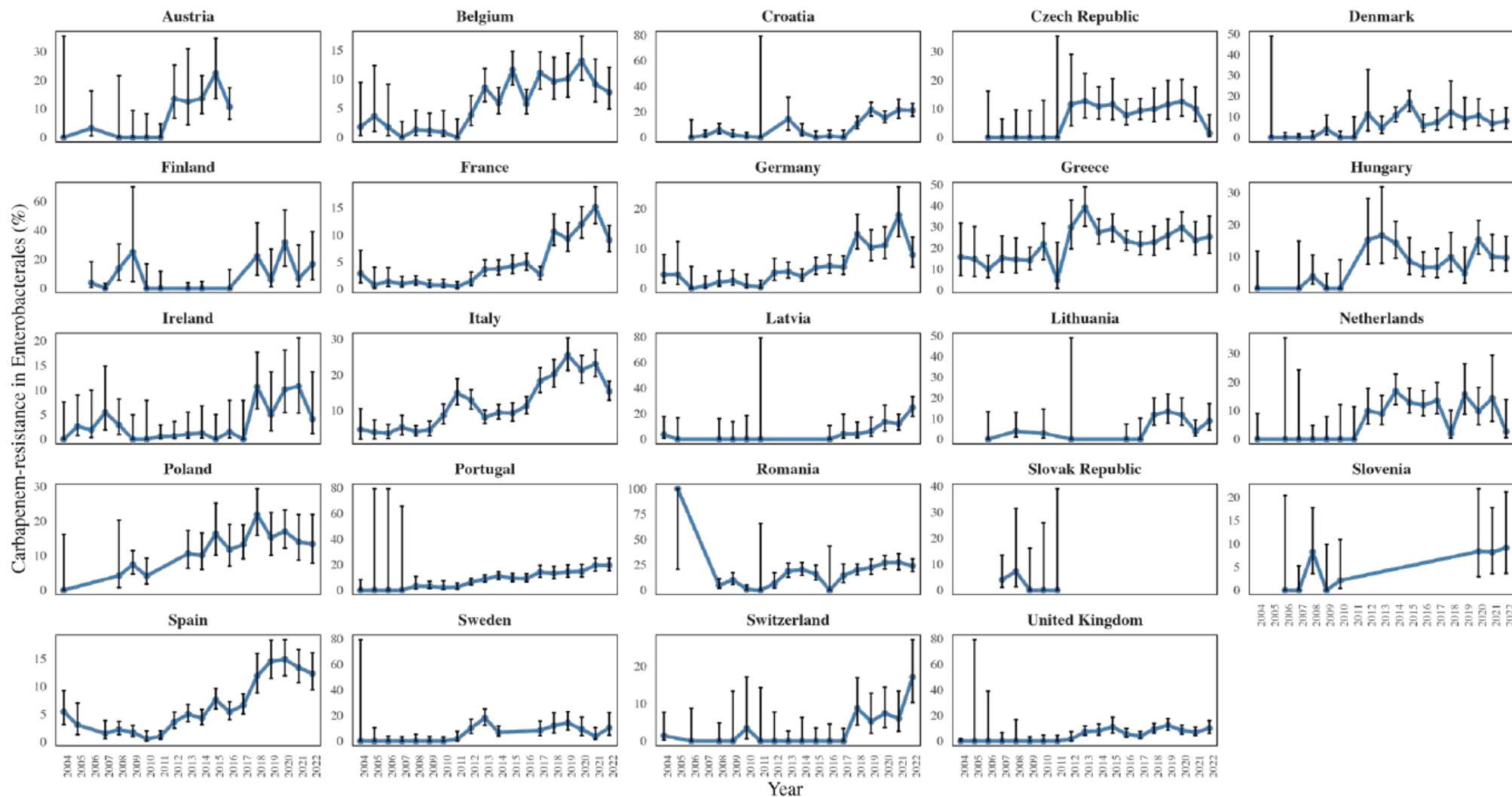
Notes: 3GCRE= 3rd generation cephalosporin-resistance. GAM= Generalised additive models. DID= Daily defined doses per 1,000 inhabitants. We included 24 European countries, following panel A. Panel C: Dark grey lines are for predicted CRE among remaining European countries. For Panel E-F: change (break) points are indicated in yellow dashed lines for second derivate breakpoints (resistance acceleration/deaccelerations) and in blue dashed lines for first derivate (increasing or decreasing trends) change points. Panel F: solid red line indicates 0% growth rate. Panel H presents antibiotic usage (3 generation cephalosporins) and its relationship with growth rates from GAMs.

Figure S3. Findings from spatio-temporal GAM analyses of 3GCRE growth rates and change points across US-states (N=45) and New York as an exemplar, 2004-2022.



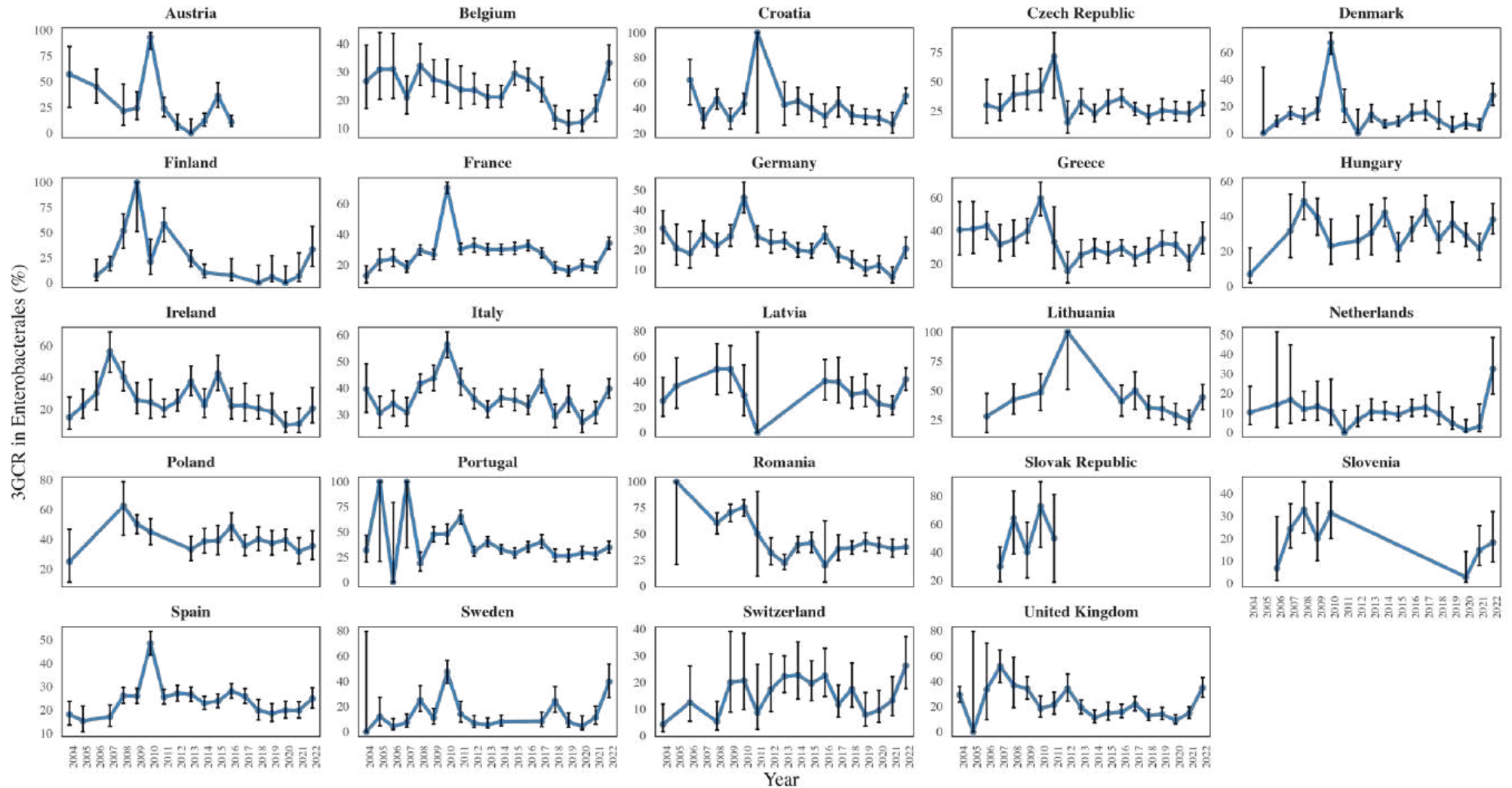
Notes: 3GCRE= Third-generation cephalosporin resistant Enterobacterales. GAM= Generalised additive models. DID= Daily defined doses per 1,000 inhabitants. We included 45 states within the United States of America (US), following panel A. Panel C: Dark grey lines are for predicted 3GCRE among remaining European countries. For Panel E-F: change (break) points are indicated in yellow dashed lines for second derivate breakpoints (resistance acceleration/deaccelerations) and in blue dashed lines for first derivate (increasing or decreasing trends) change points. Panel F: solid red line indicates 0% growth rate. *Panel H presents overall estimated antibiotic usage (AMU) in DID in the US, descriptively, obtained from the Global Burden of Disease study. State-specific estimates were not available.

Figure S4. Observed carbapenem-resistance in **European countries**, before applying GAMs



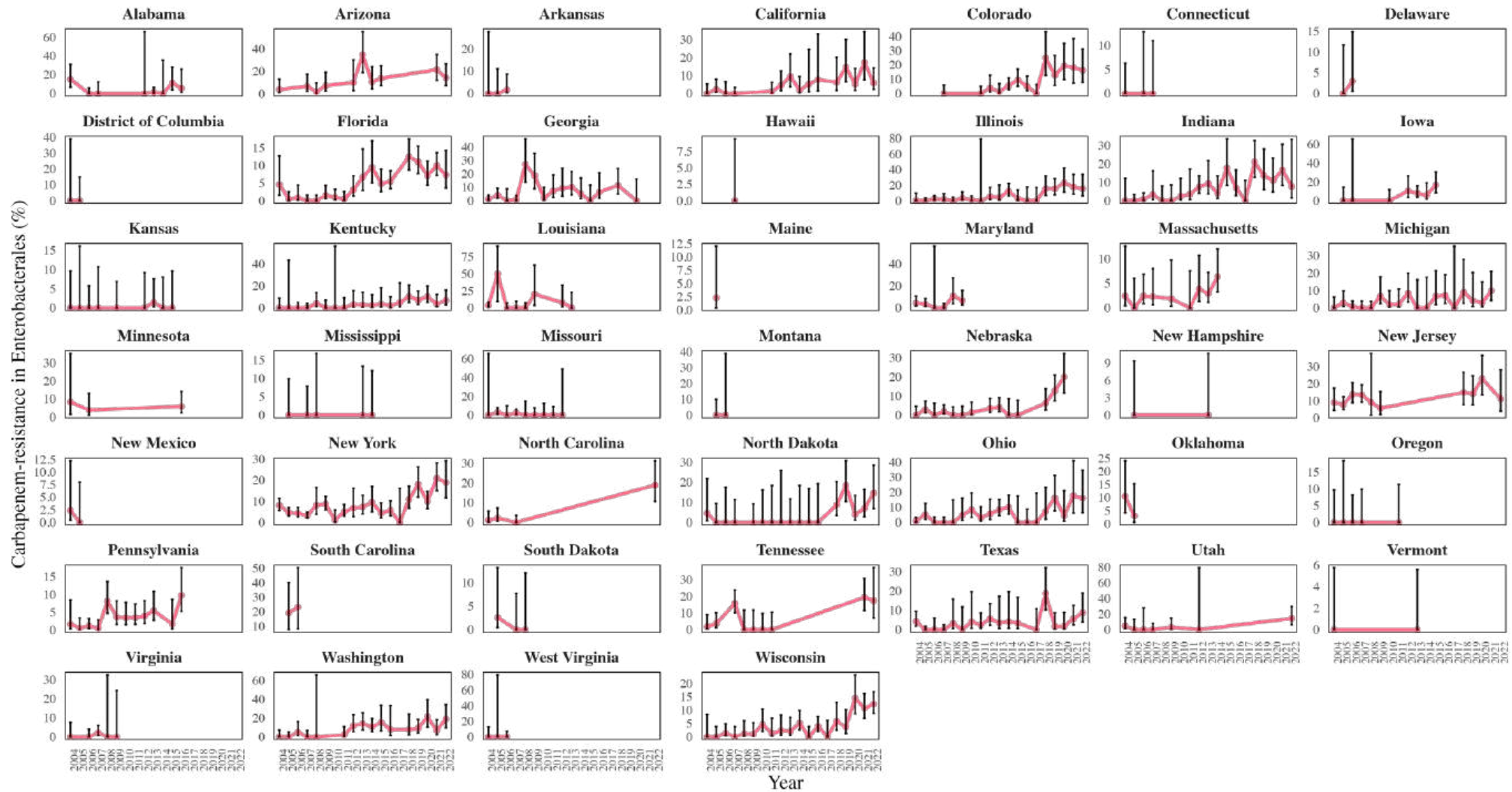
Notes: GAM= Generalised additive model. Black error bars represent binomial confidence intervals, estimated using number of cultures per territory/year.

Figure S5. Observed 3rd generation cephalosporin resistance in **European countries**, before applying GAMs



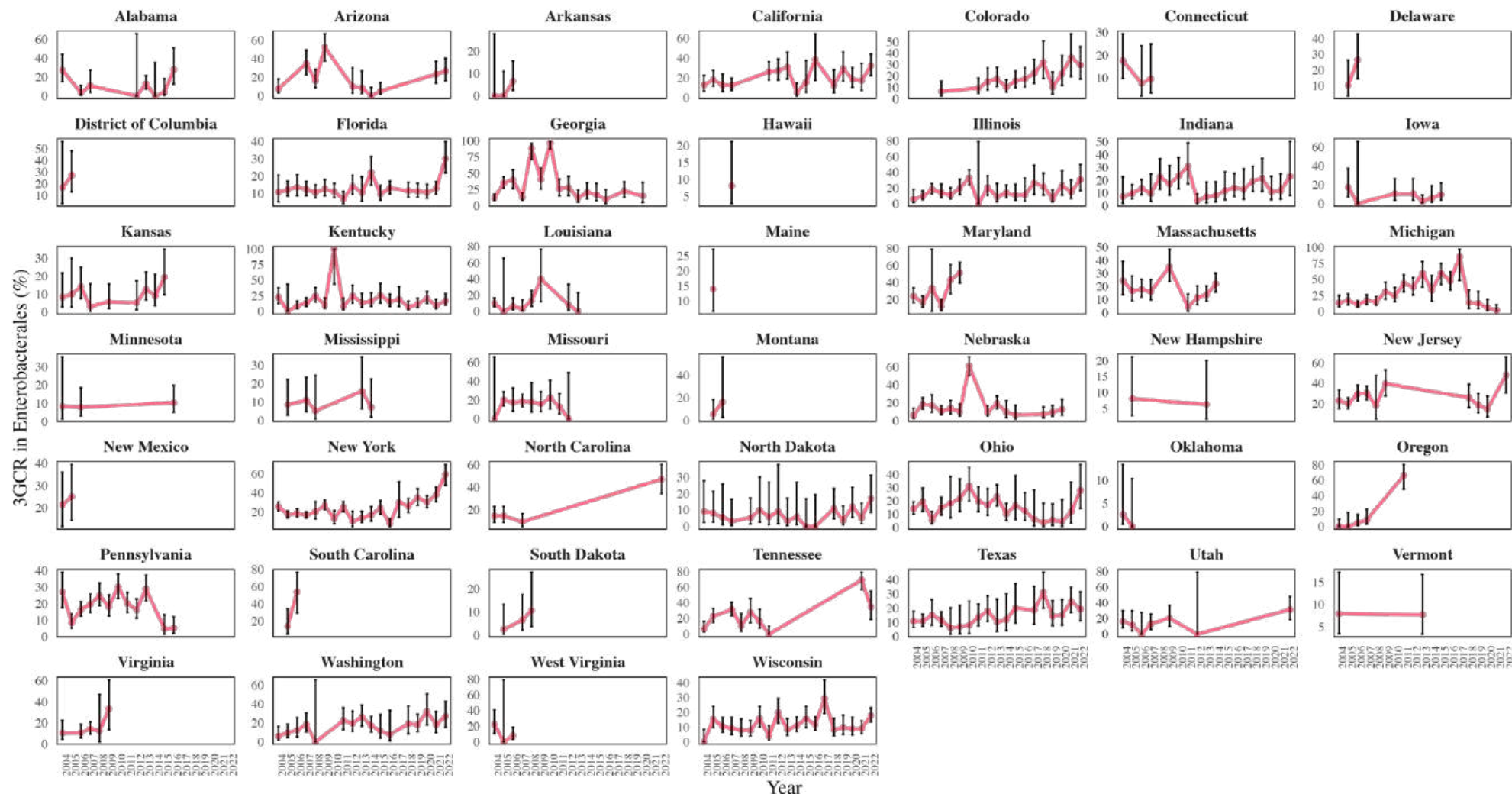
Notes: GAM= Generalised additive model. Black error bars represent binomial confidence intervals, estimated using number of cultures per territory/year. 3GCR= Third-generation cephalosporin-resistance.

Figure S6. Observed values for carbapenem-resistance in Enterobacterales across US-states



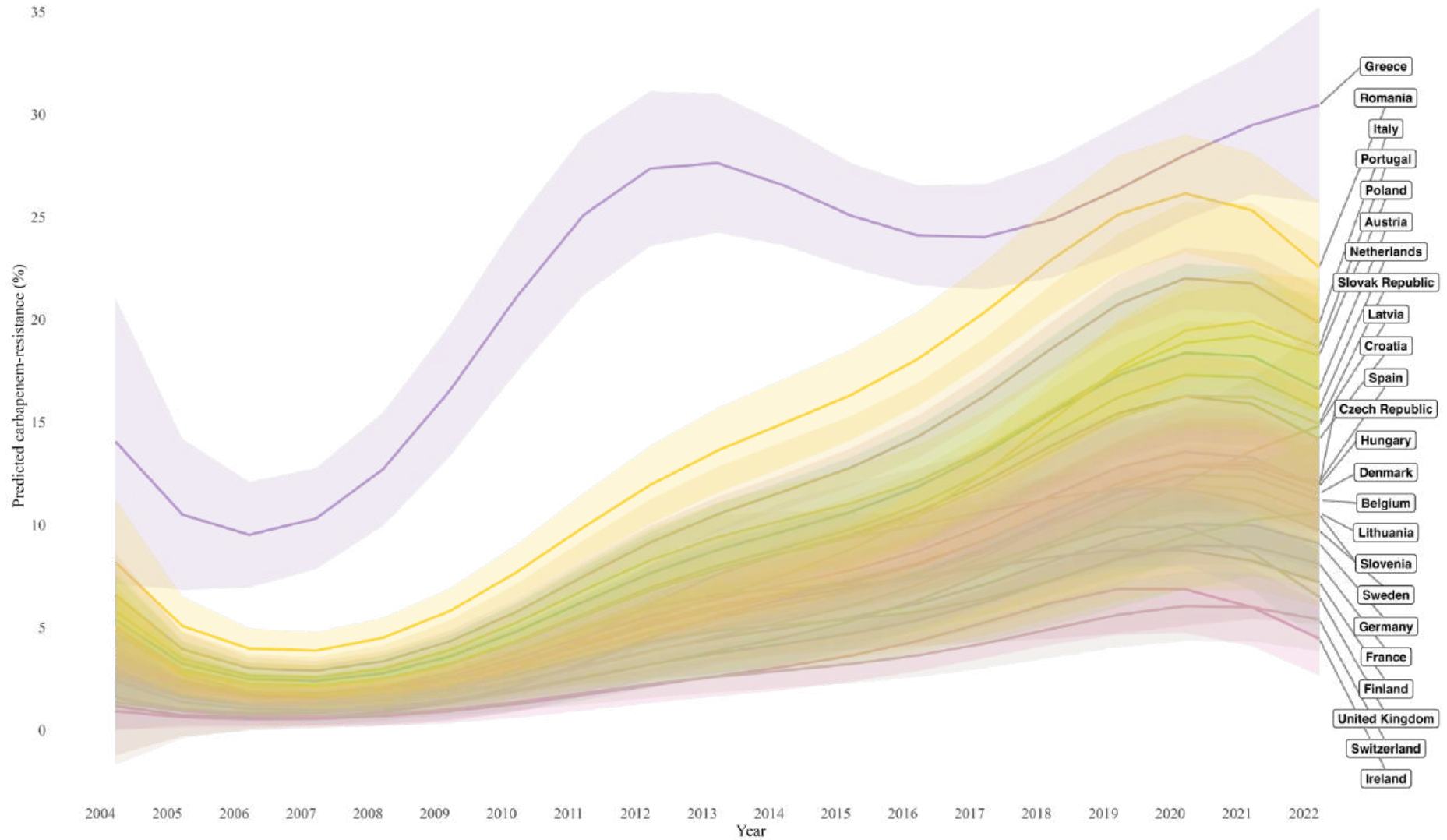
Notes: Hawaii was removed from further analyses due to island (no spatial correlation). Black error bars represent binomial confidence intervals, estimated using number of cultures per territory/year.

Figure S7. Observed values for third-generation cephalosporin-resistance in Enterobacterales across US states



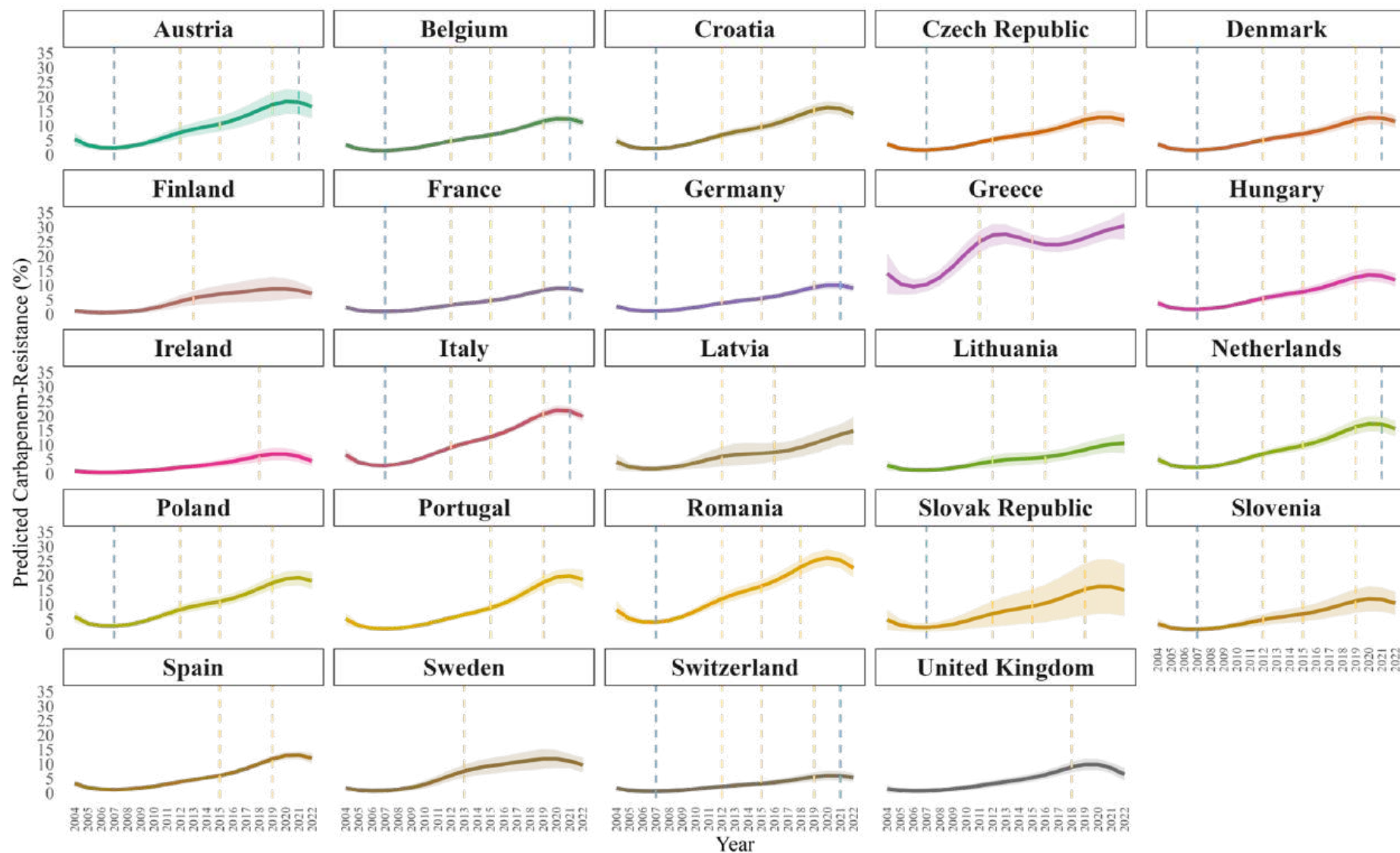
Notes: Hawaii was removed from further analyses due to island (no spatial correlation). Black error bars represent binomial confidence intervals, estimated using number of cultures per territory/year. 3GCR= Third-generation cephalosporin-resistance.

Figure S8. Estimated carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, aggregated**



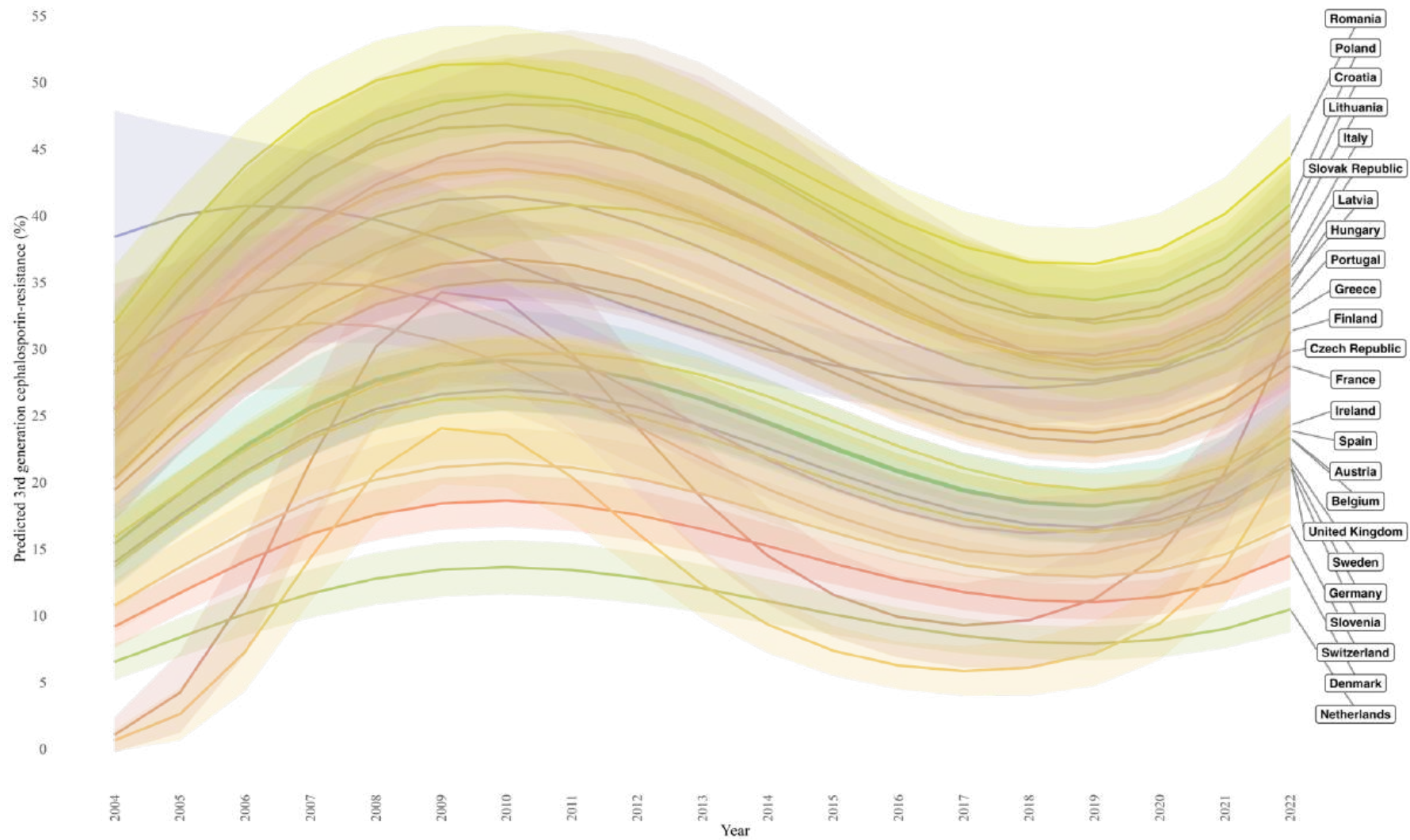
Notes: GAM= Generalised additive model.

Figure S9. Estimated carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, by country**



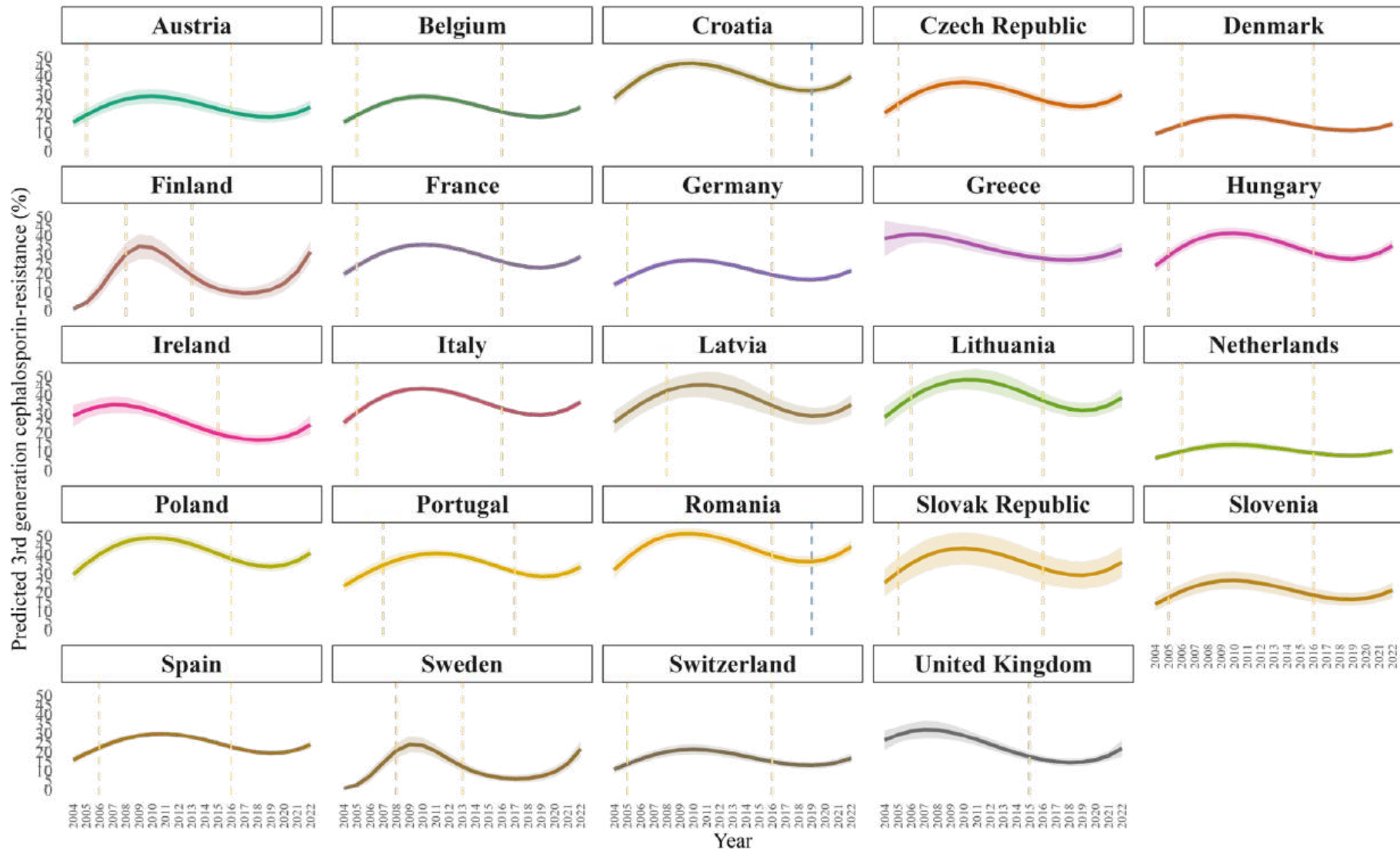
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa).

Figure S10. Estimated 3rd generation cephalosporin -resistance in Enterobacterales using a spatiotemporal GAM across **European countries**, aggregated



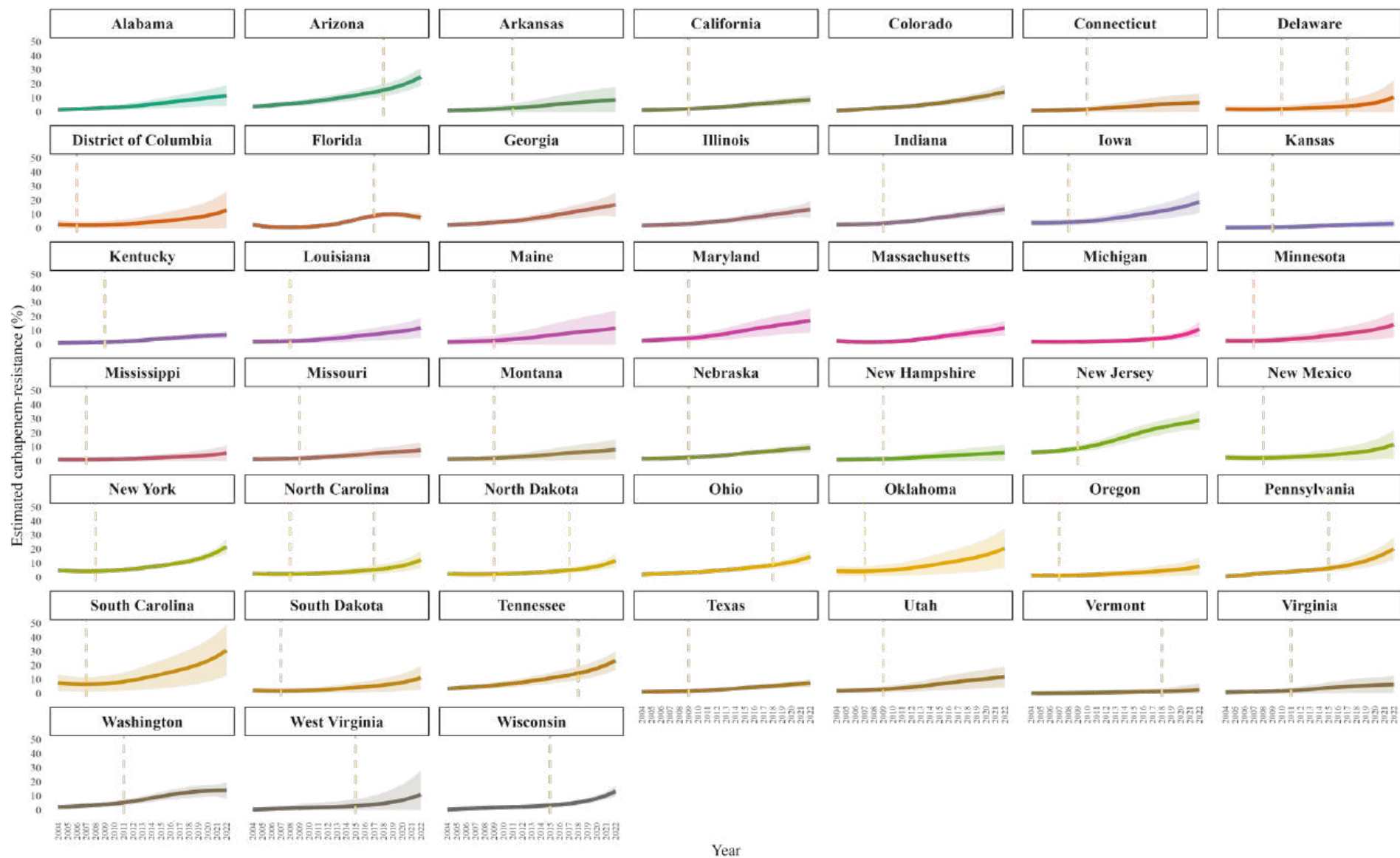
Notes: GAM= Generalised additive model.

Figure S11. Estimated 3rd generation cephalosporin -resistance in Enterobacterales using a spatiotemporal GAM across **European countries**, by country



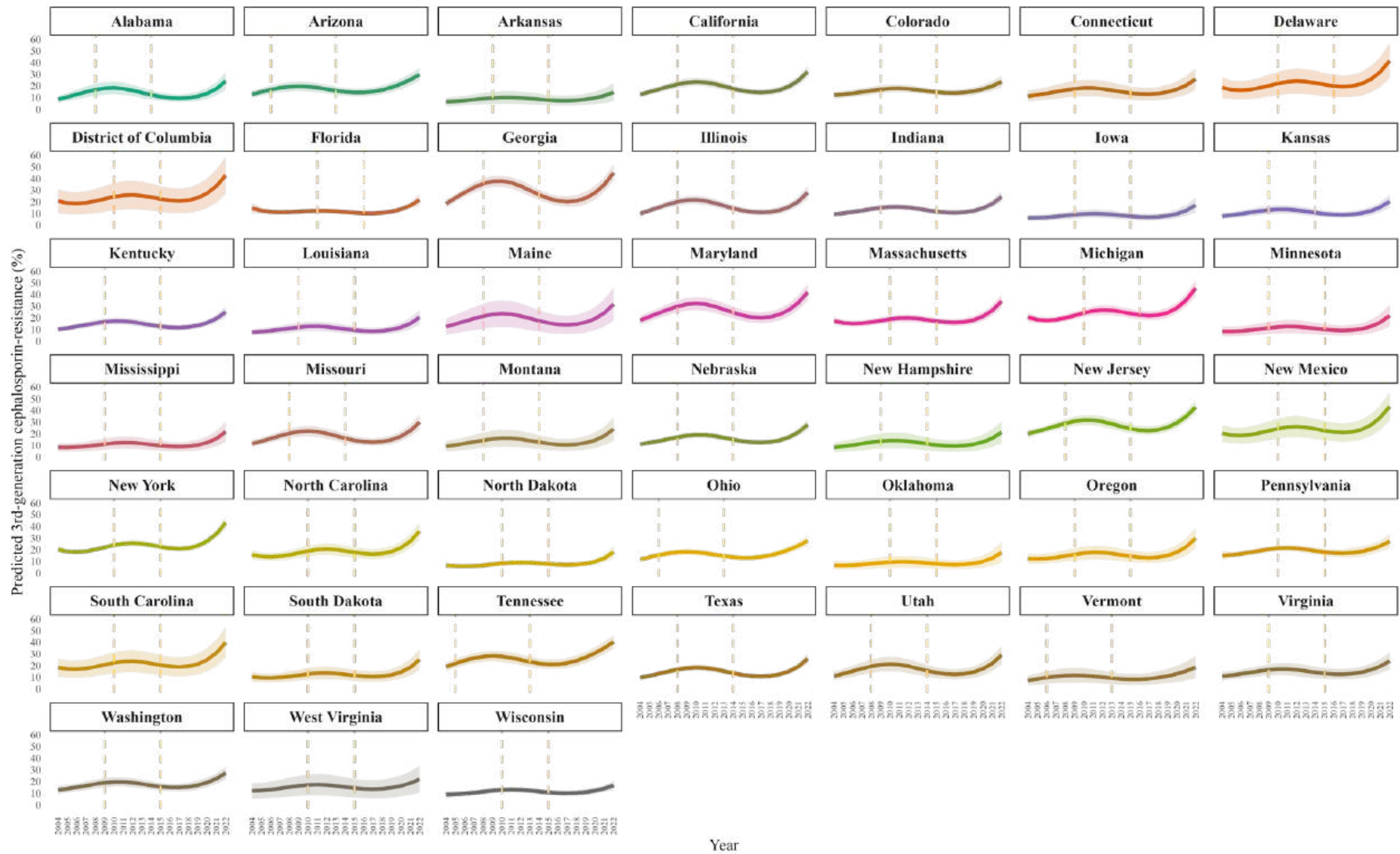
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa).

Figure S12. Estimated carbapenem-resistance in Enterobacteriales using a spatiotemporal GAM across **US states**



Notes: GAM= Generalised additive model. Dashed lines indicate change points (when either the 95% confidence interval or average touch zero).

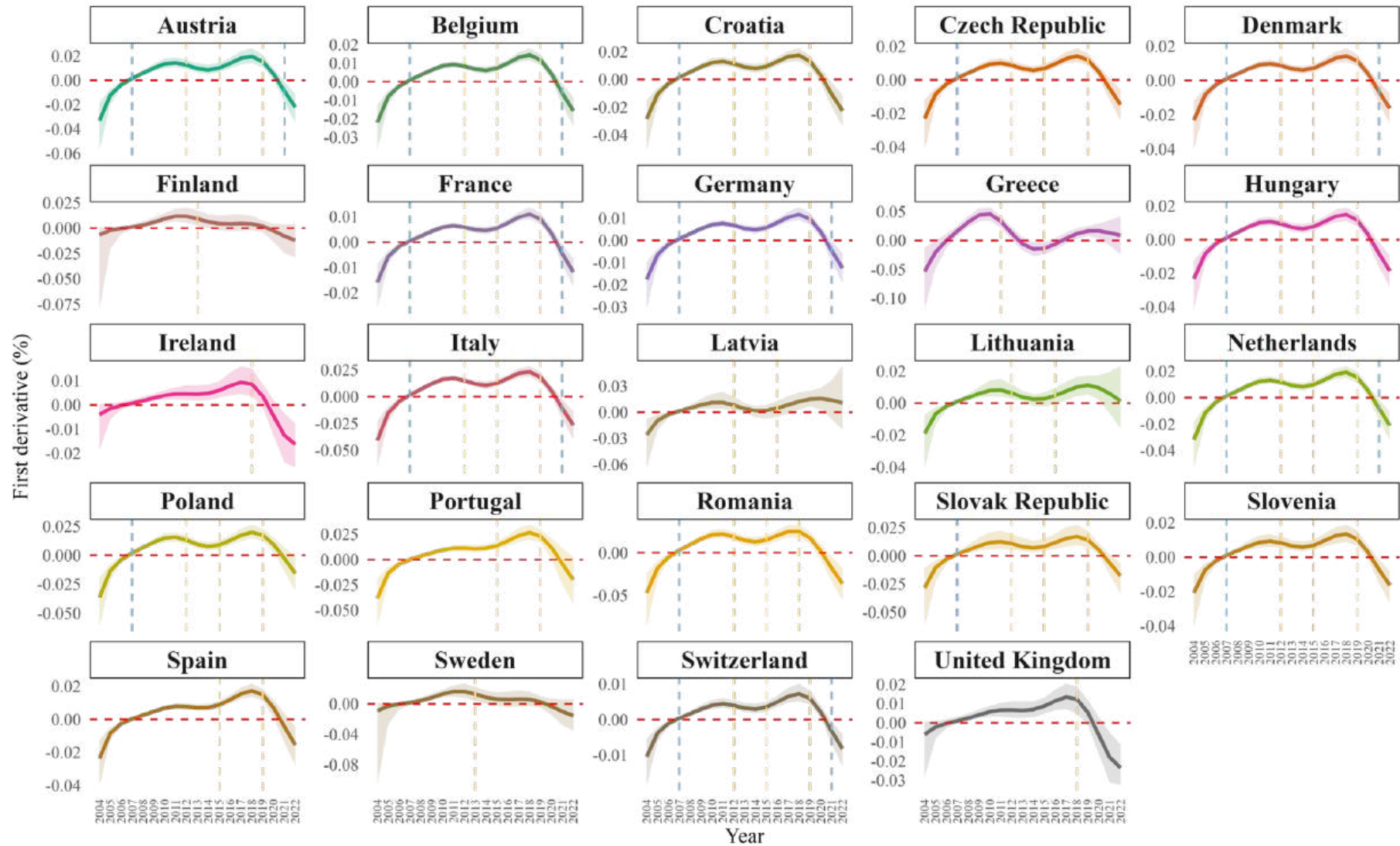
Figure S13. Estimated 3rd generation cephalosporin resistance in Enterobacteriales using a spatiotemporal GAM across US states



Notes: GAM= Generalised additive model. Dashed lines indicate change points (when either the 95% confidence interval or average touch zero).

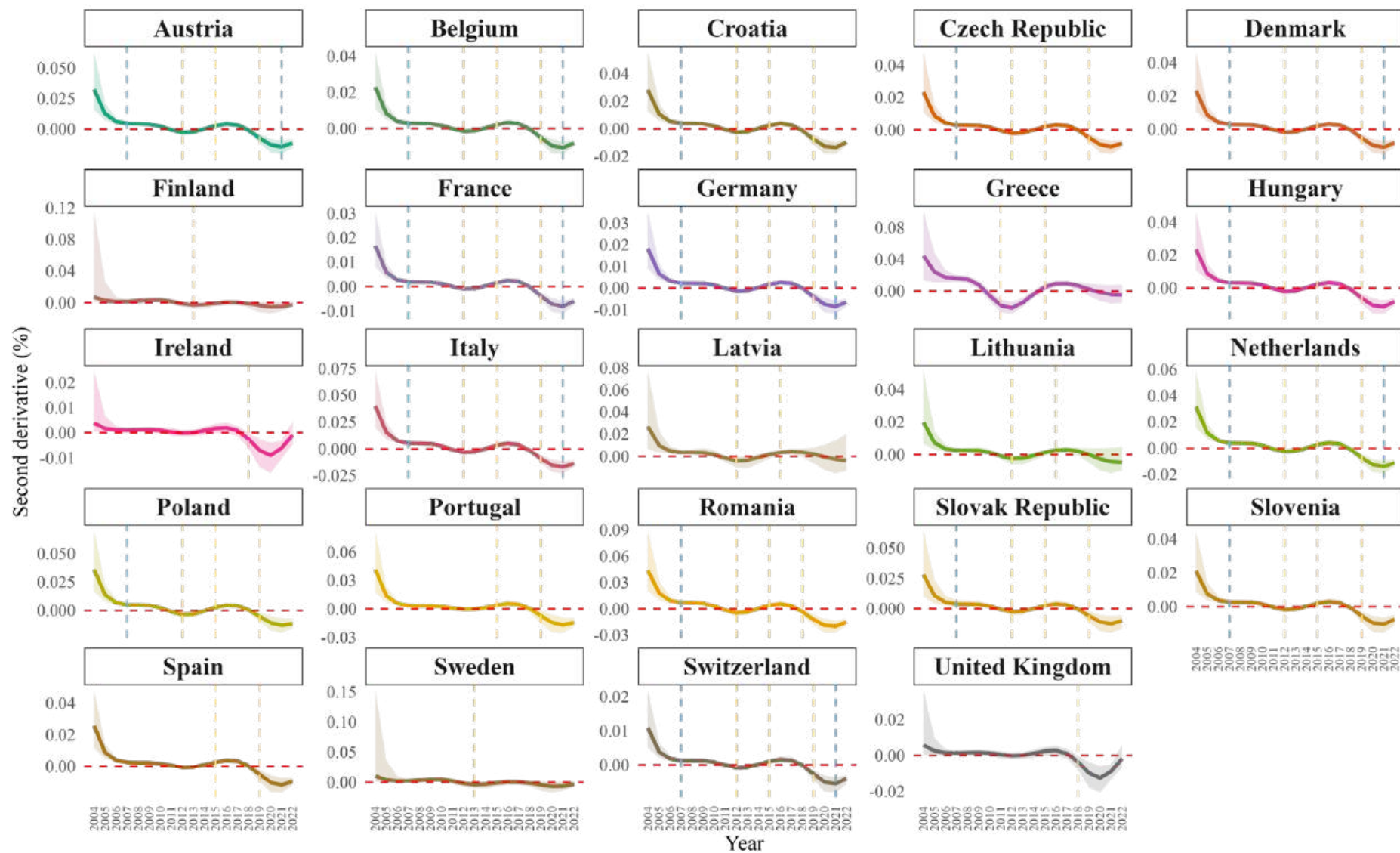
I. Carbapenem-resistance (derivatives, growth rates), Europe

Figure S14. First derivative of predicted carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across **European countries**



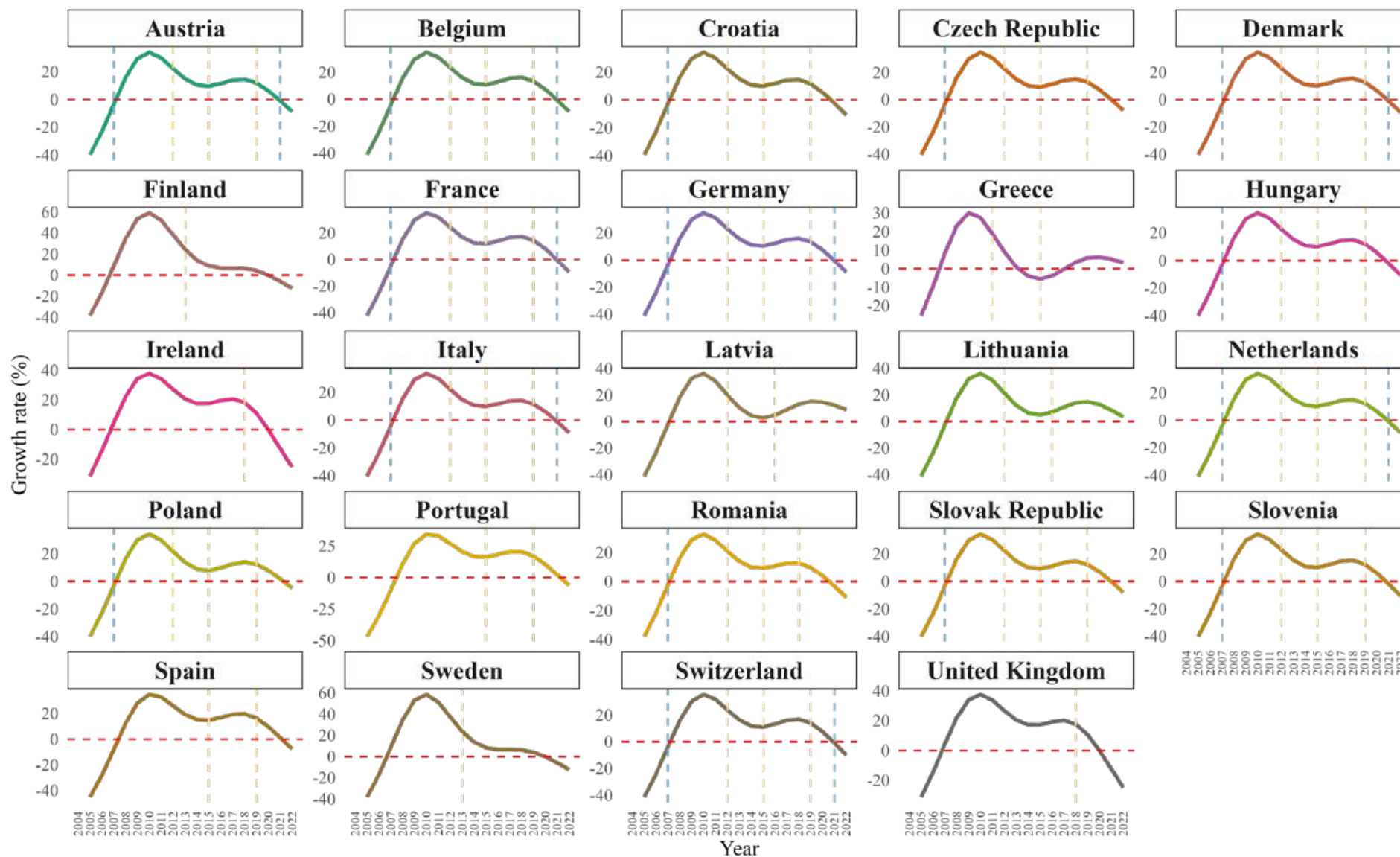
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero change.

Figure S15. Second derivative of predicted carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across European countries



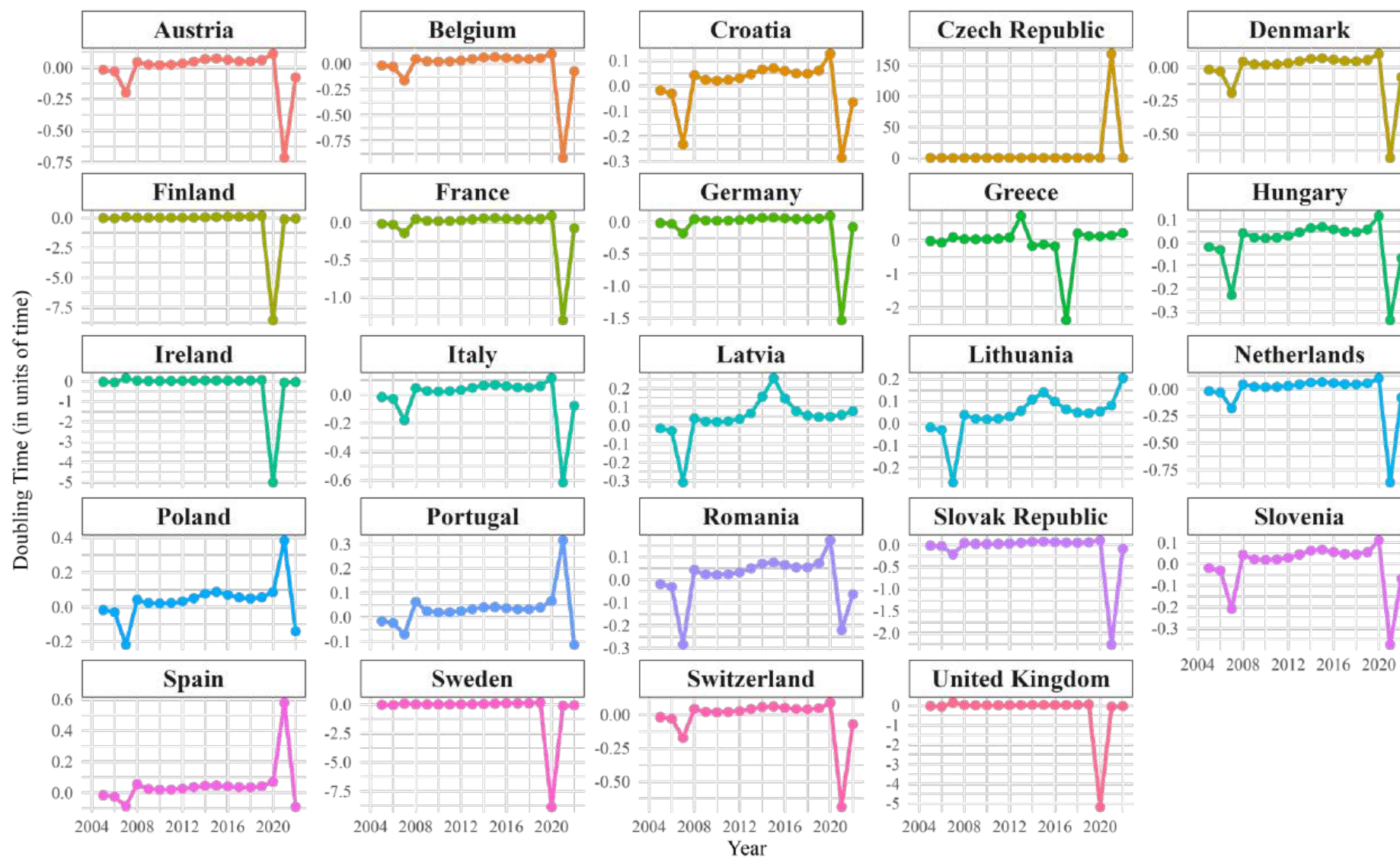
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero change.

Figure S16. Growth rates of predicted carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across European countries



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

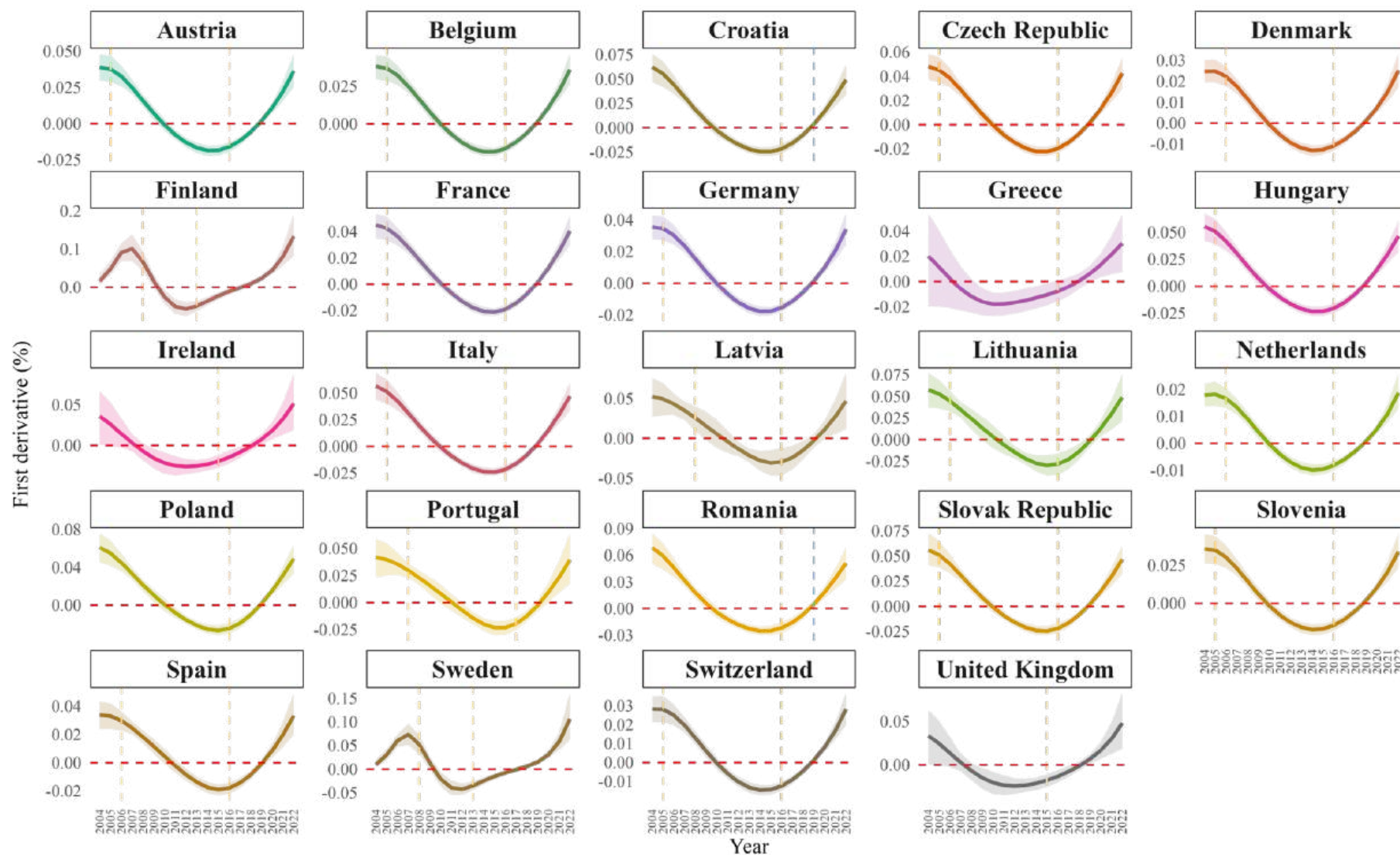
Figure S17. Doubling times for the growth rates of predicted carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across European countries



Notes: GAM= Generalised additive model.

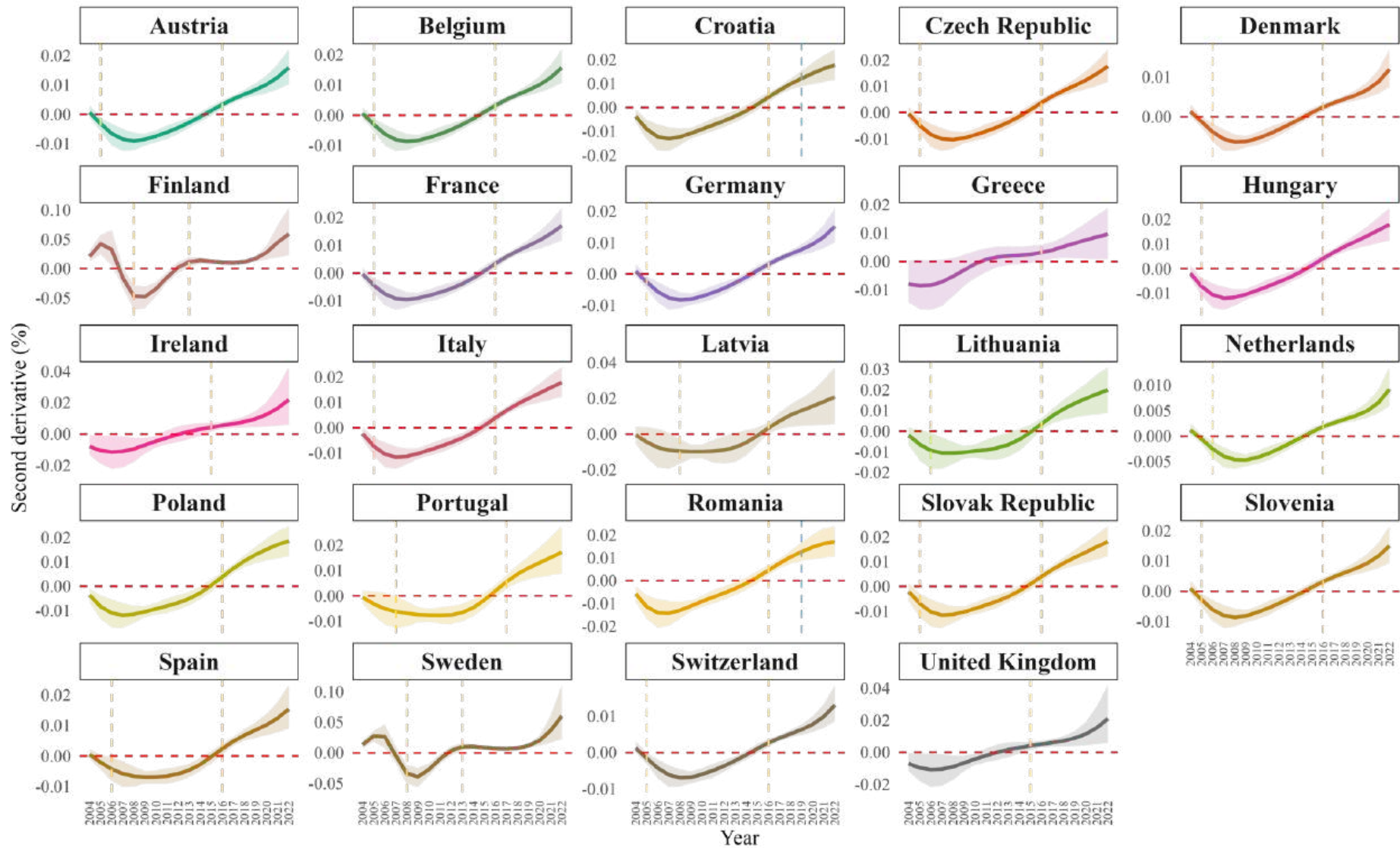
II. 3rd generation cephalosporin resistance (derivatives, growth rates), **Europe**

Figure S18. First derivative of predicted 3rd generation cephalosporin resistance in Enterobacterales using a spatiotemporal GAM across European countries



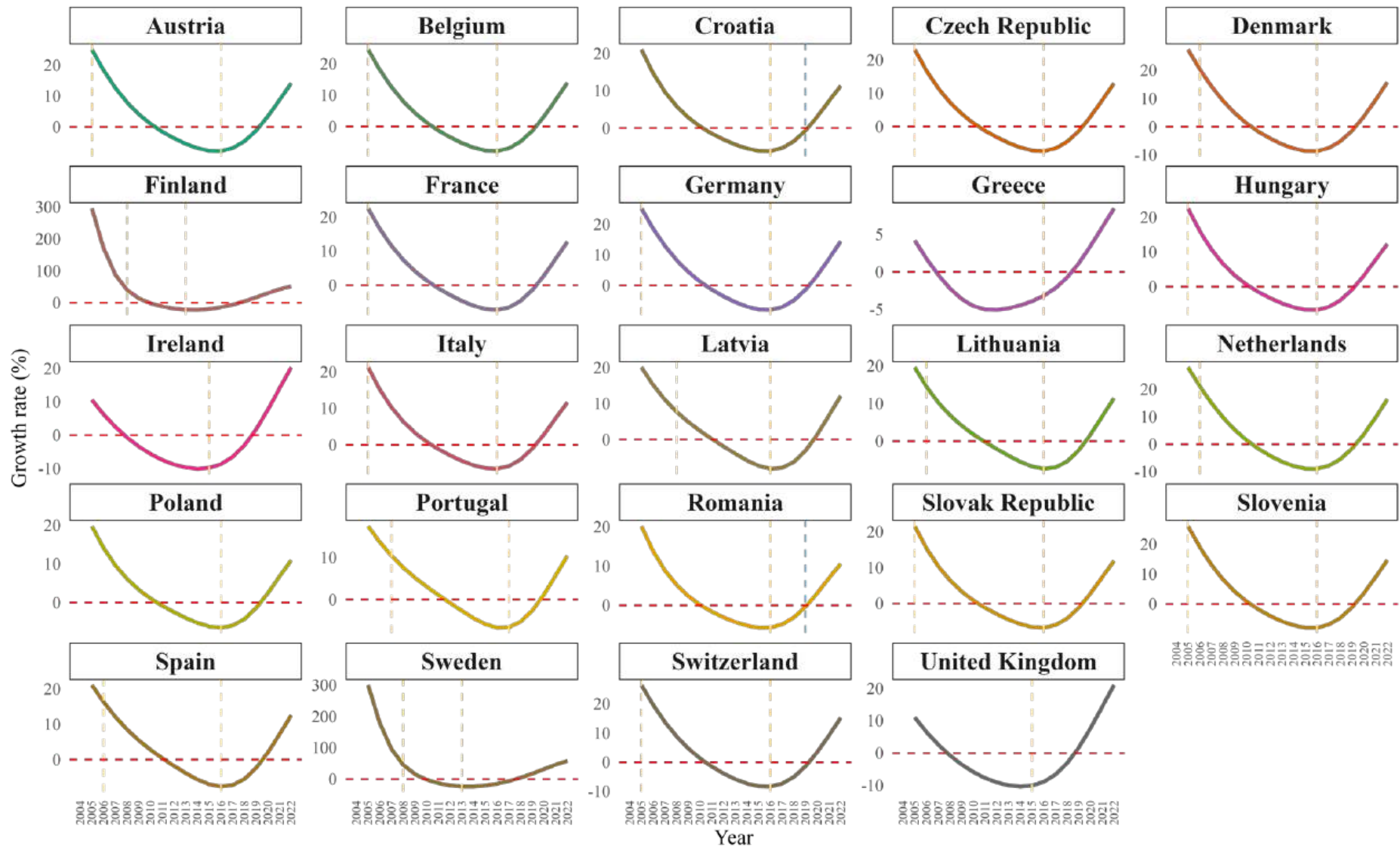
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero change.

Figure S19. Second derivative of predicted 3rd generation cephalosporin resistance in Enterobacterales using a spatiotemporal GAM across European countries



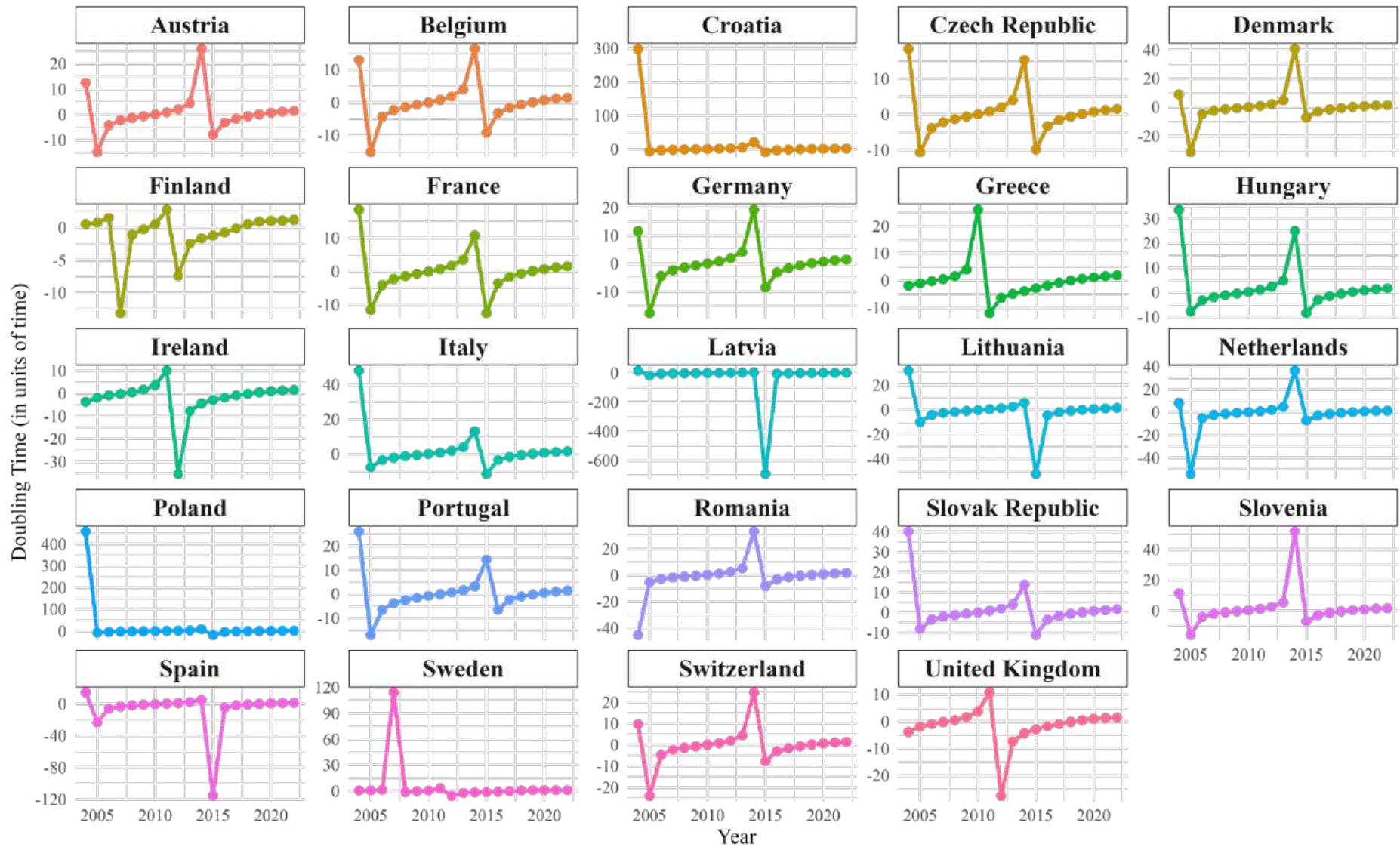
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero change.

Figure S20. Growth rates of predicted 3rd generation cephalosporin resistance in Enterobacterales using a spatiotemporal GAM across European countries



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

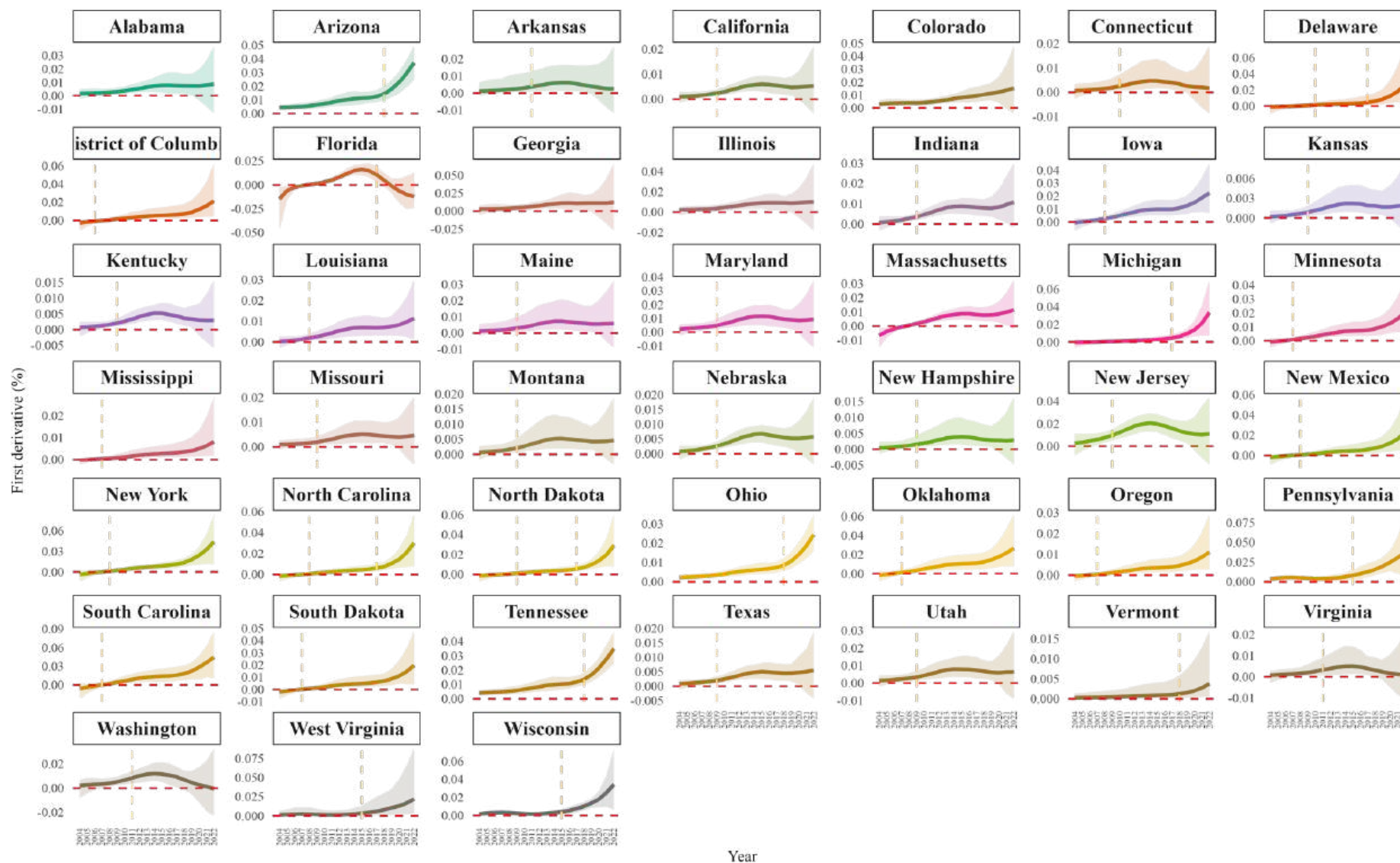
Figure S21. Doubling times for the growth rates of predicted 3rd generation cephalosporin resistance in Enterobacterales using a spatiotemporal GAM across European countries



Notes: GAM= Generalised additive model.

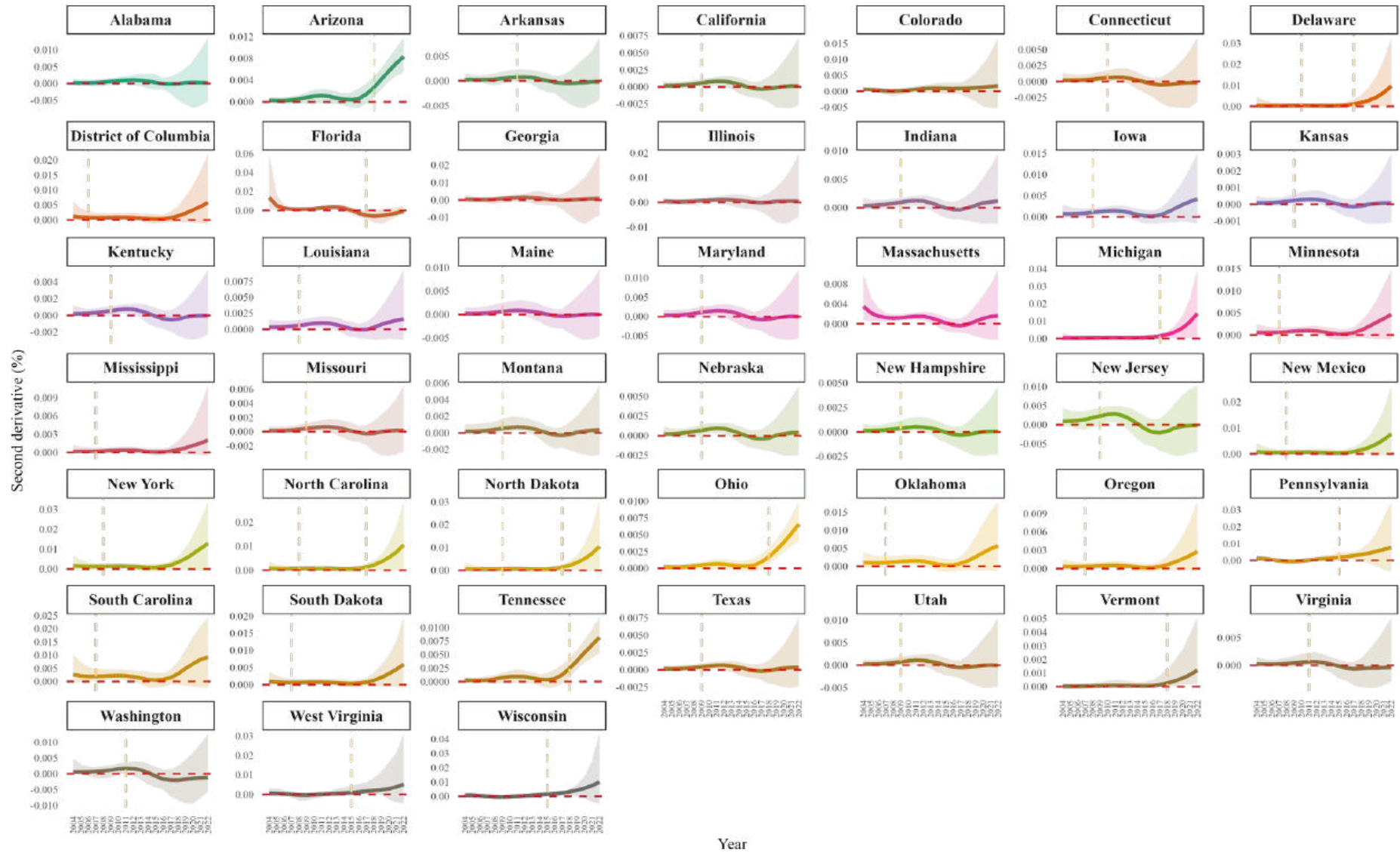
III. Carbapenem-resistance (derivatives, growth rates), US states

Figure S22. First derivative of predicted carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across US states



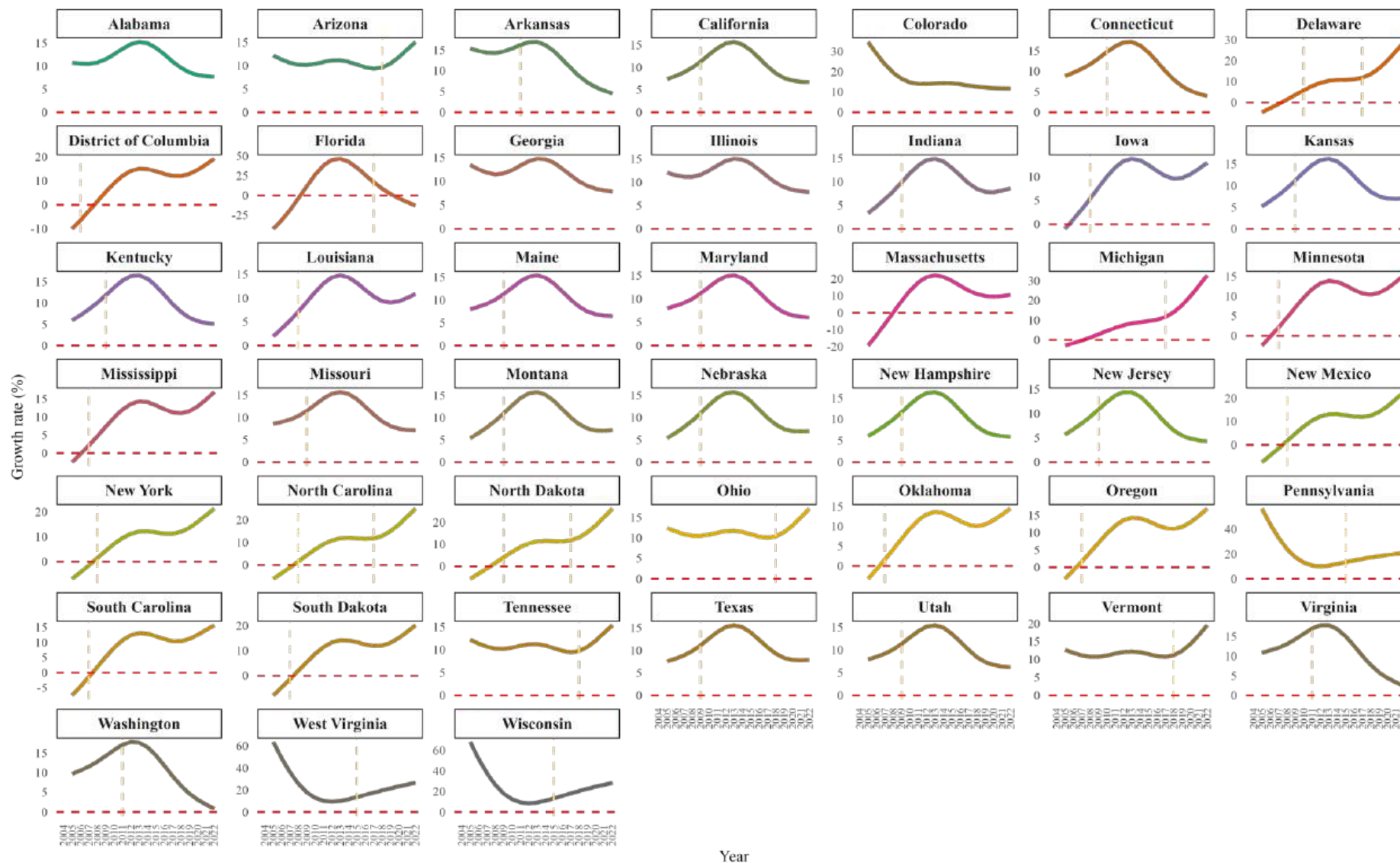
Notes: GAM= Generalised additive model. Dashed lines indicate change points (when either the 95% confidence interval or average touch zero). The red horizontal dashed line means zero change.

Figure S23. Second derivative of predicted carbapenem-resistance in Enterobacteriales using a spatiotemporal GAM across US states



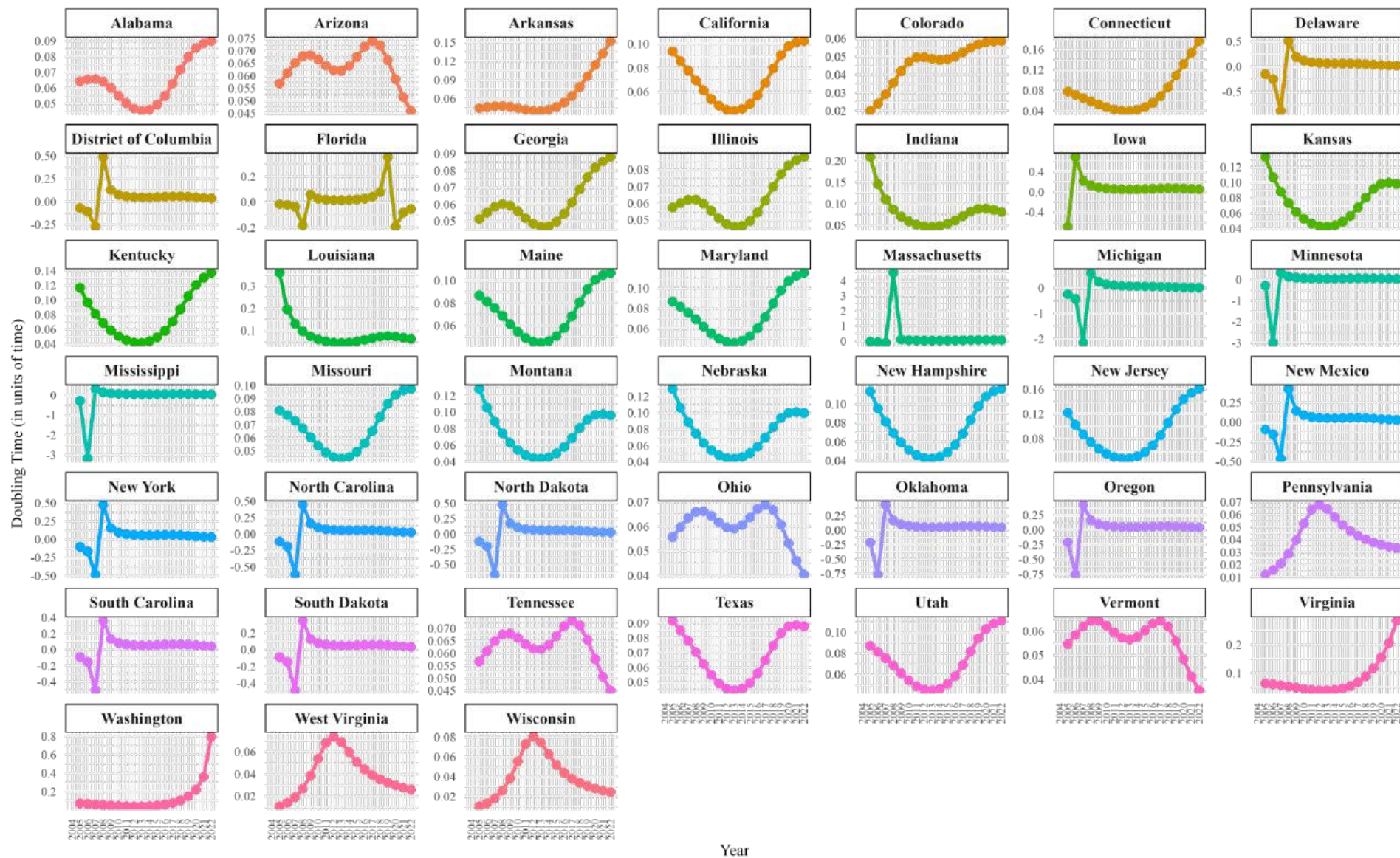
Notes: GAM= Generalised additive model. Dashed lines indicate change points (when either the 95% confidence interval or average touch zero). The red horizontal dashed line means zero change.

Figure S24. Growth rates of predicted carbapenem-resistance in Enterobacteriales using a spatiotemporal GAM across US states



Notes: GAM= Generalised additive model. Dashed lines indicate change points (when either the 95% confidence interval or average touch zero). The red horizontal dashed line means zero growth.

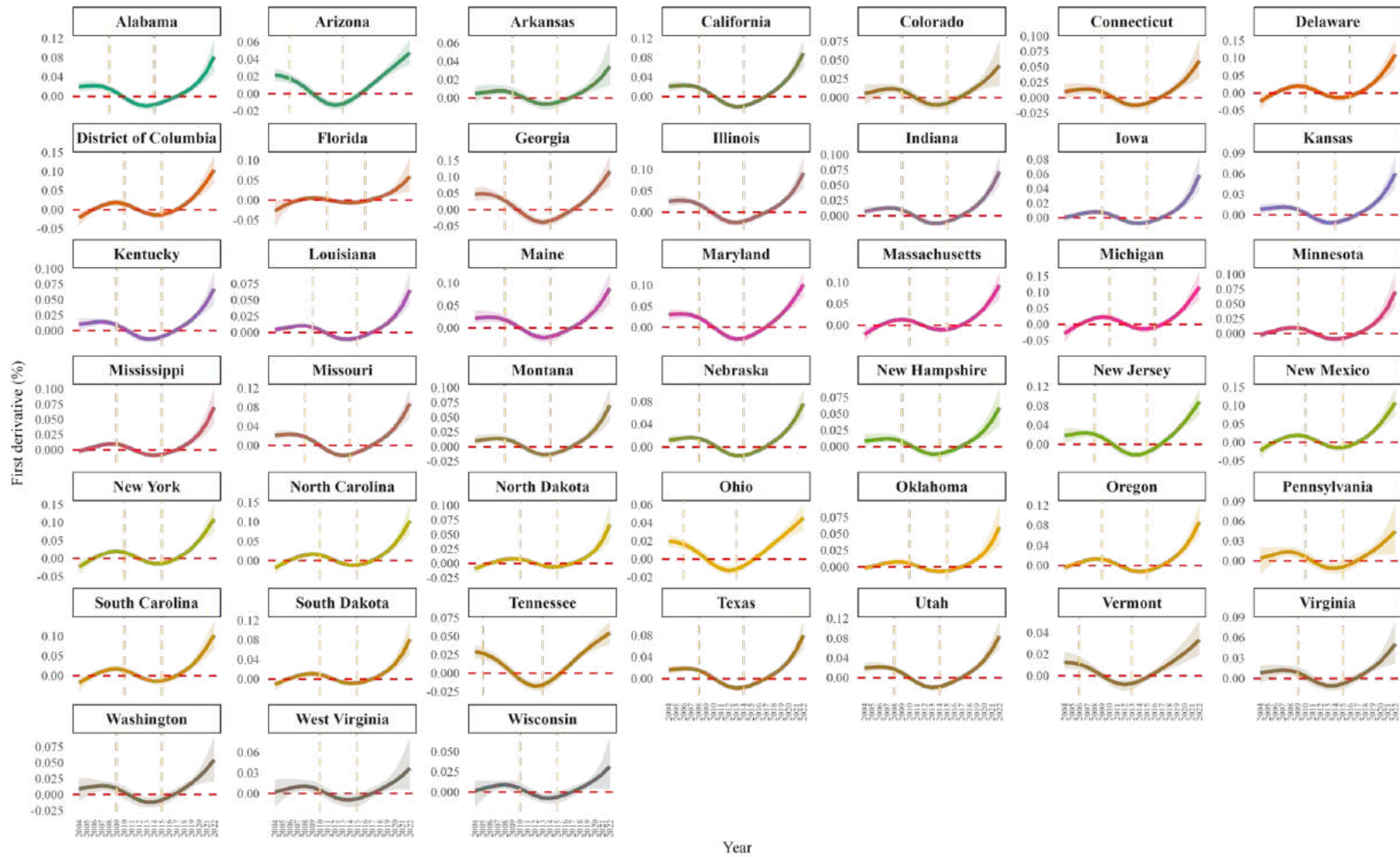
Figure S25. Doubling times for the growth rates of predicted carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across US states



Notes: GAM= Generalised additive model. Dashed lines indicate change points (when either the 95% confidence interval or average touch zero).

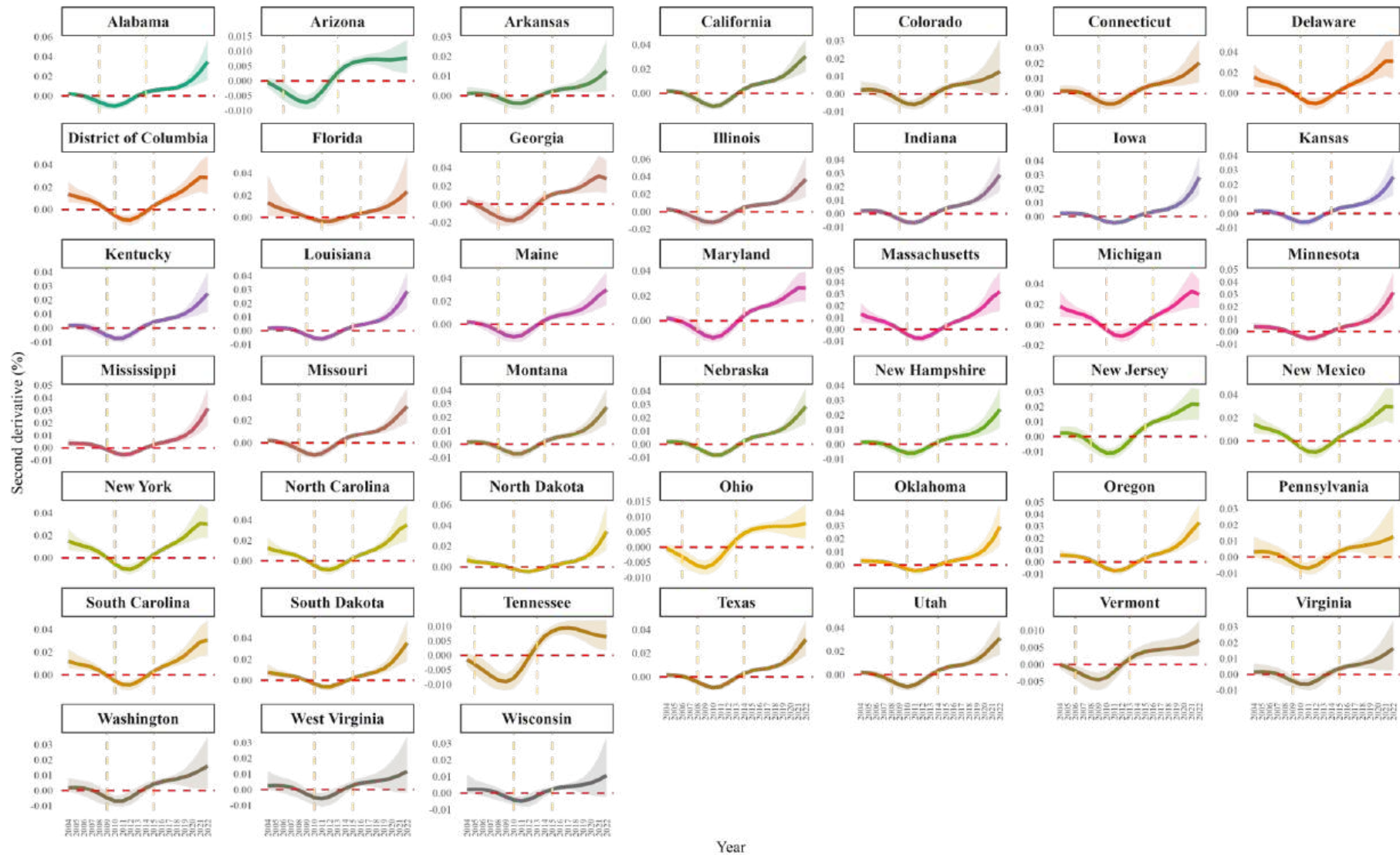
IV. 3rd generation cephalosporin resistance (derivatives, growth rates), US states

Figure S26. First derivative of predicted 3rd generation cephalosporin resistance in Enterobacterales using a spatiotemporal GAM across US states



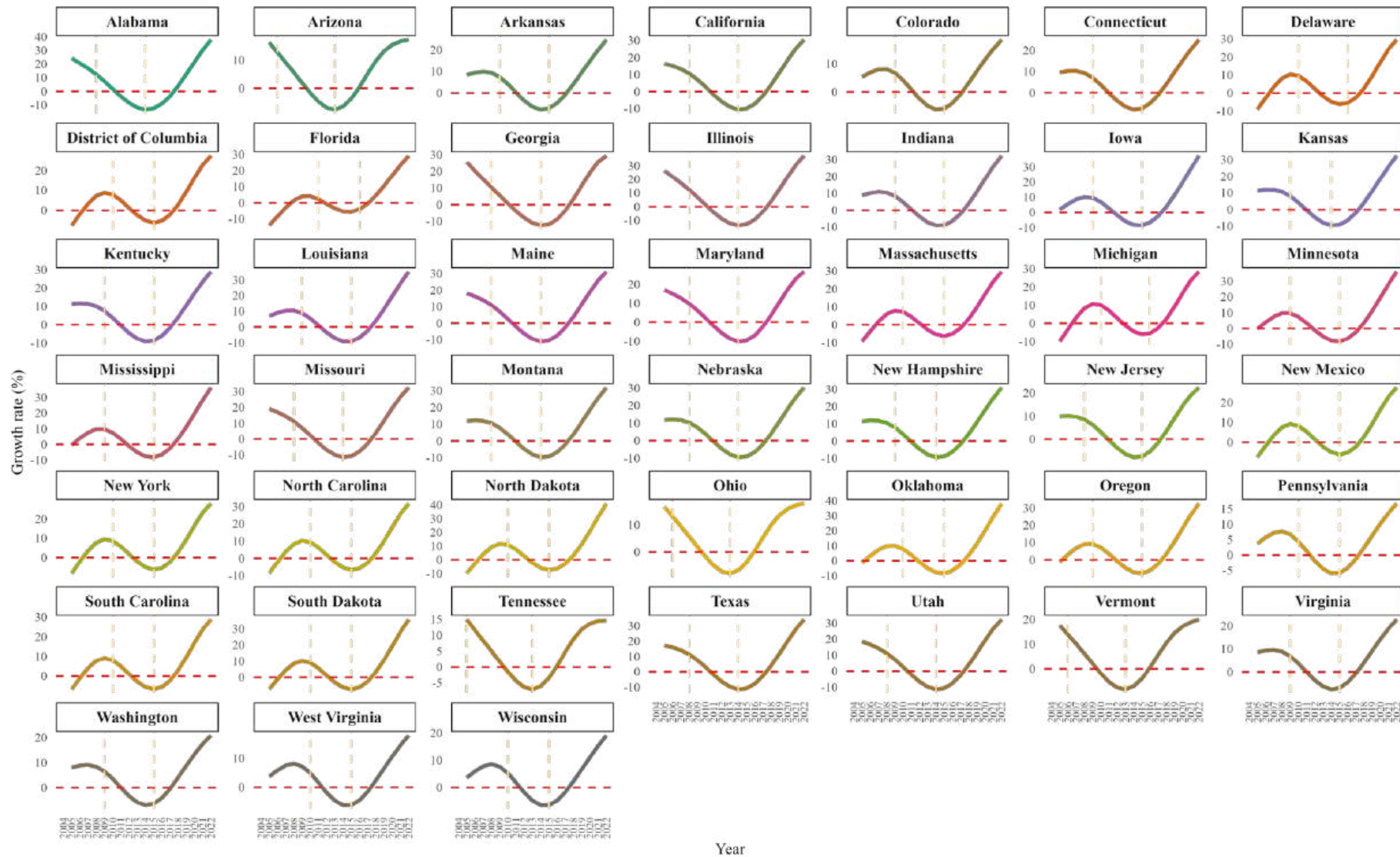
Notes: GAM= Generalised additive model. Dashed lines indicate change points (when either the 95% confidence interval or average touch zero). The red horizontal dashed line means zero change.

Figure S27. Second derivative of predicted 3rd generation cephalosporin resistance in Enterobacterales using a spatiotemporal GAM across US states



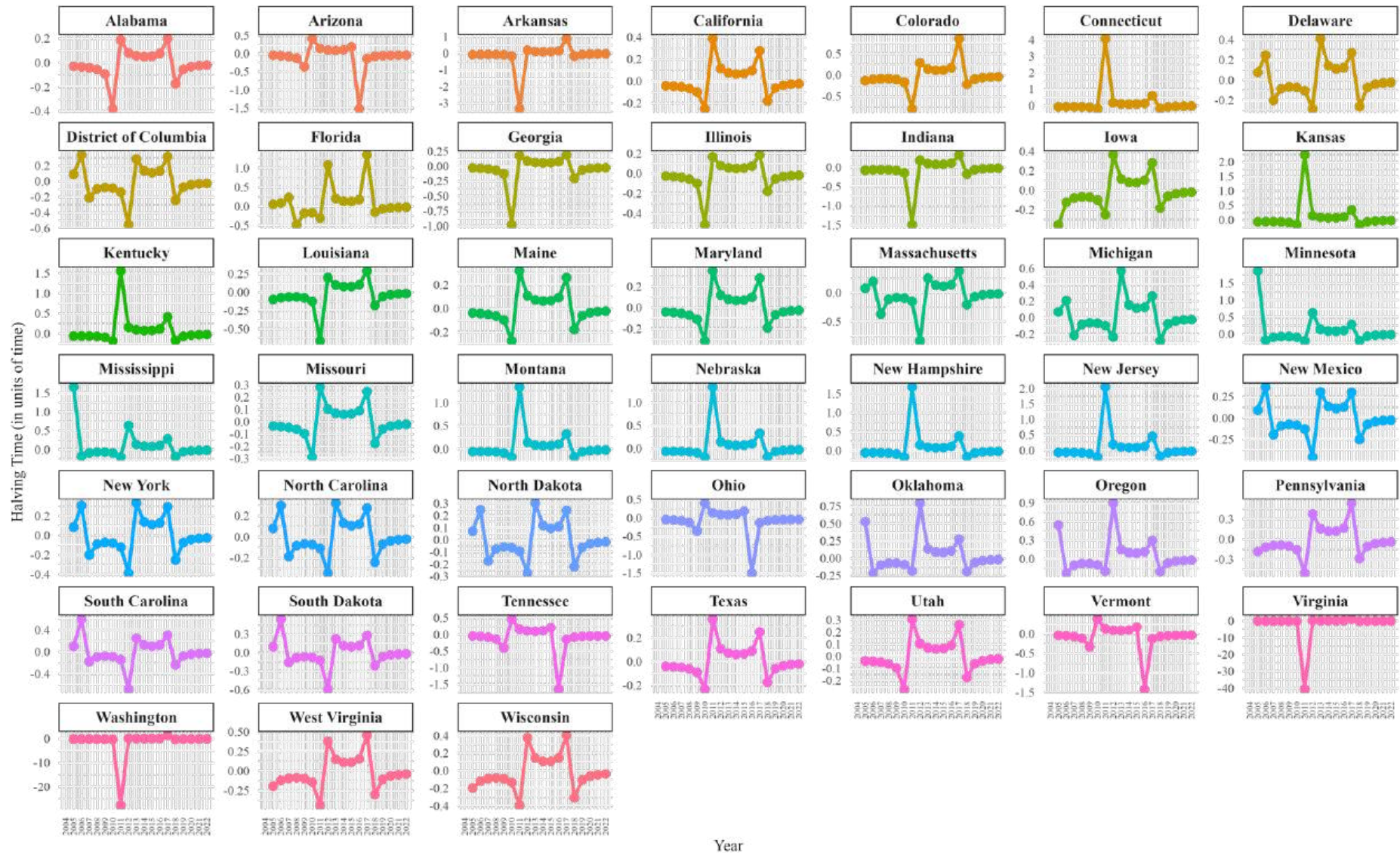
Notes: GAM= Generalised additive model. Dashed lines indicate change points (when either the 95% confidence interval or average touch zero). The red horizontal dashed line means zero change.

Figure S28. Growth rates of predicted 3rd generation cephalosporin resistance in Enterobacterales using a spatiotemporal GAM across US states



Notes: GAM= Generalised additive model. Dashed lines indicate change points (when either the 95% confidence interval or average touch zero). The red horizontal dashed line means zero growth.

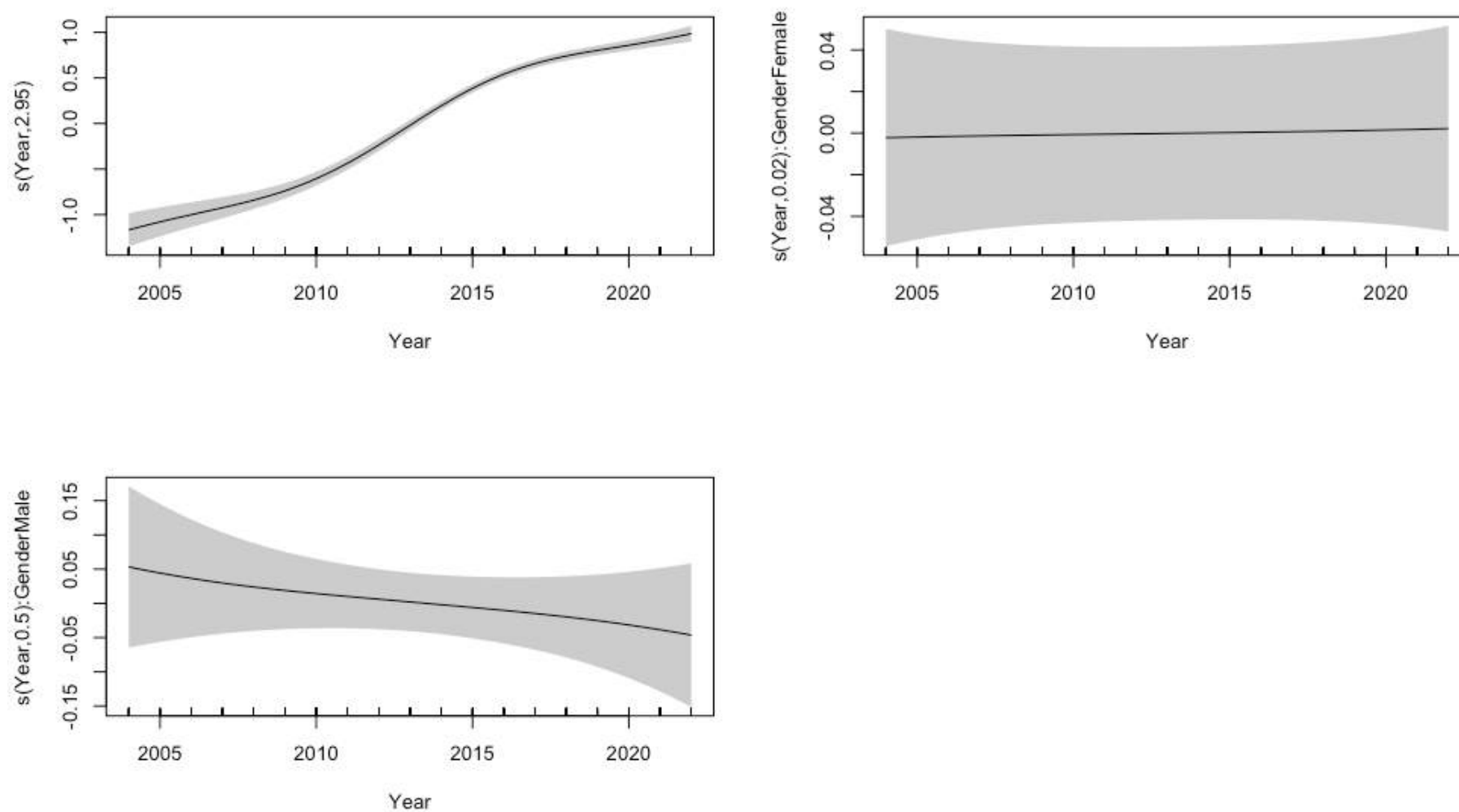
Figure S29. Doubling times for the growth rates of predicted 3rd generation cephalosporin resistance in Enterobacterales using a spatiotemporal GAM across US states



Notes: GAM= Generalised additive model. Dashed lines indicate change points (when either the 95% confidence interval or average touch zero).

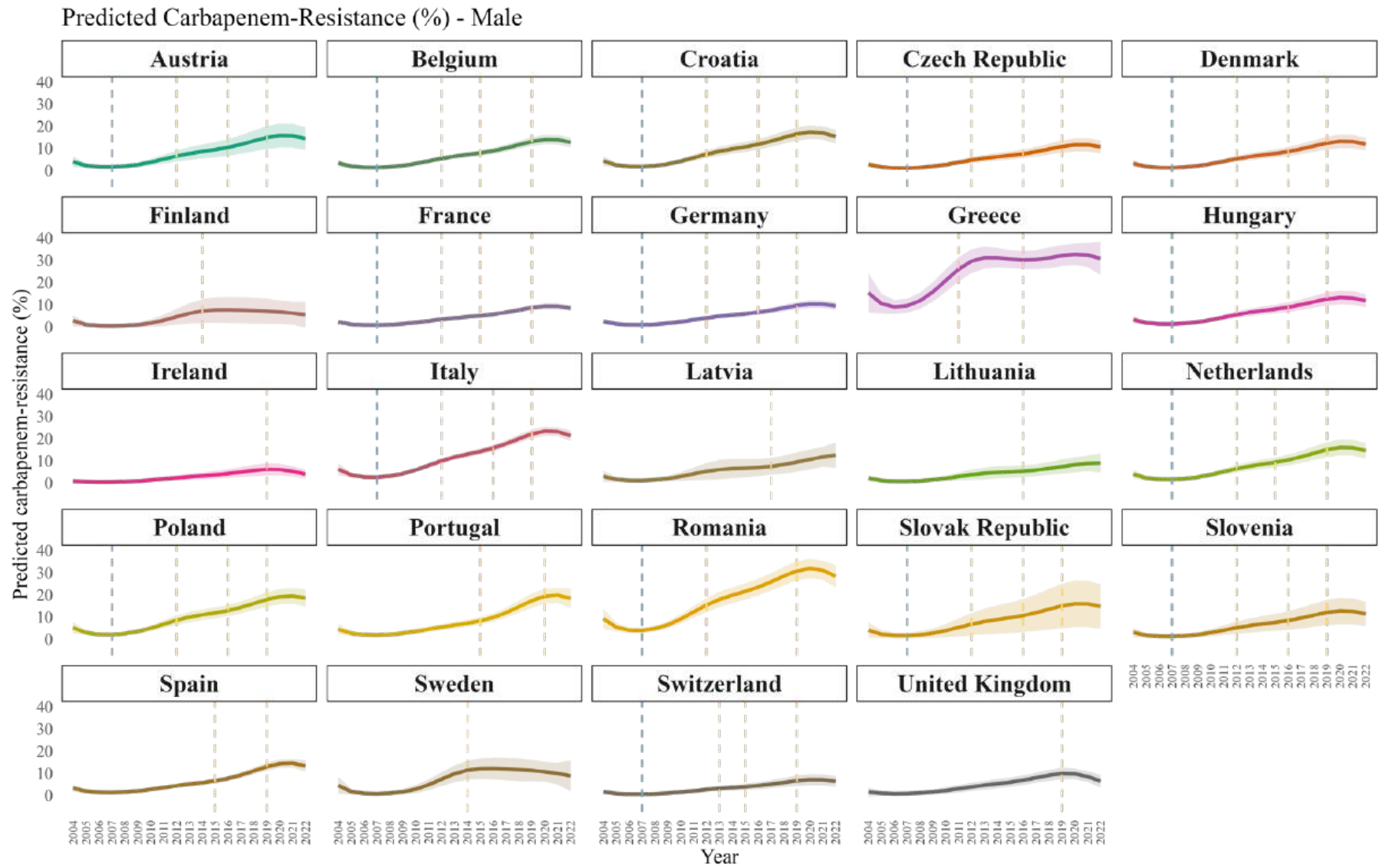
**V. Subgroup analysis by biological sex for carbapenem-resistance,
Europe and the US**

Figure S30. Estimated carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across European countries using biological sex and time interactions



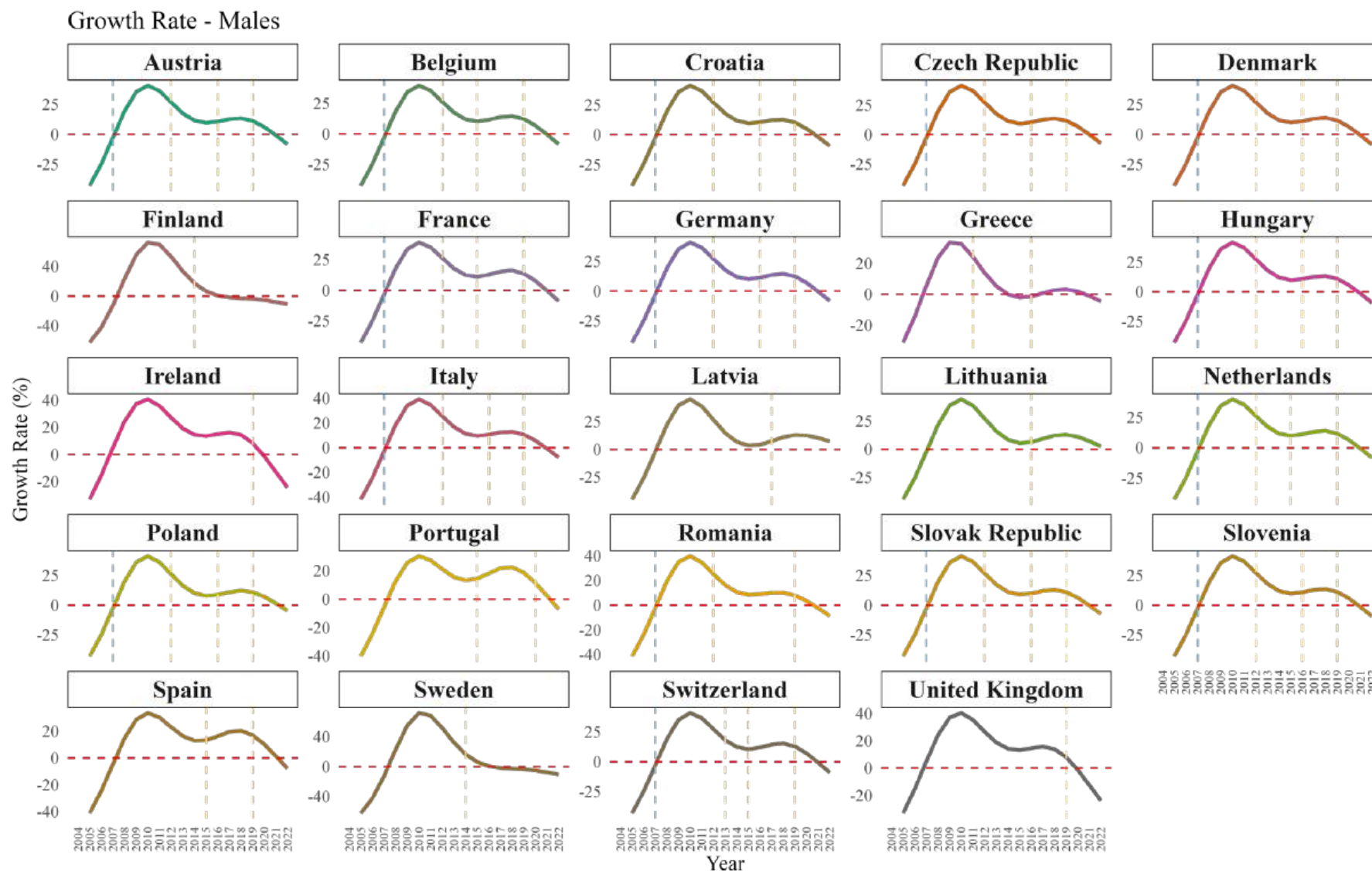
Notes: GAM= Generalised additive models.

Figure S31. Estimated carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, males**



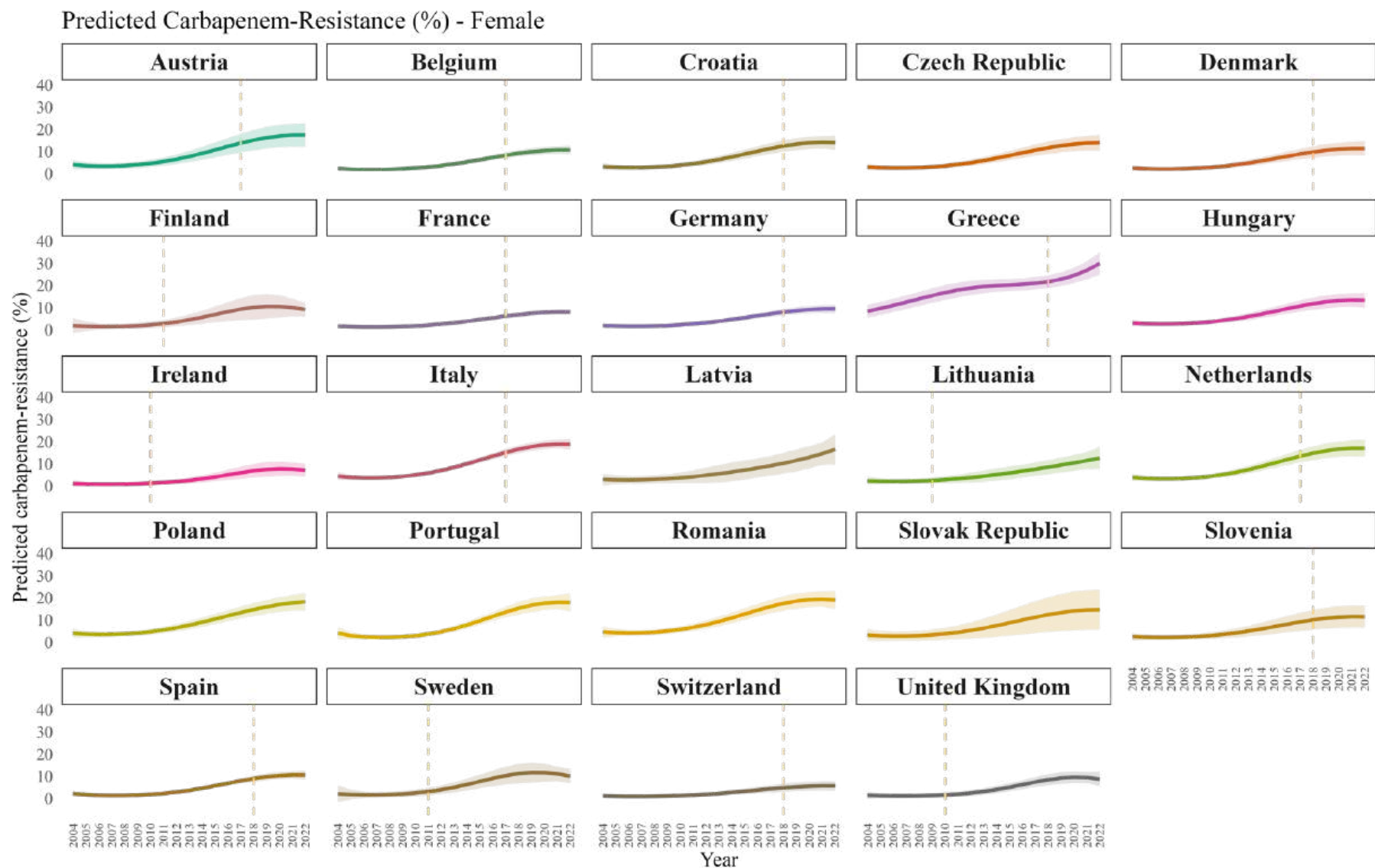
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa).

Figure S32. Growth rate of carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, males**



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

Figure S33. Predicted carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, females**



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa).

Figure S34. Growth rate of carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, females**

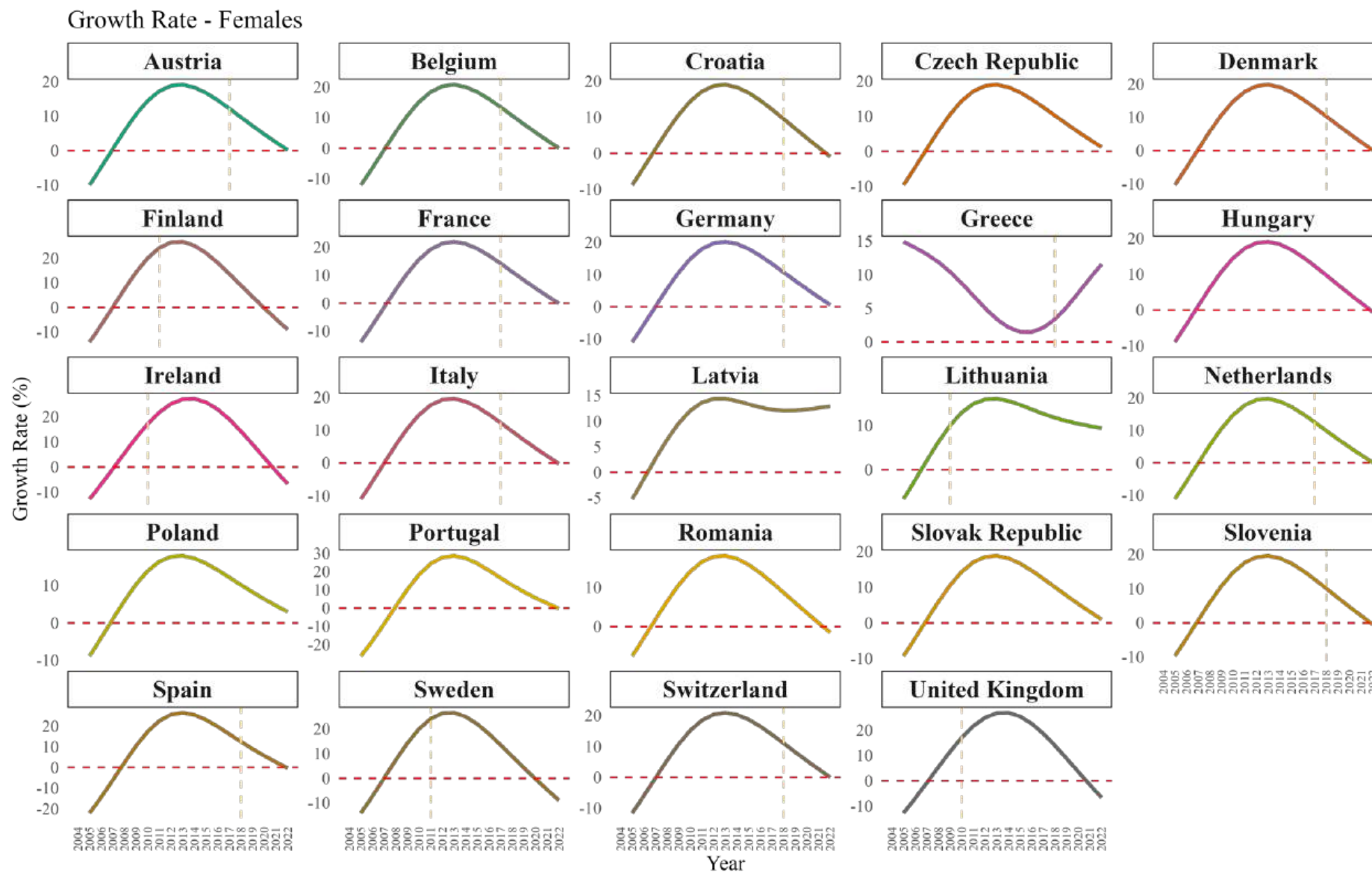
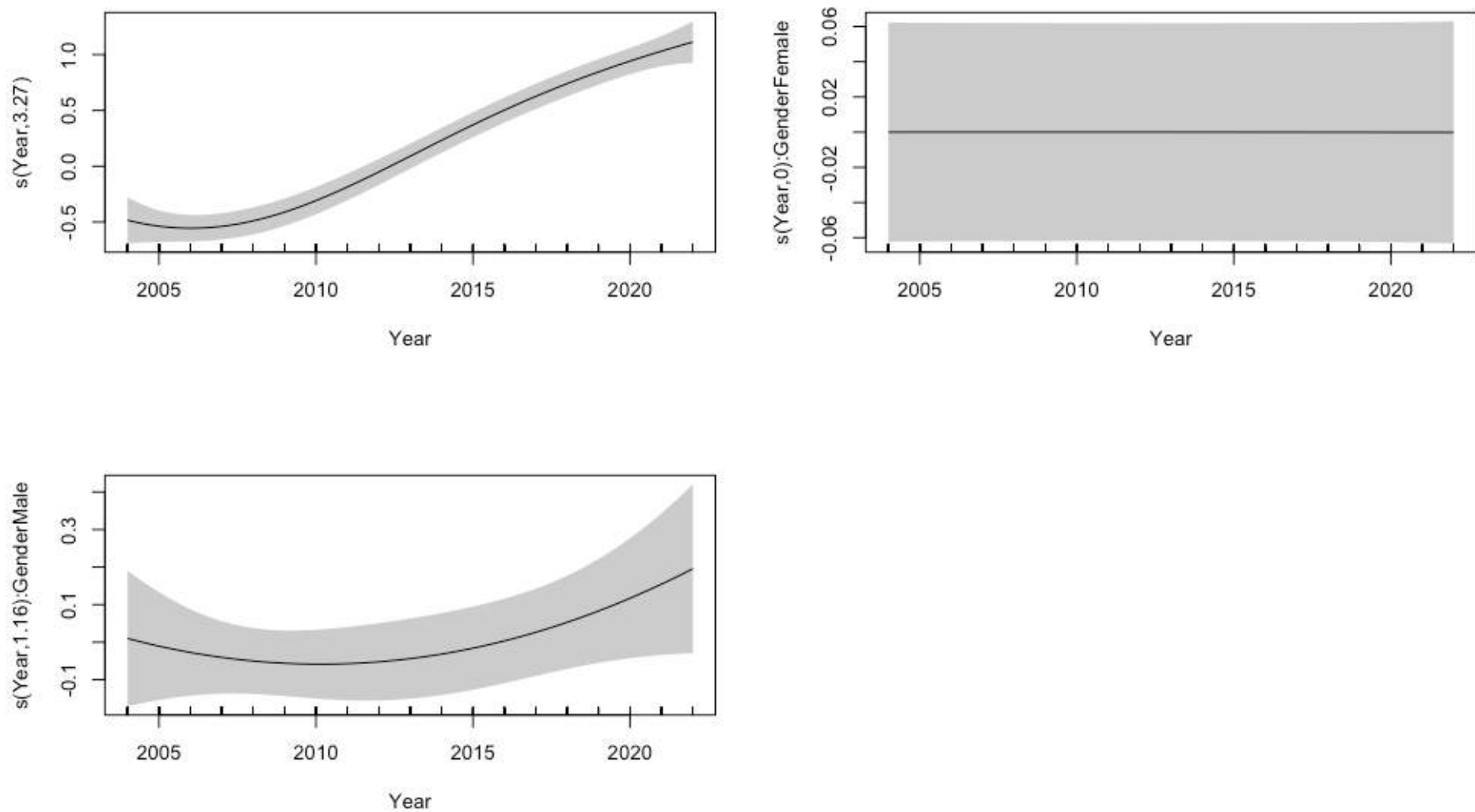


Figure S35. Estimated carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across US-states using biological sex and time interactions



Notes: GAM= Generalised additive models.

Figure S36. Estimated carbapenem-resistance in Enterobacteriales using a spatiotemporal GAM across **US states, males**

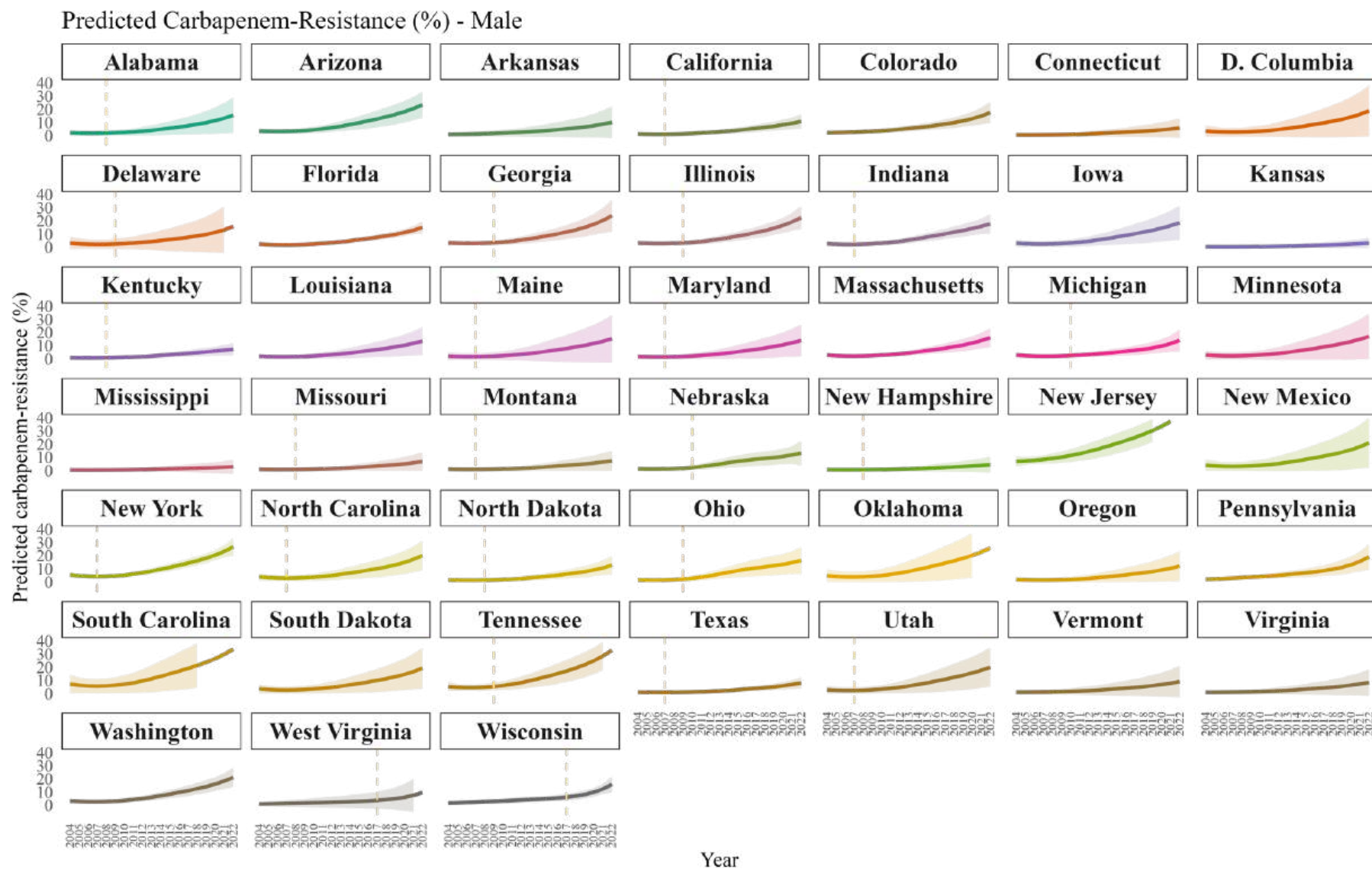
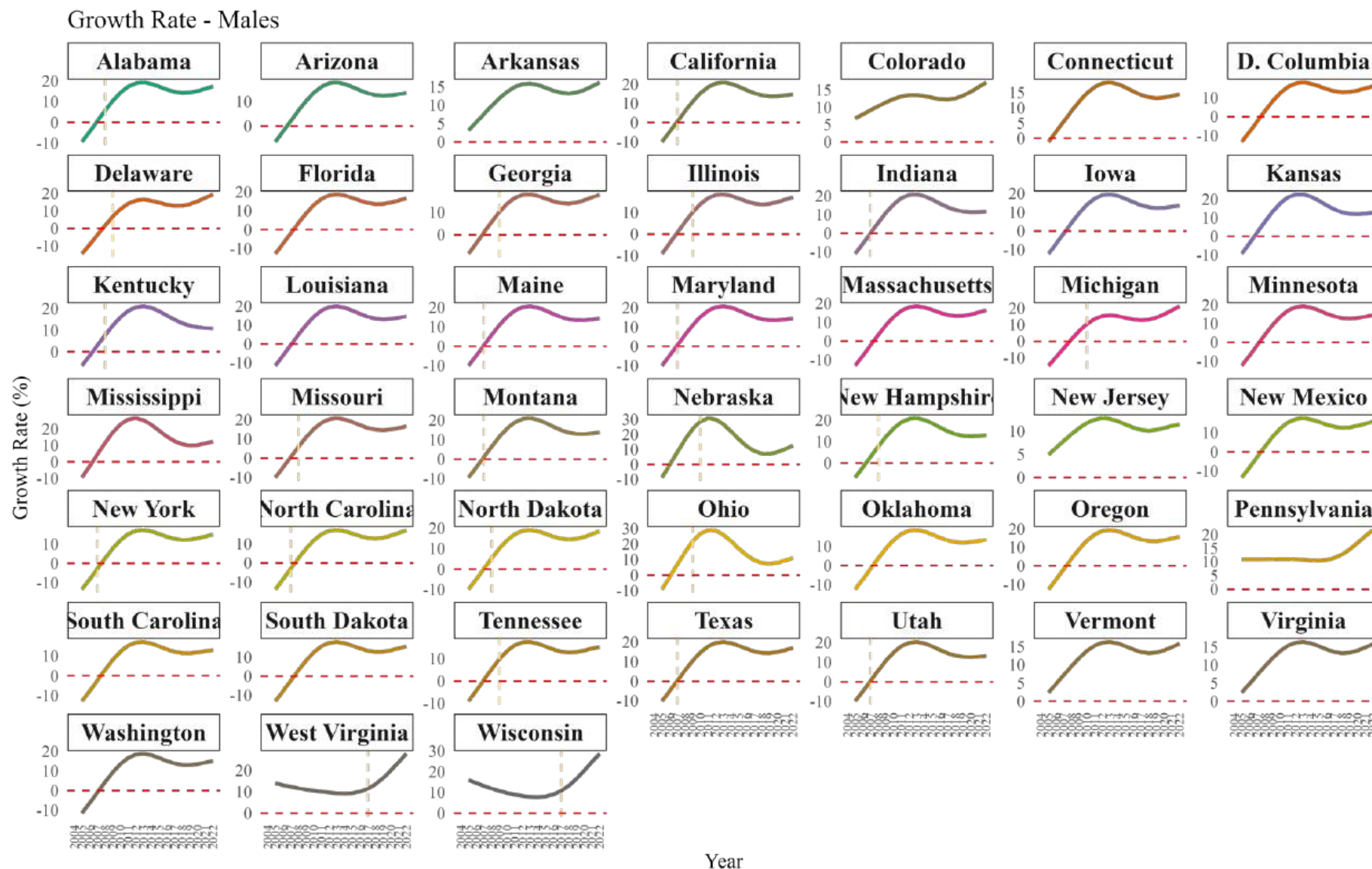
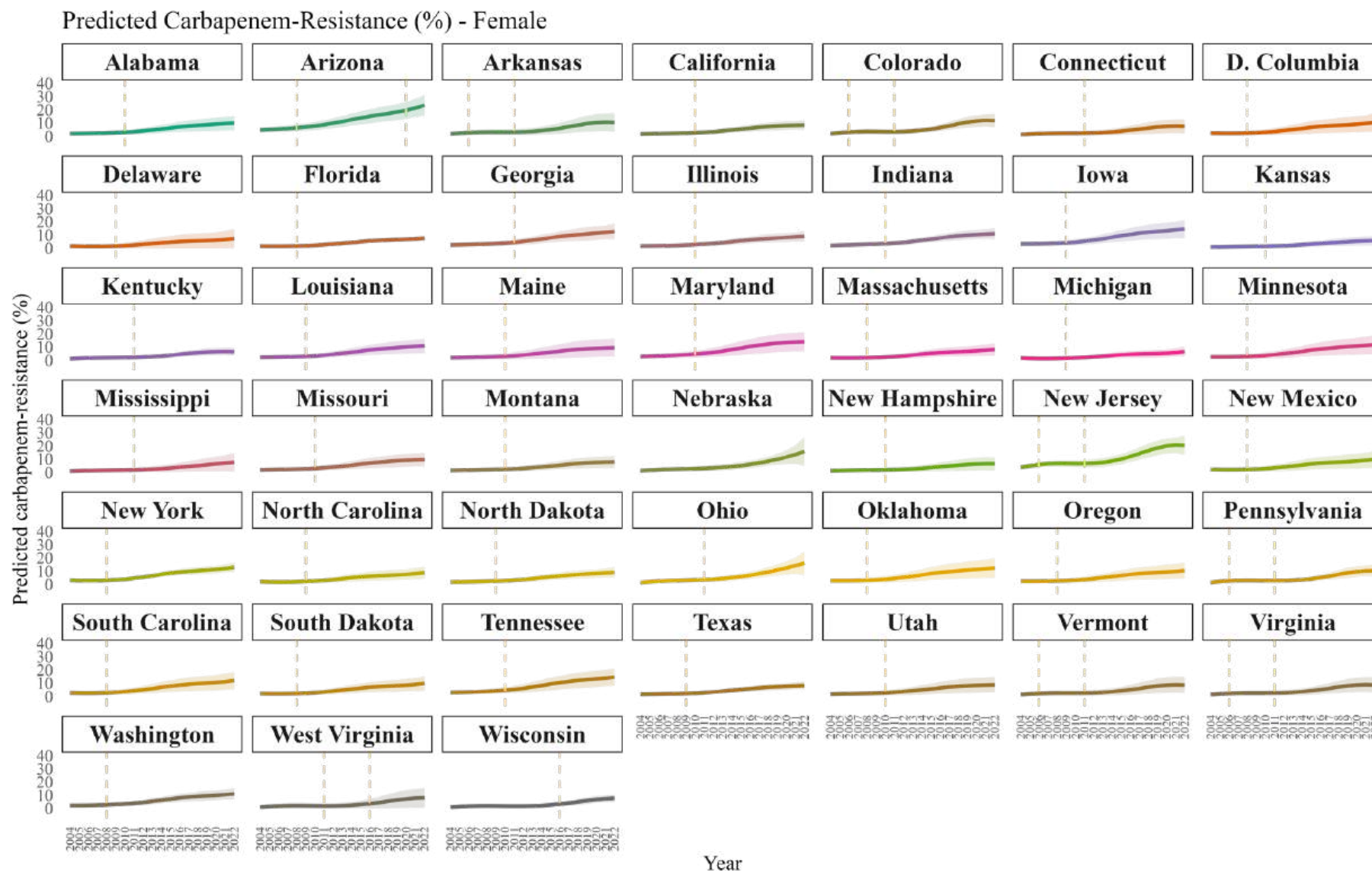


Figure S37. Growth rate of carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across **US-states, males**



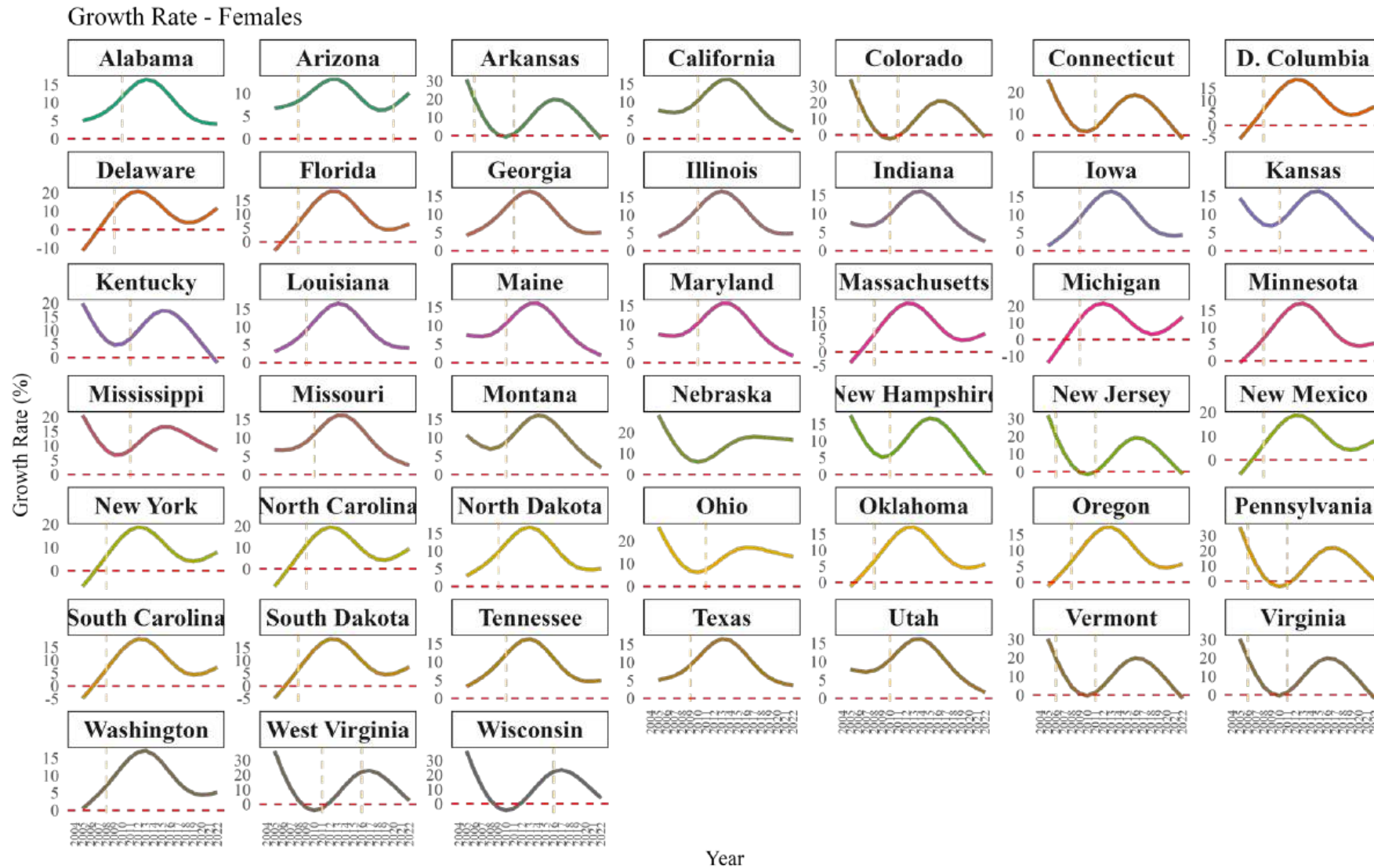
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

Figure S38. Estimated carbapenem-resistance in Enterobacteriales using a spatiotemporal GAM across **US states, females**



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa).

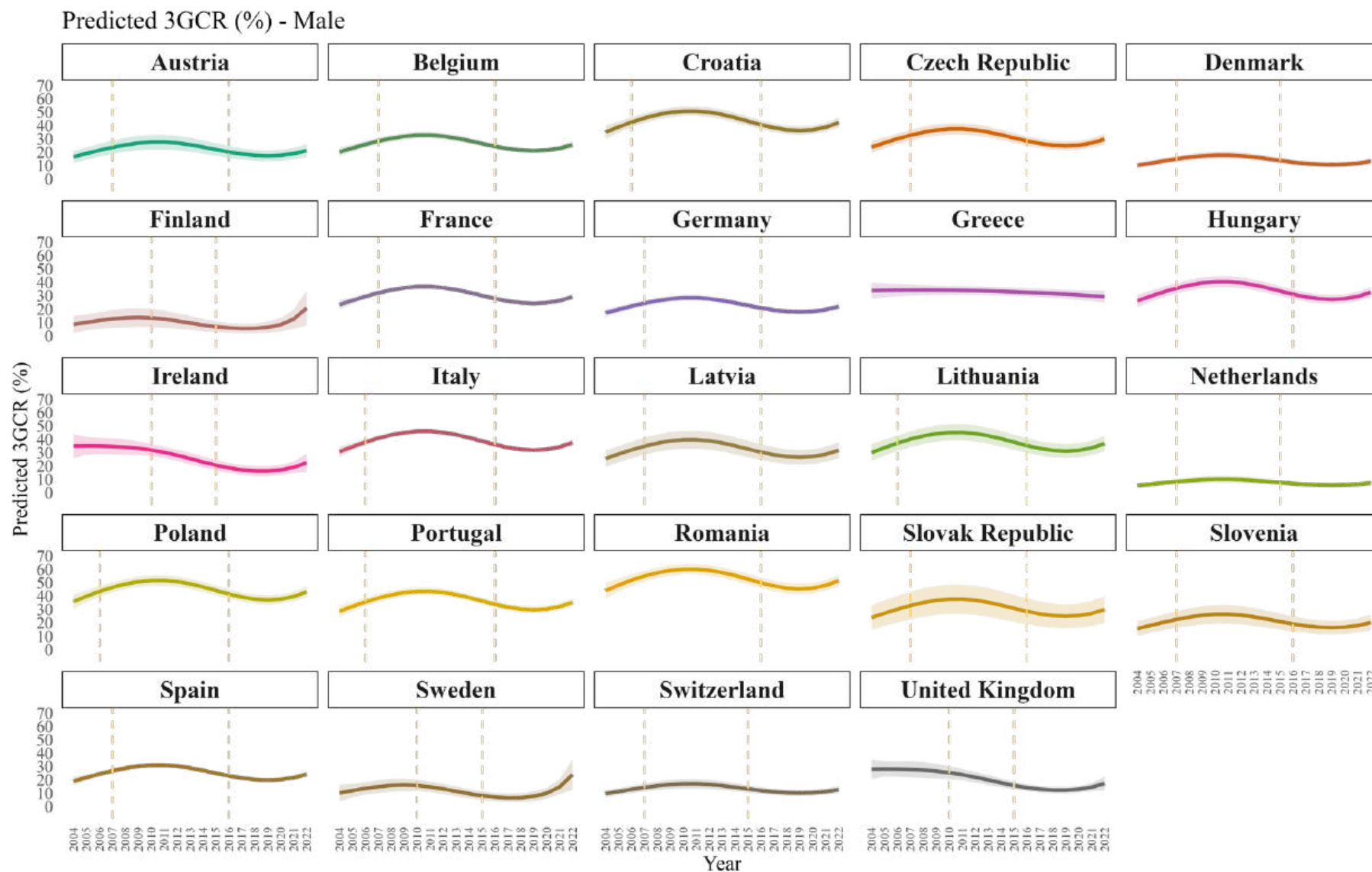
Figure S39. Growth rate of carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across **US-states, females**



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero change.

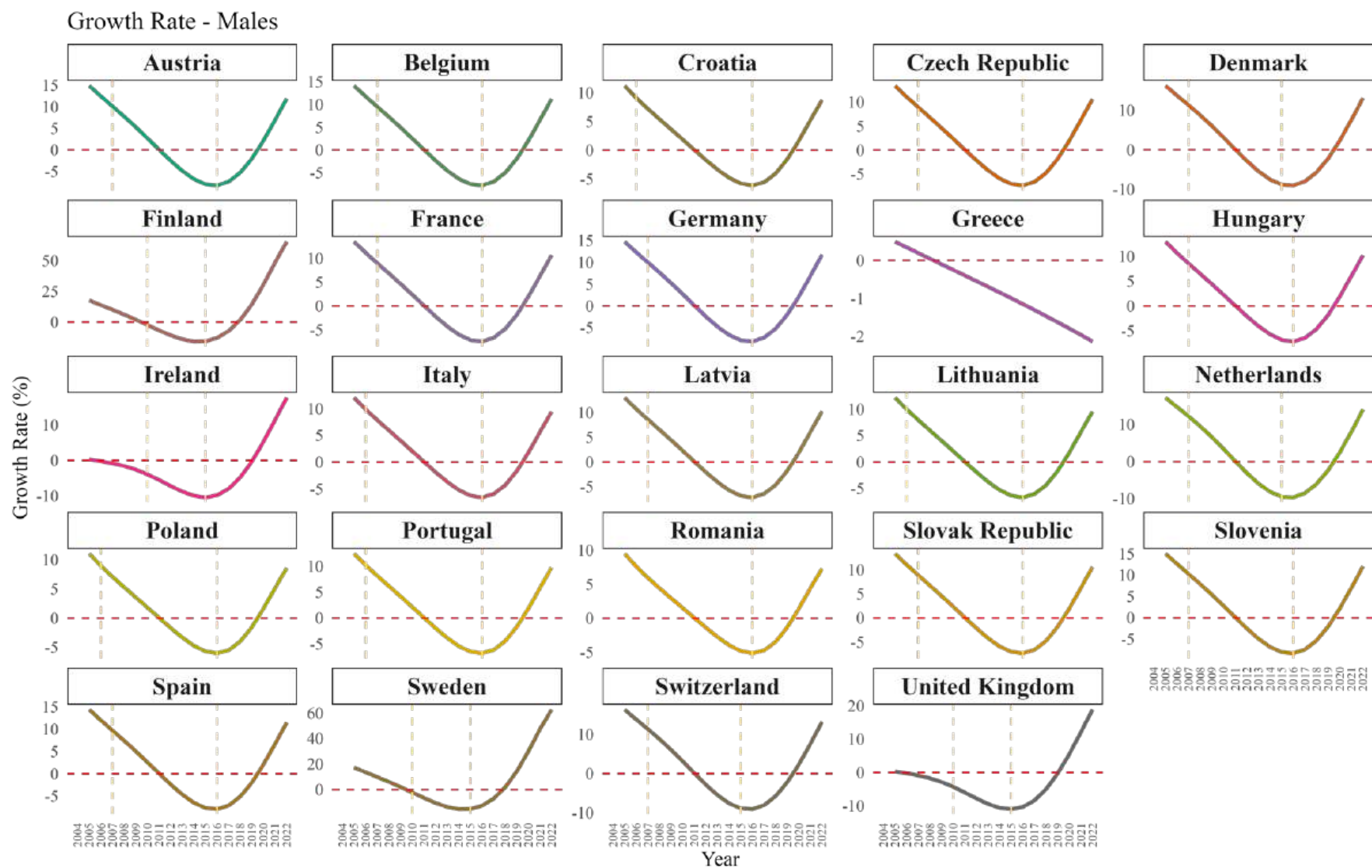
VI. Subgroup analysis by biological sex for 3rd generation cephalosporin resistance, **Europe and the US**

Figure S40. Predicted 3GCR in Enterobacteriales using a spatiotemporal GAM across **European countries, males**



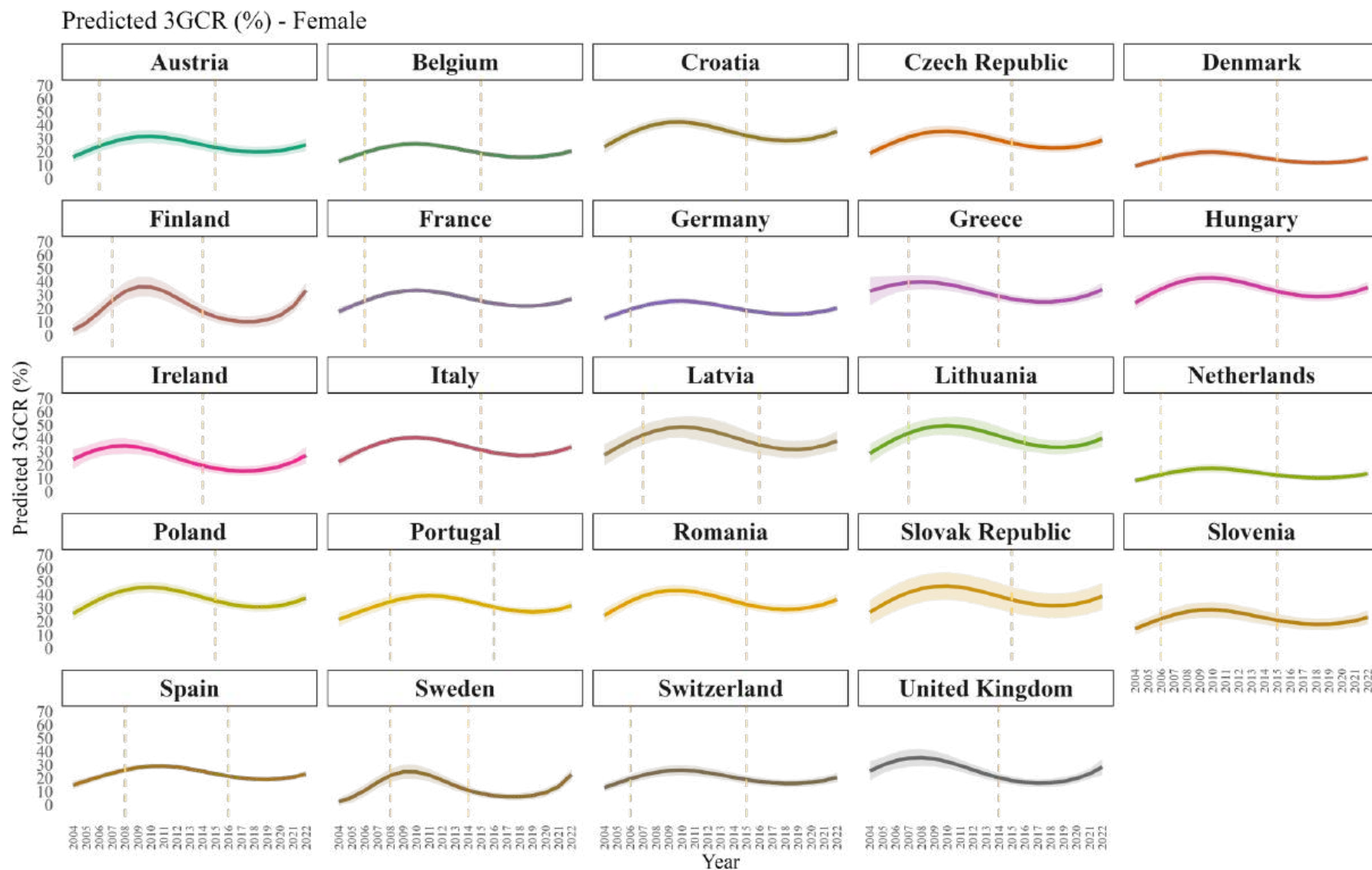
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). 3GCR= Third generation cephalosporin resistance.

Figure S41. Growth rate of third generation cephalosporin-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, males**



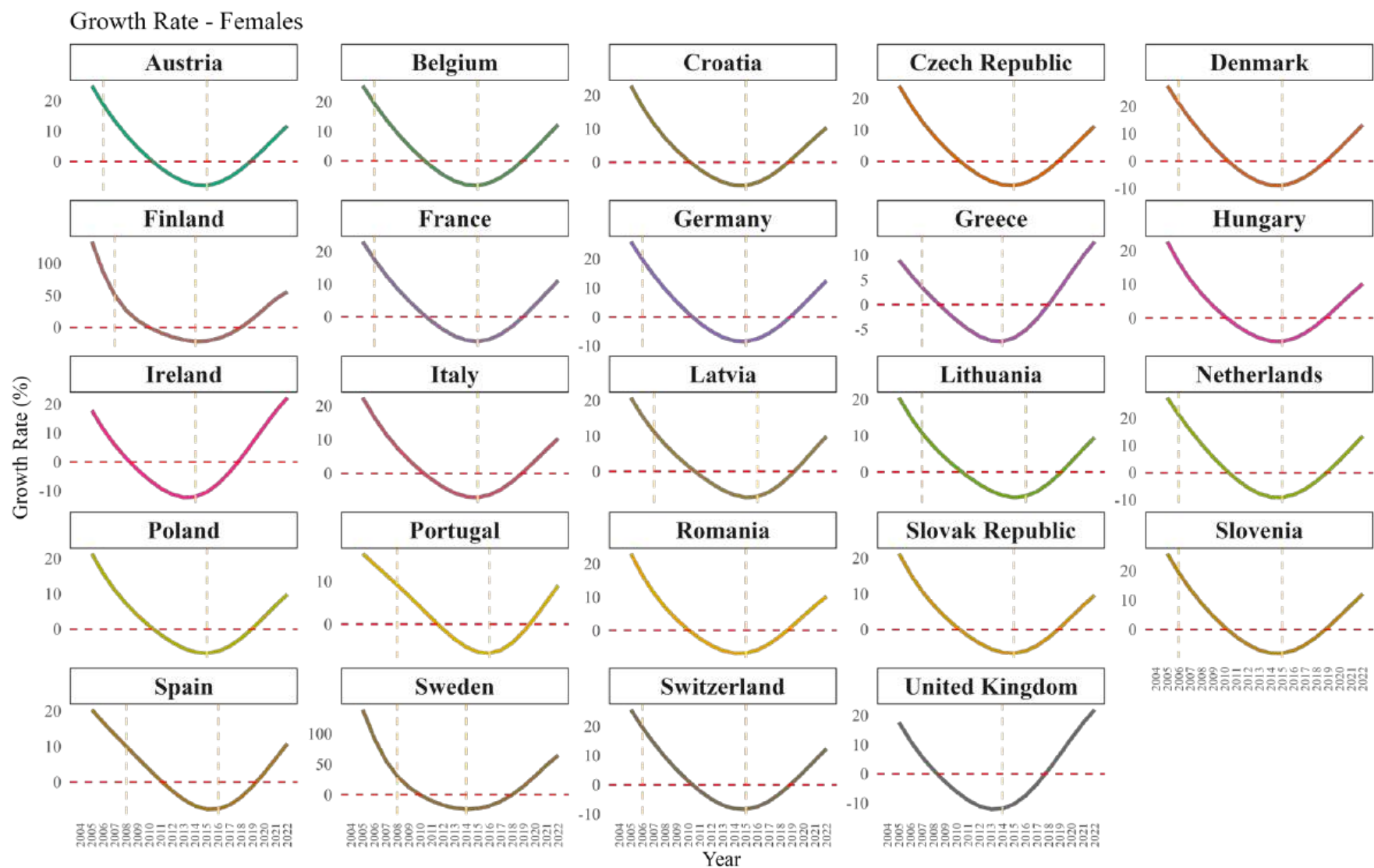
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

Figure S42. Predicted 3GCR in Enterobacteriales using a spatiotemporal GAM across **European countries, females**



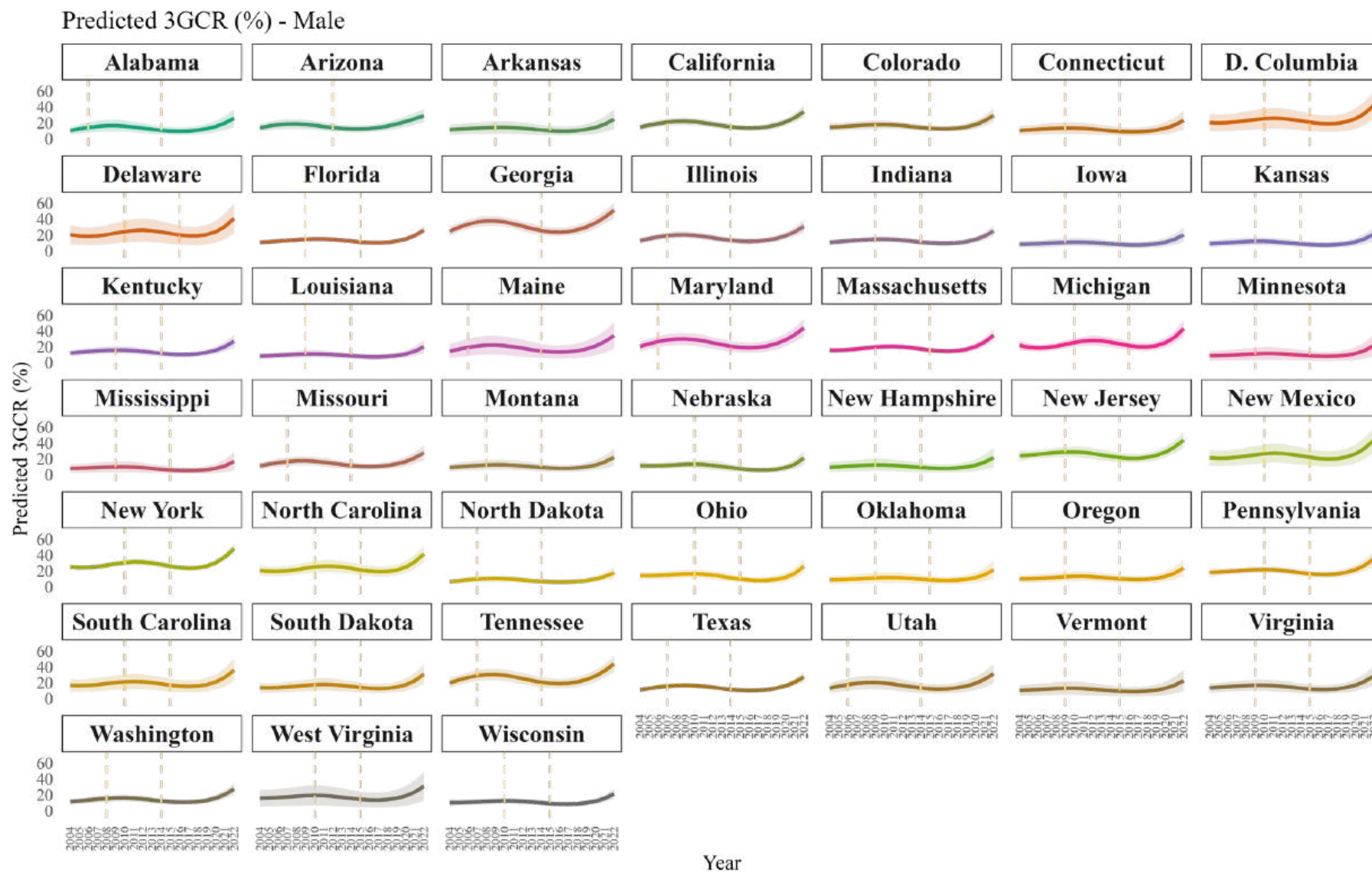
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). 3GCR= Third generation cephalosporin resistance.

Figure S43. Growth rate of third generation cephalosporin-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, females**



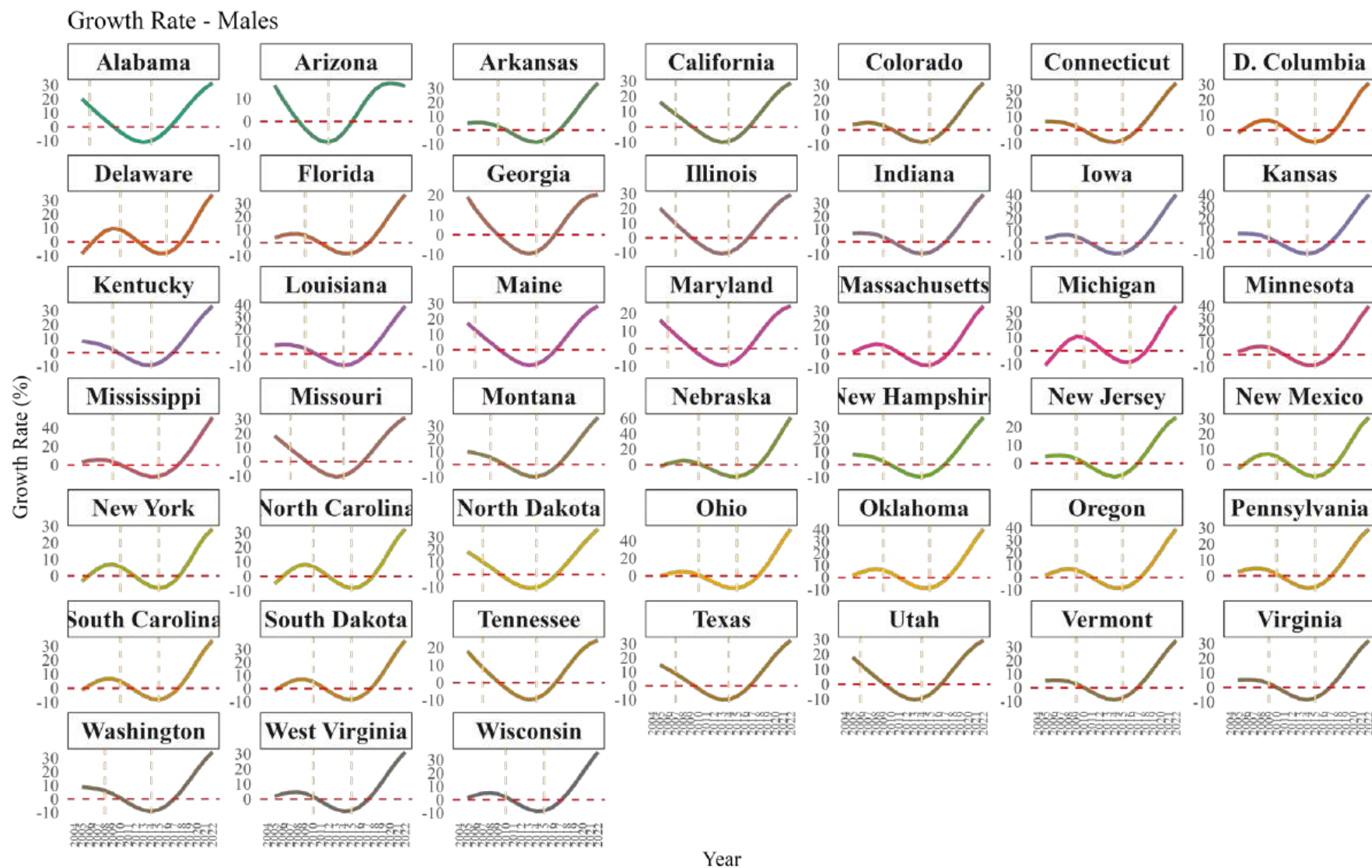
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

Figure S44. Predicted 3GCR in Enterobacteriales using a spatiotemporal GAM across US states, males



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). 3GCR= Third generation cephalosporin resistance.

Figure S45. Growth rate of third-generation cephalosporin-resistance in Enterobacteriales using a spatiotemporal GAM across **US-states, males**



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

Figure S46. Predicted 3GCR in Enterobacteriales using a spatiotemporal GAM across US states, females

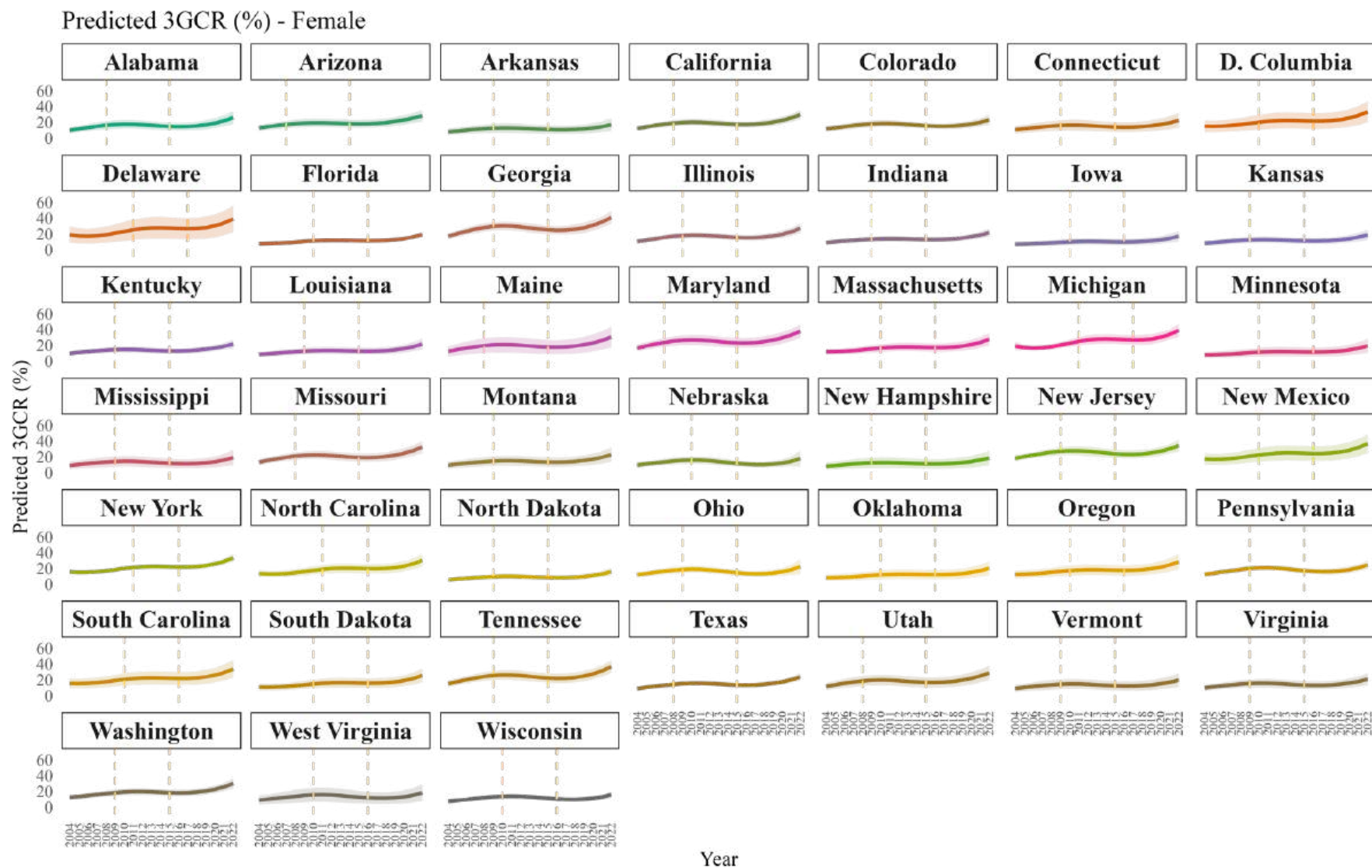
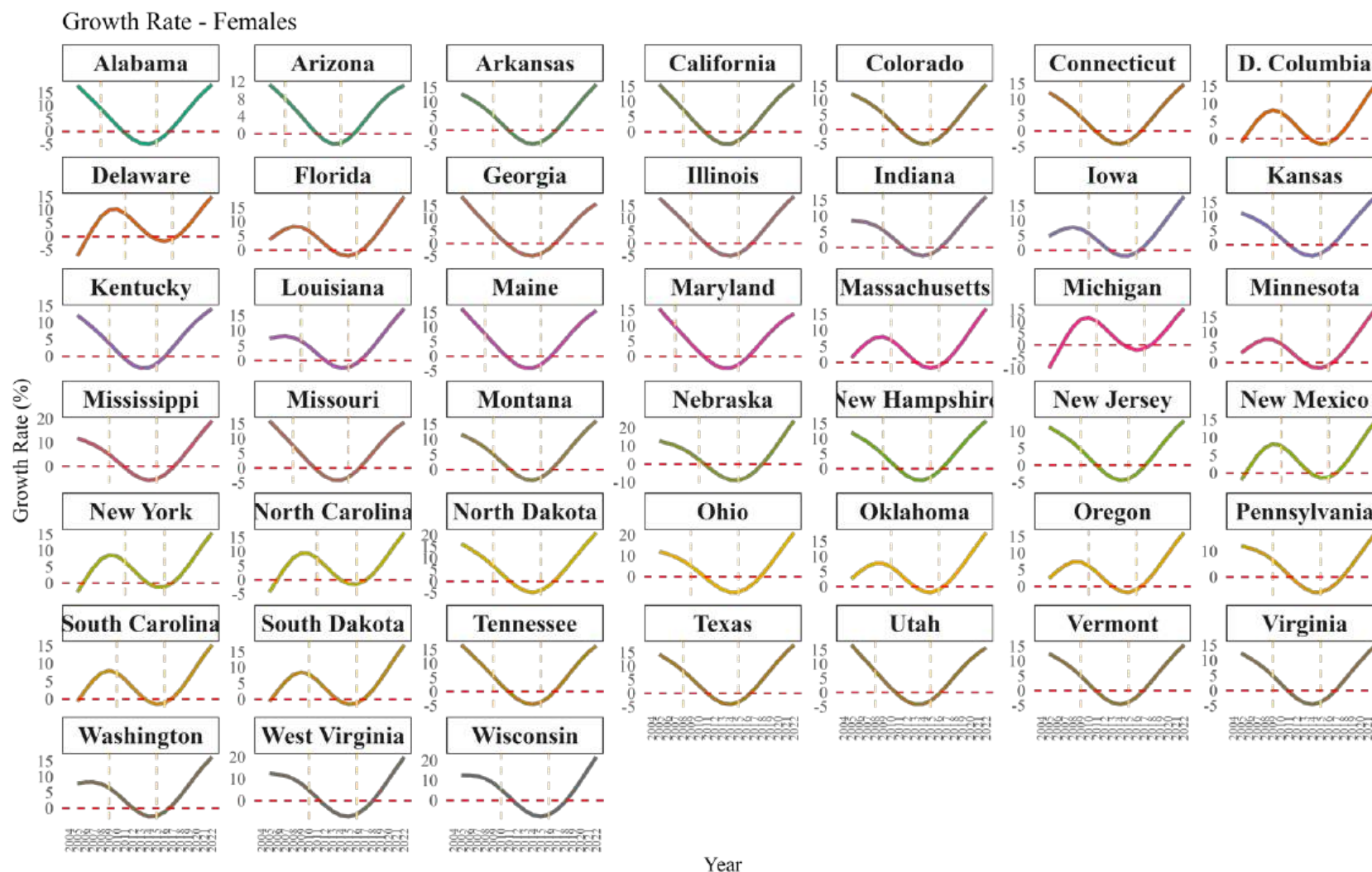
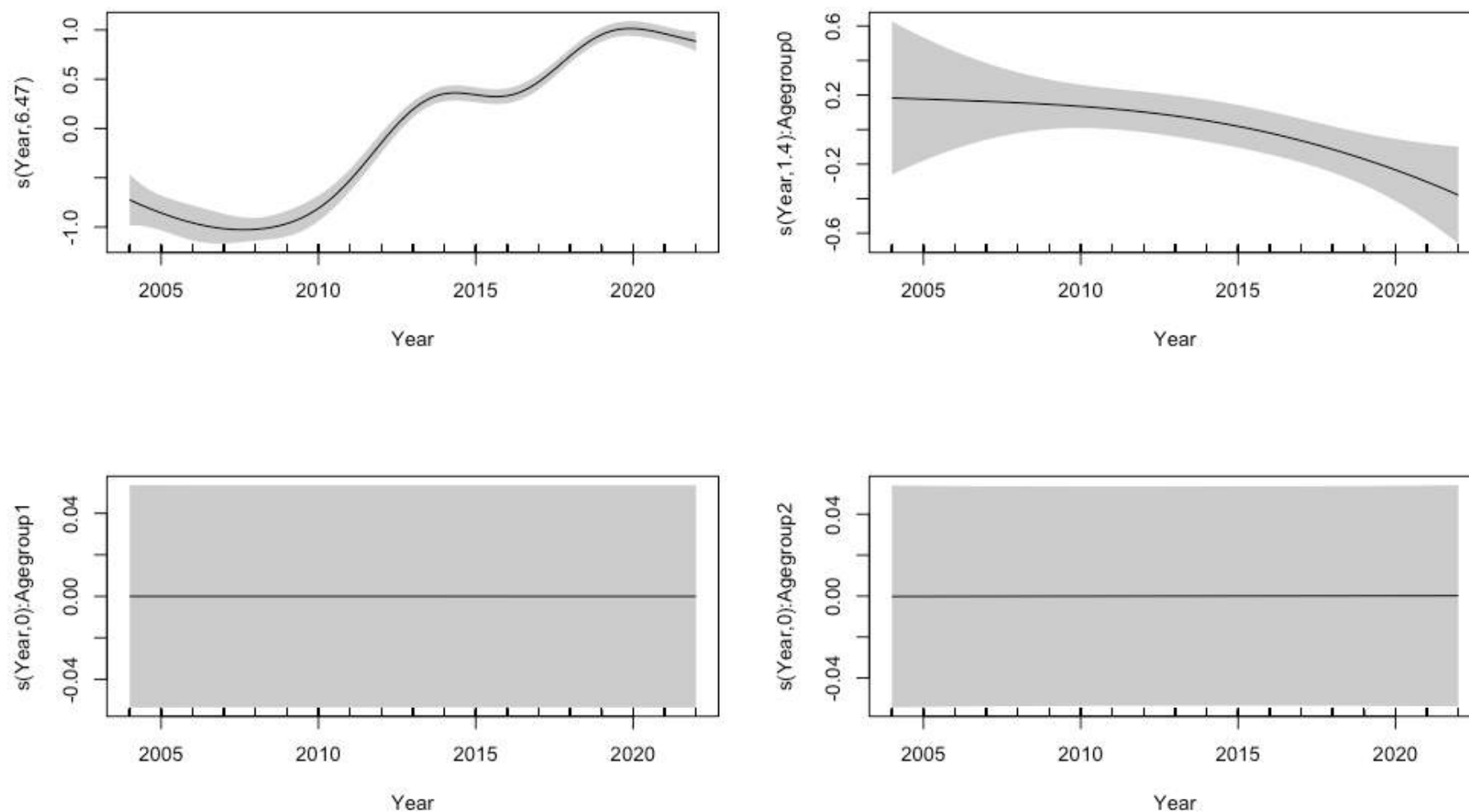


Figure S47. Growth rate of third-generation cephalosporin-resistance in Enterobacterales using a spatiotemporal GAM across **US-states, females**



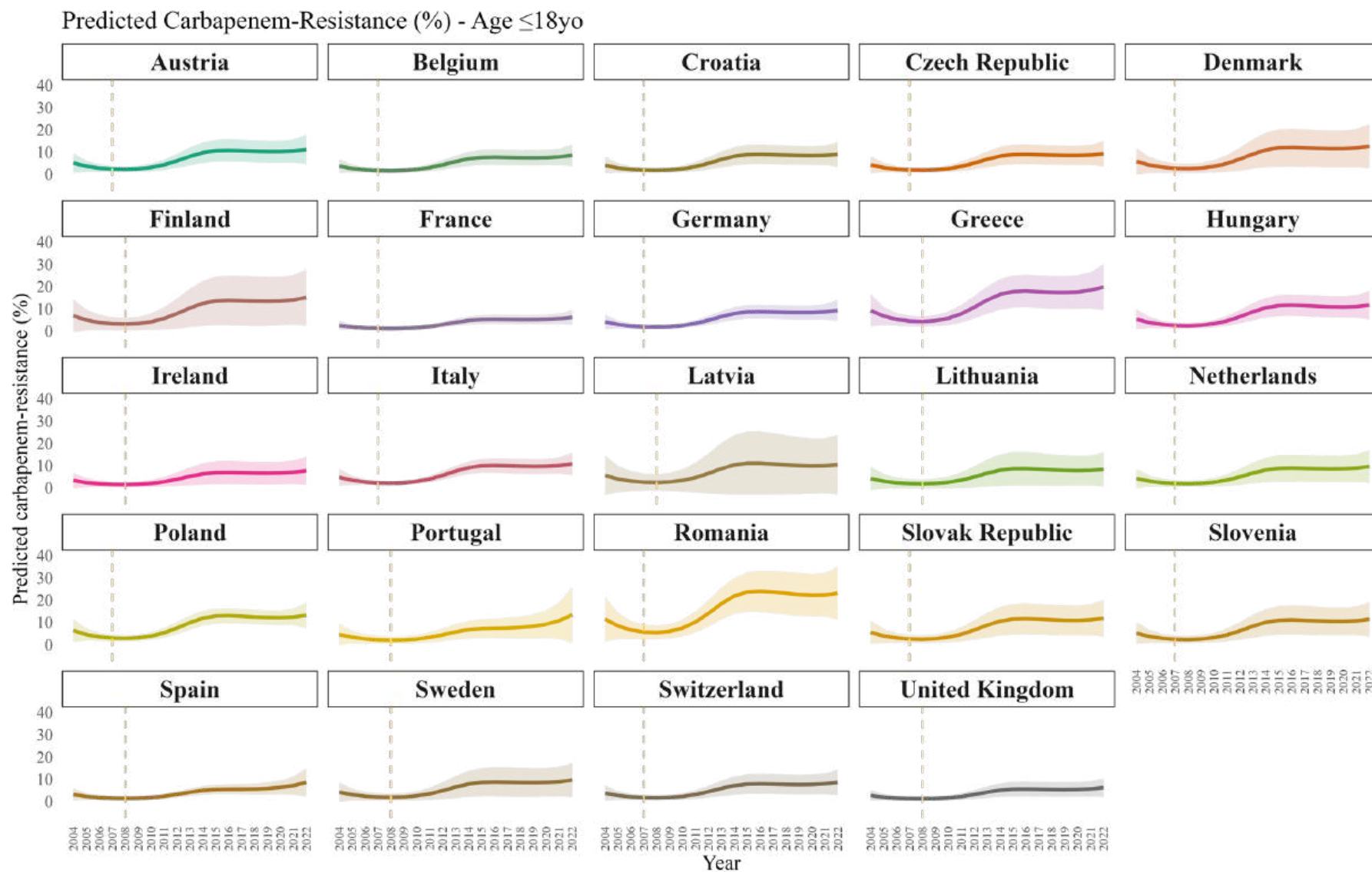
VII. Subgroup analysis by age-groups for carbapenem-resistance, Europe and the US

Figure S48. Estimated carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across European countries using age-group and time interactions



Notes: GAM=Generalised additive models. Age group 0 is <18 years, Age group 1 is 19–64 years, and Age group 2 is ≥ 65 .

Figure S49. Estimated carbapenem-resistance in Enterobacteriales using a spatiotemporal GAM across **European countries, age ≤18**



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa).

Figure S50. Growth rate of carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, age ≤18**

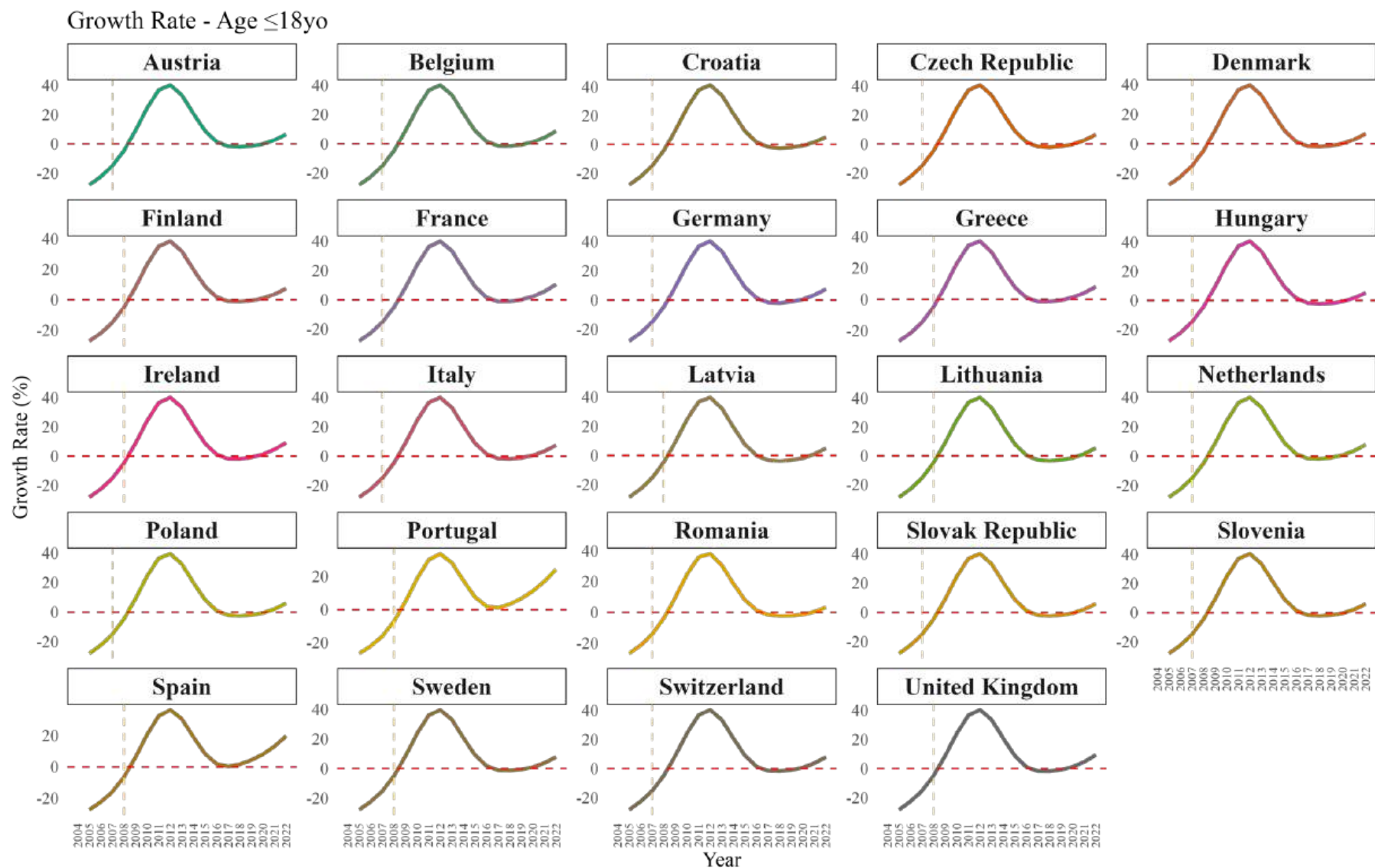
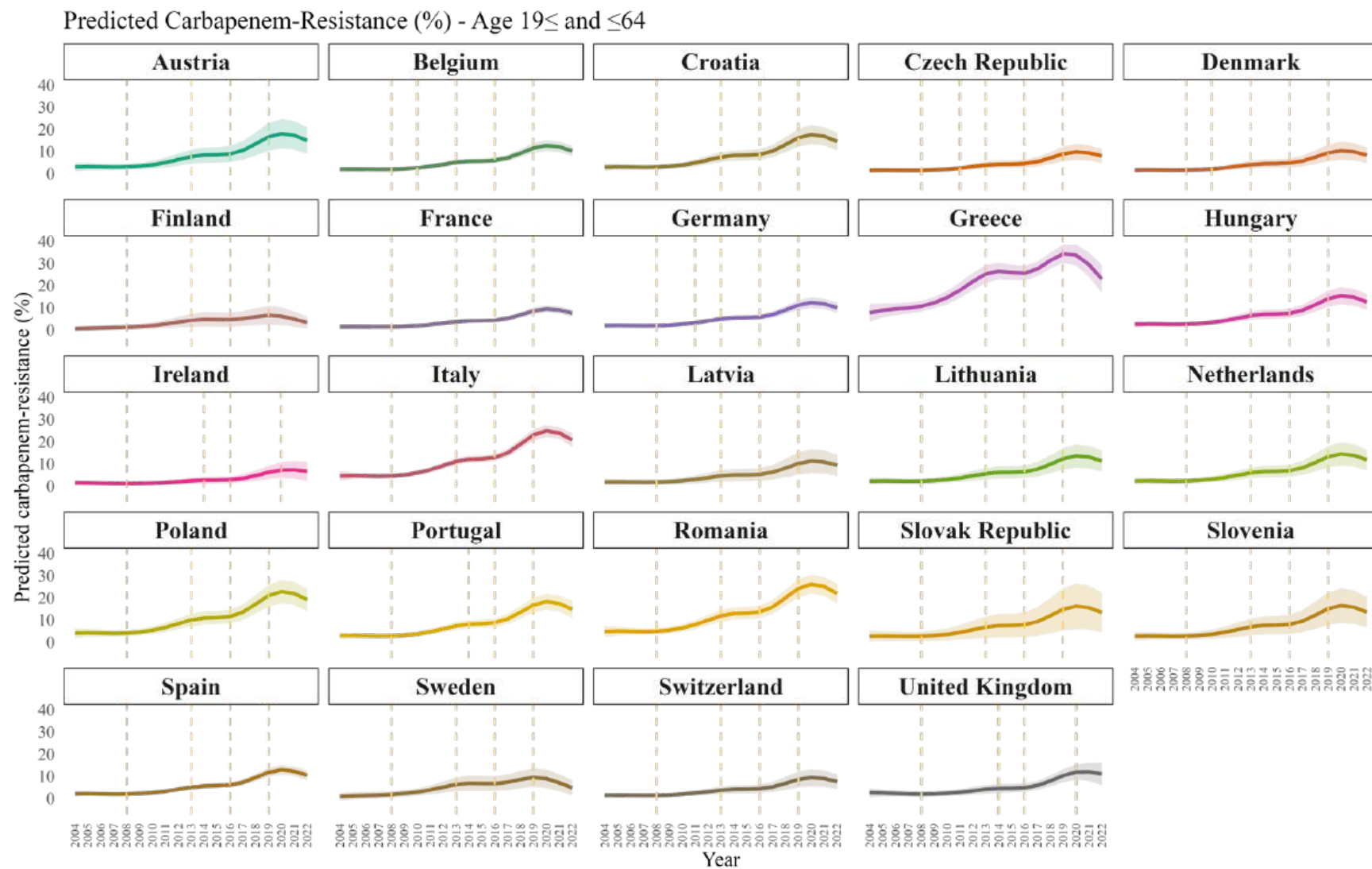


Figure S51. Estimated carbapenem-resistance in Enterobacteriales using a spatiotemporal GAM across **European countries, age 19≤ and ≤64**



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa).

Figure S52. Growth rate of carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, age 19≤ and ≤64**

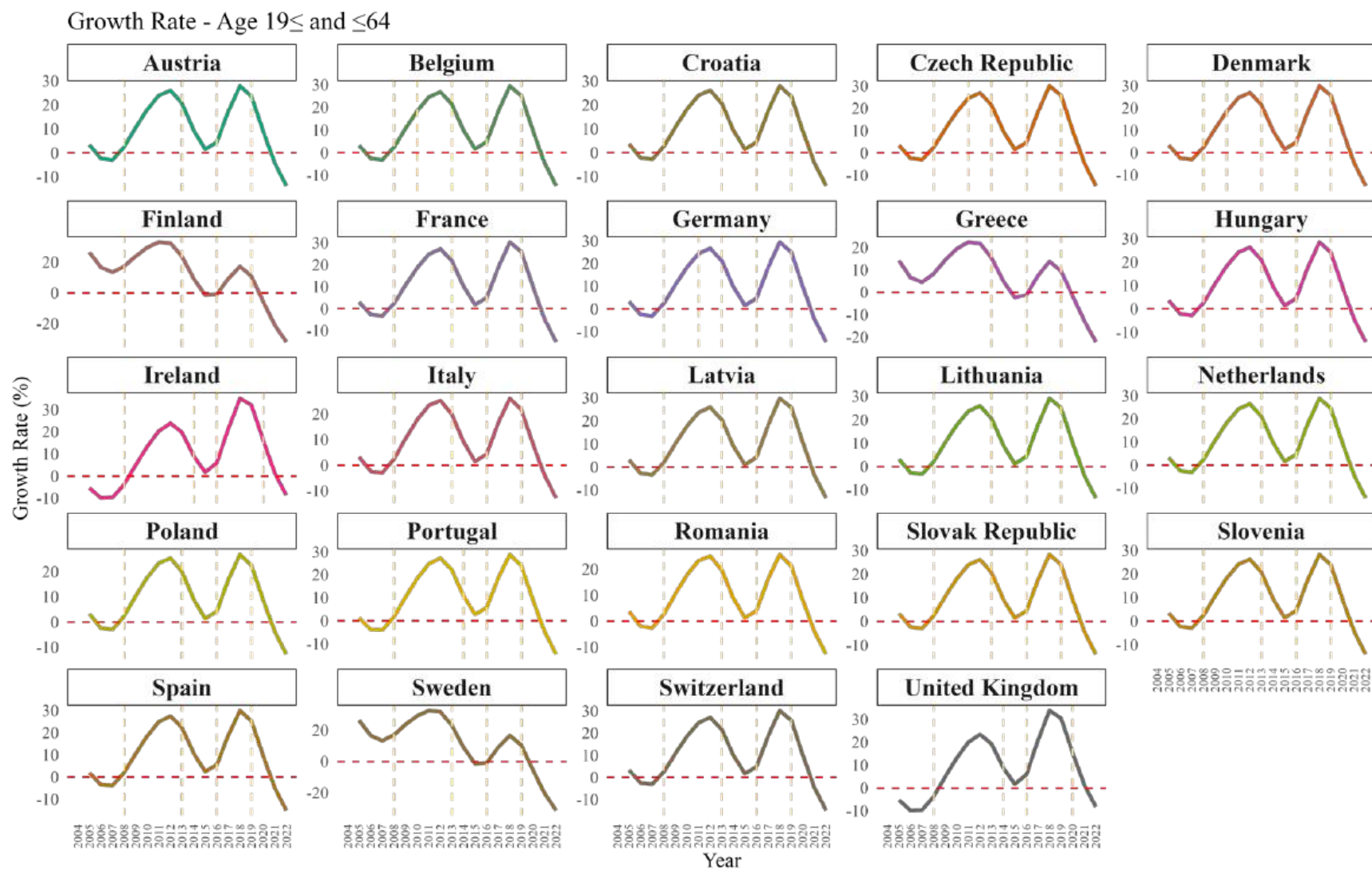
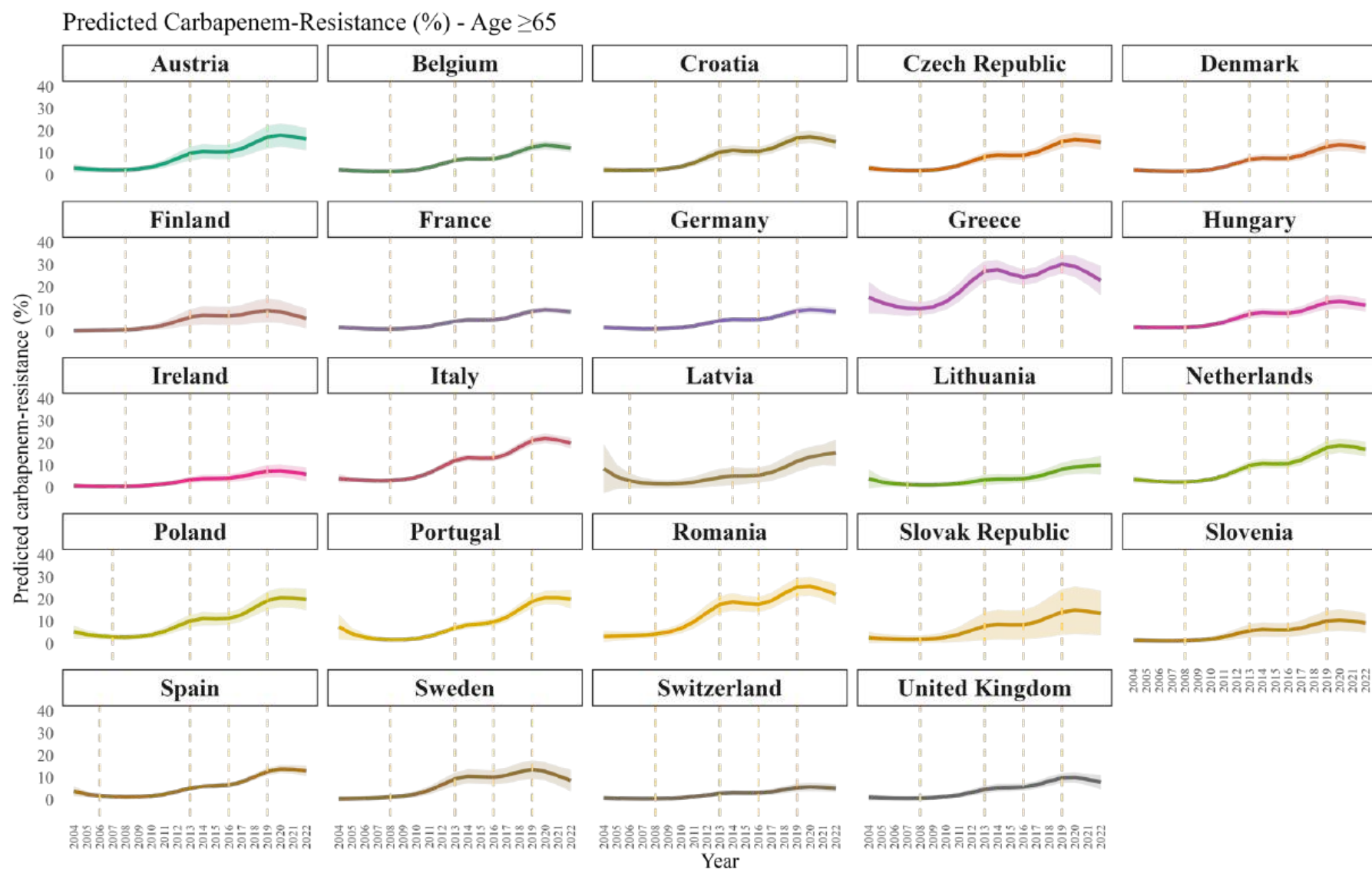


Figure S53. Estimated carbapenem-resistance in Enterobacteriales using a spatiotemporal GAM across **European countries, age ≥ 65**



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa).

Figure S54. Growth rate of carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, age ≥ 65**

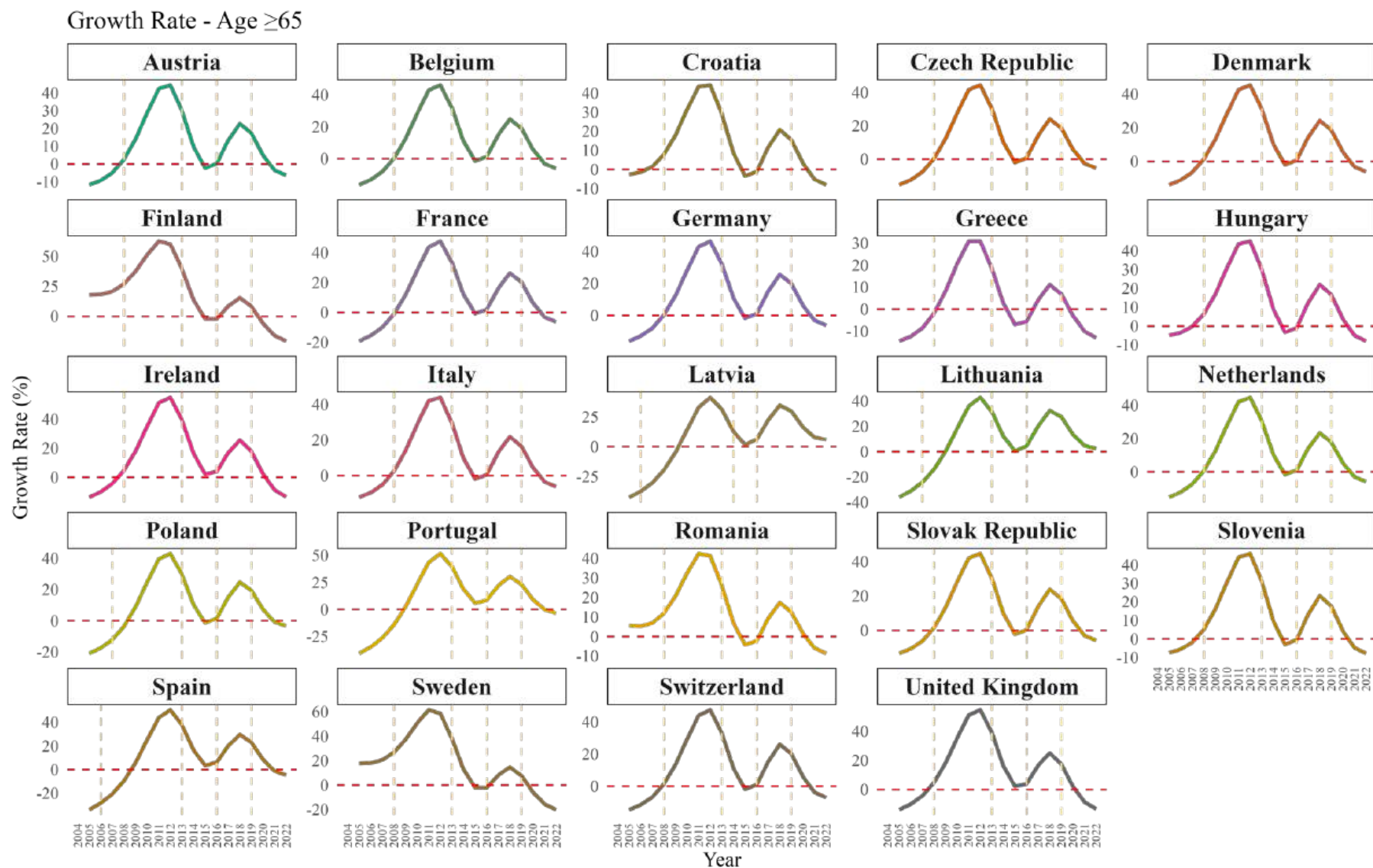
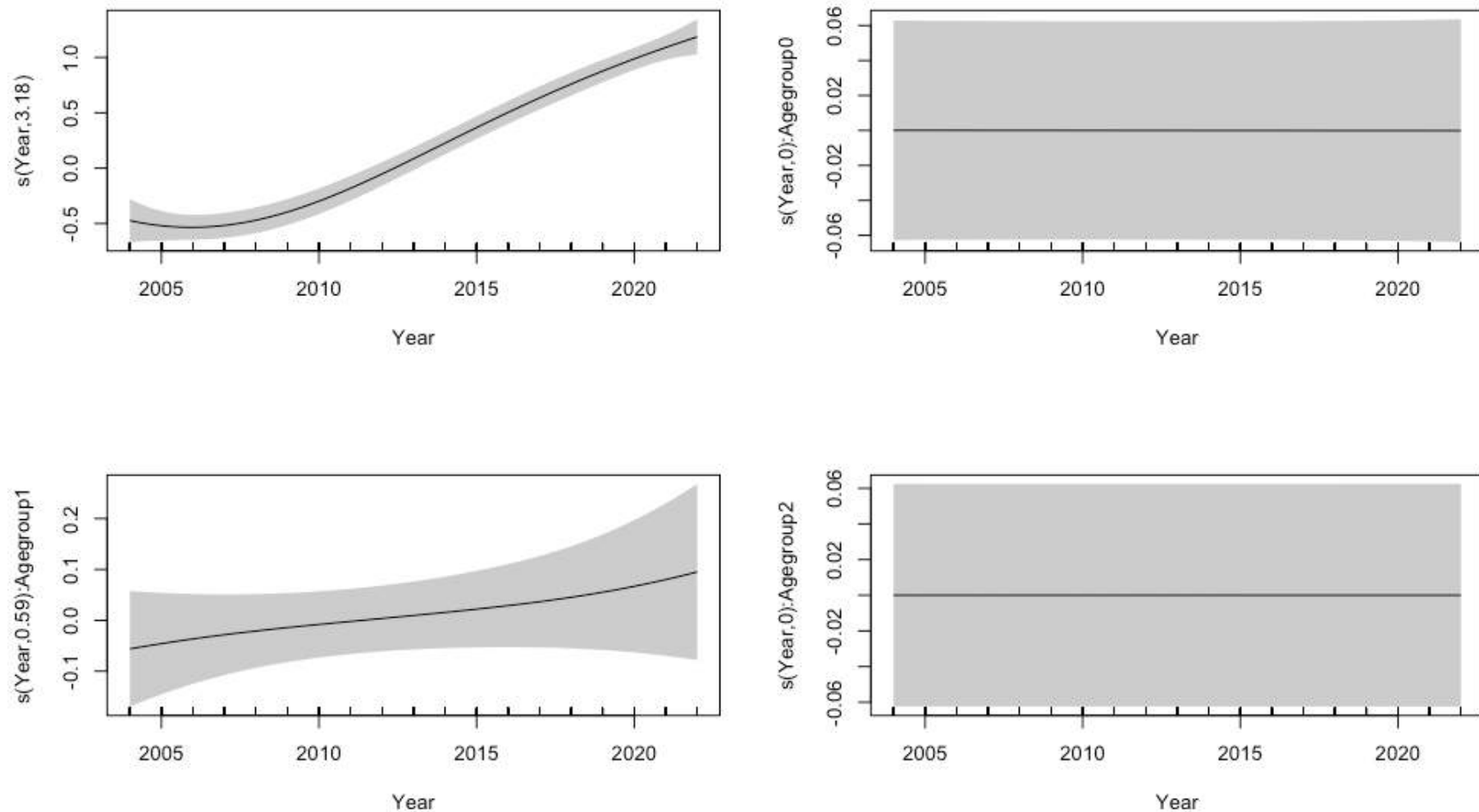
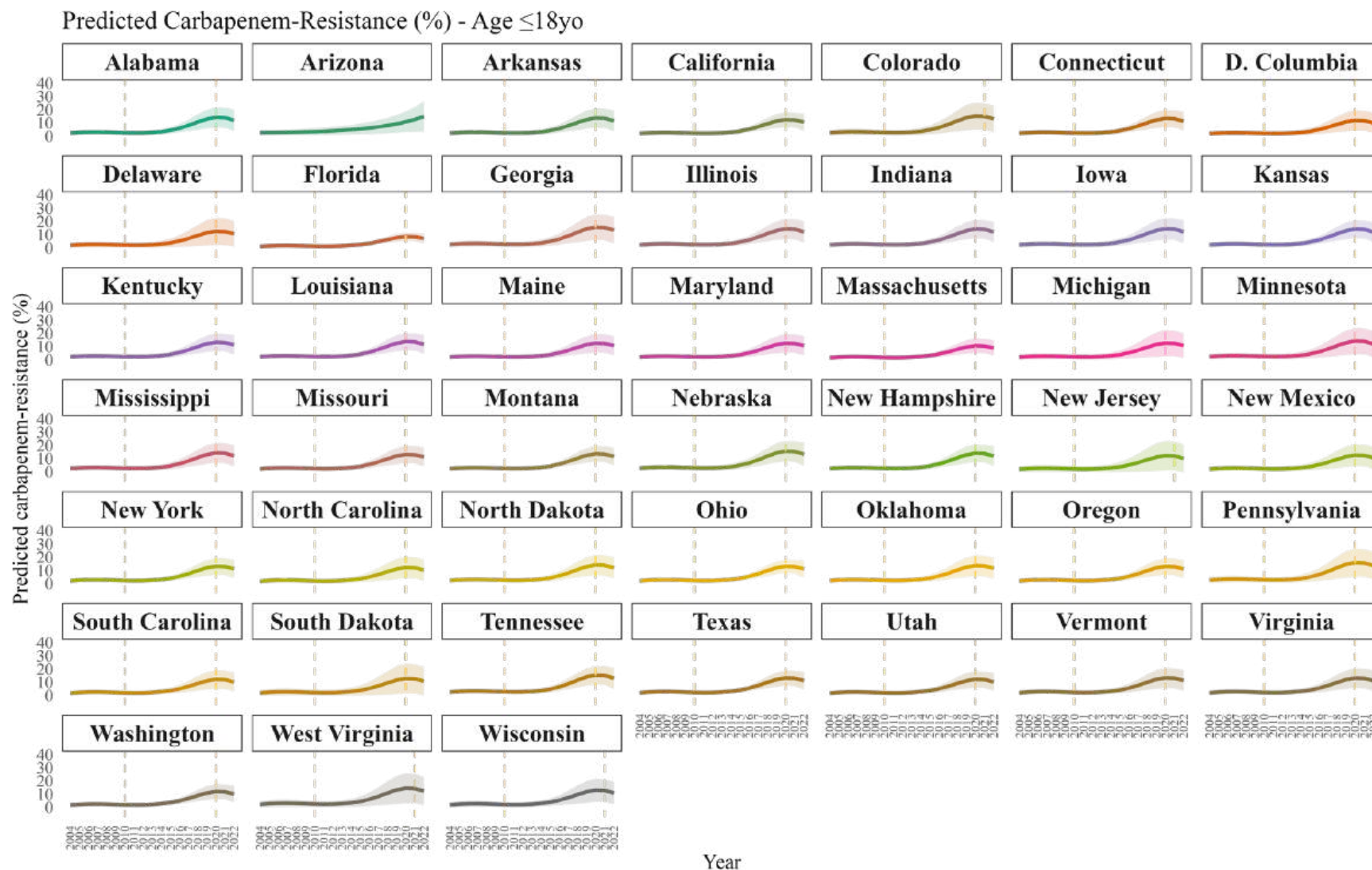


Figure S55. Estimated carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across US-states using age-group and time interactions



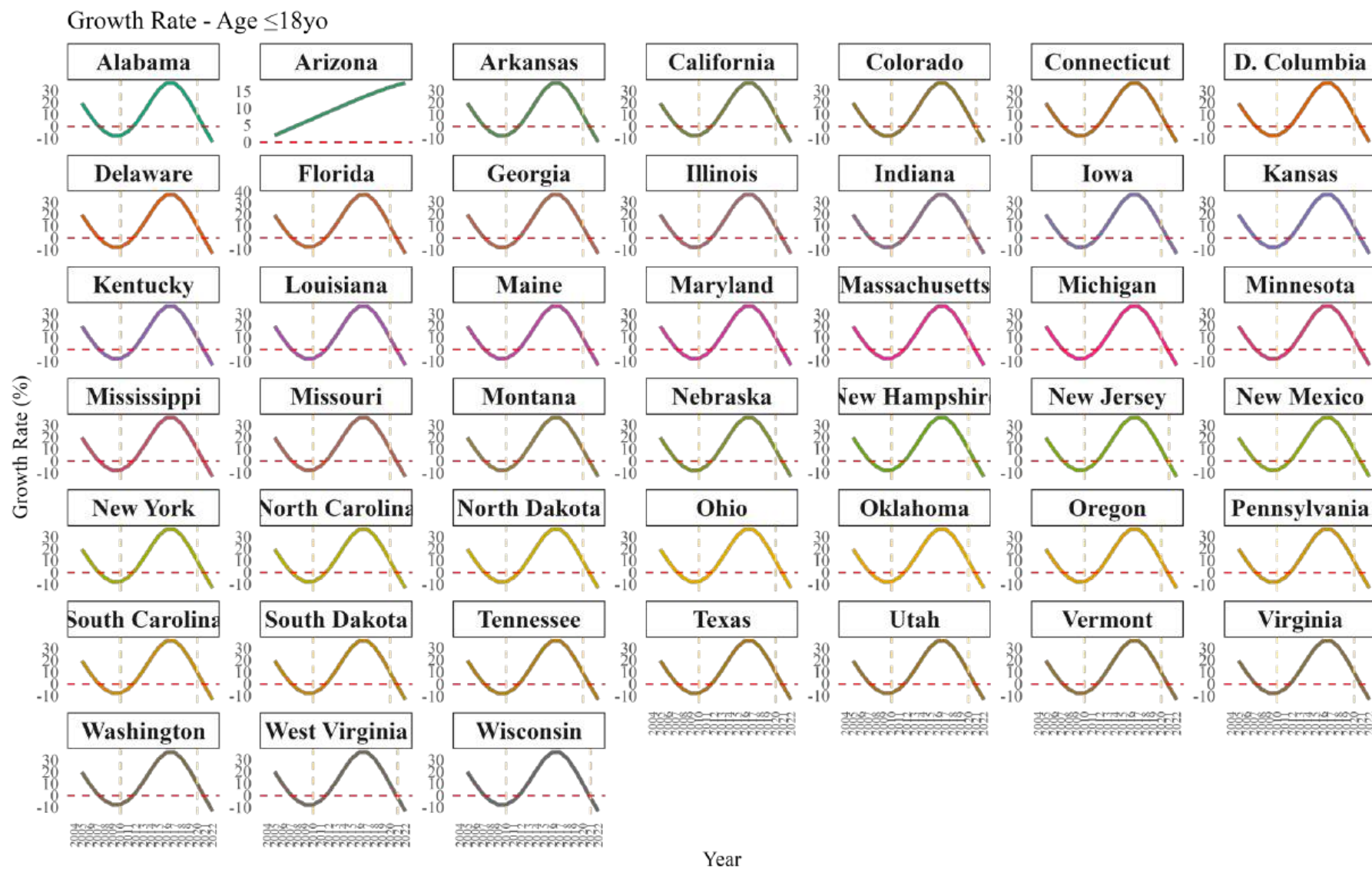
Notes: GAM=Generalised additive models. Age group 0 is <18 years, Age group 1 is 19–64 years, and Age group 2 is ≥65.

Figure S56. Estimated carbapenem-resistance in Enterobacteriales using a spatiotemporal GAM across **US states, age ≤ 18**



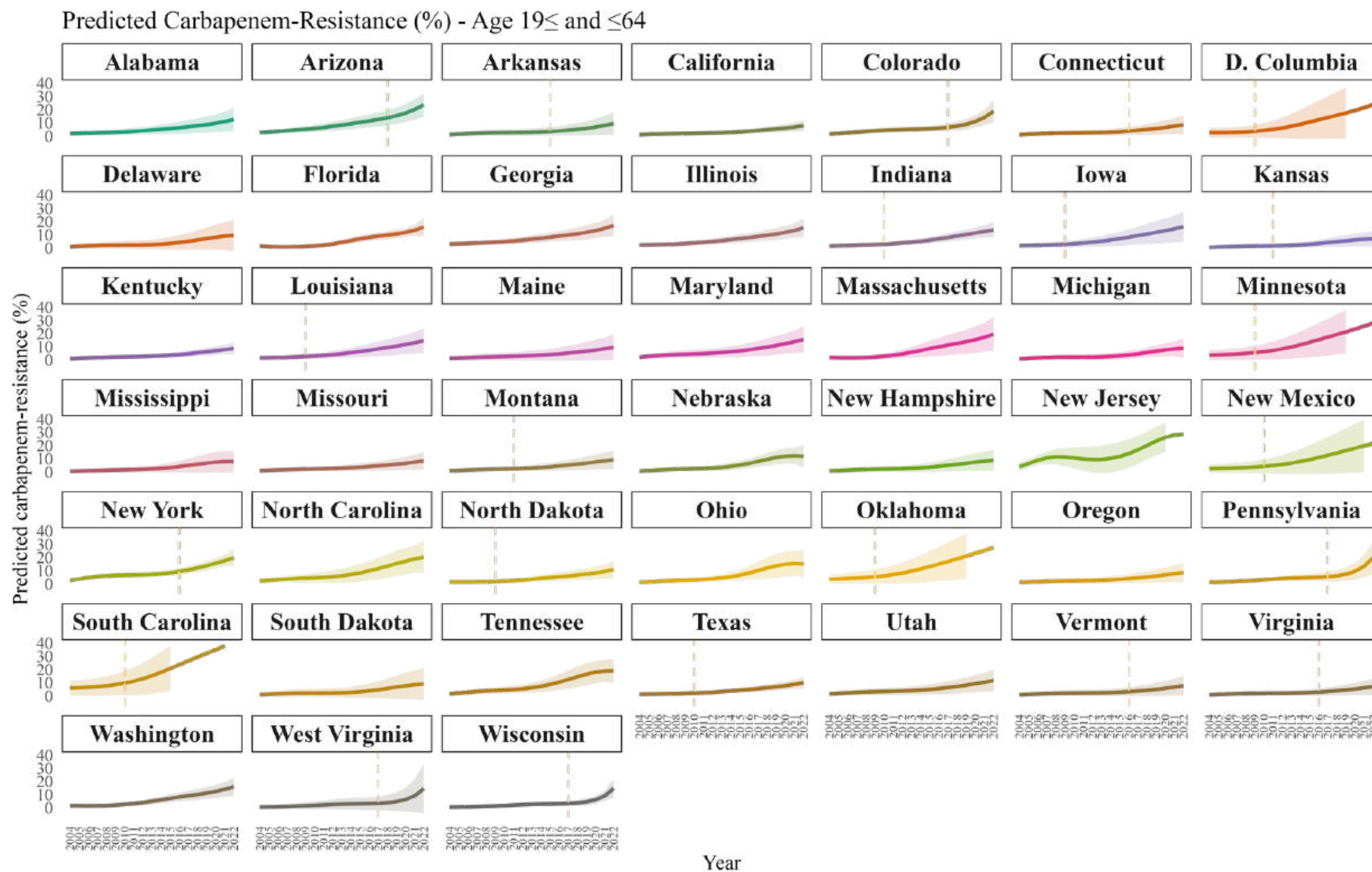
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa).

Figure S57. Growth rate of carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across US-states, age ≤ 18



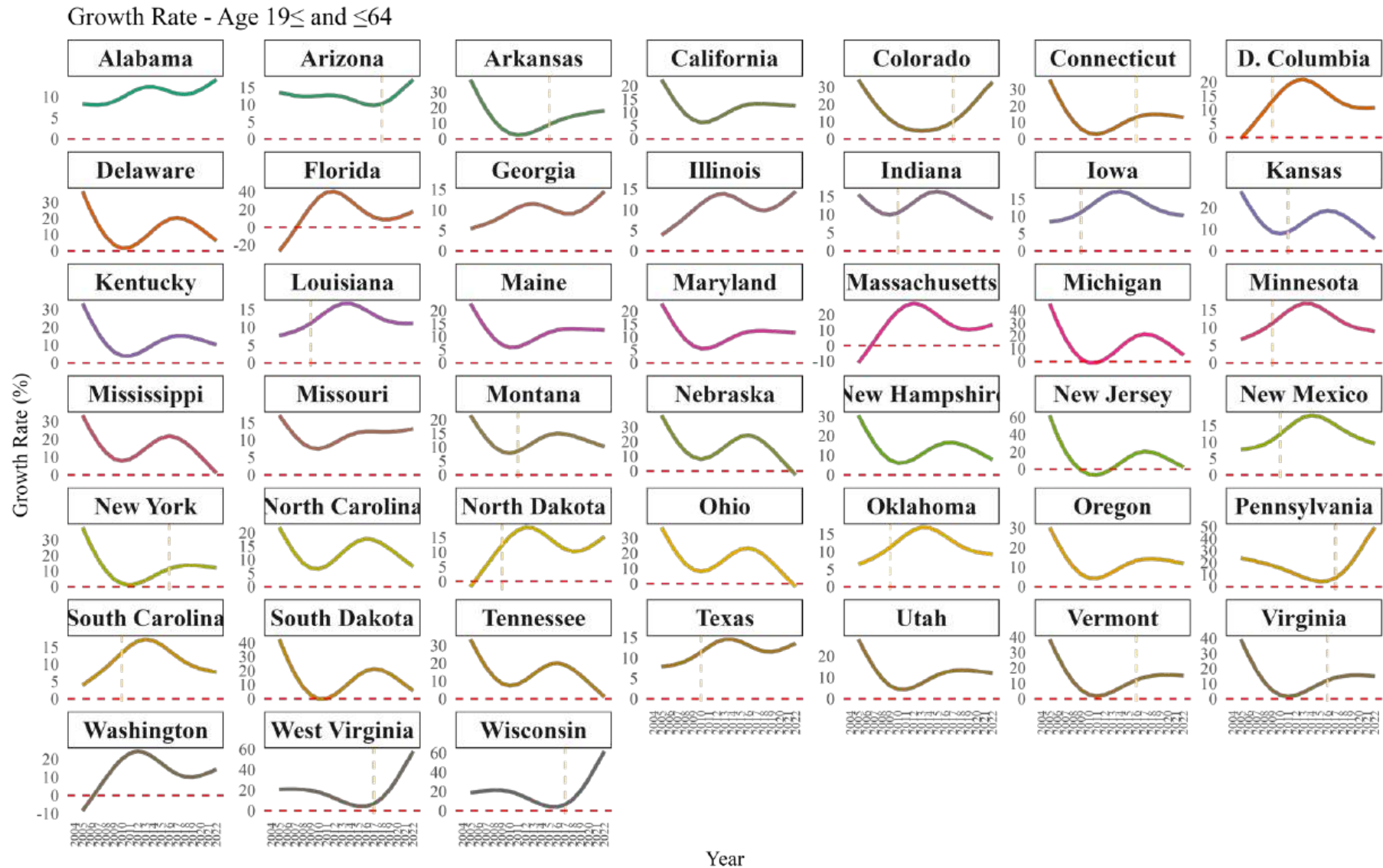
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

Figure S58. Estimated carbapenem-resistance in Enterobacteriales using a spatiotemporal GAM across **US states, age 19≤ and ≤64**



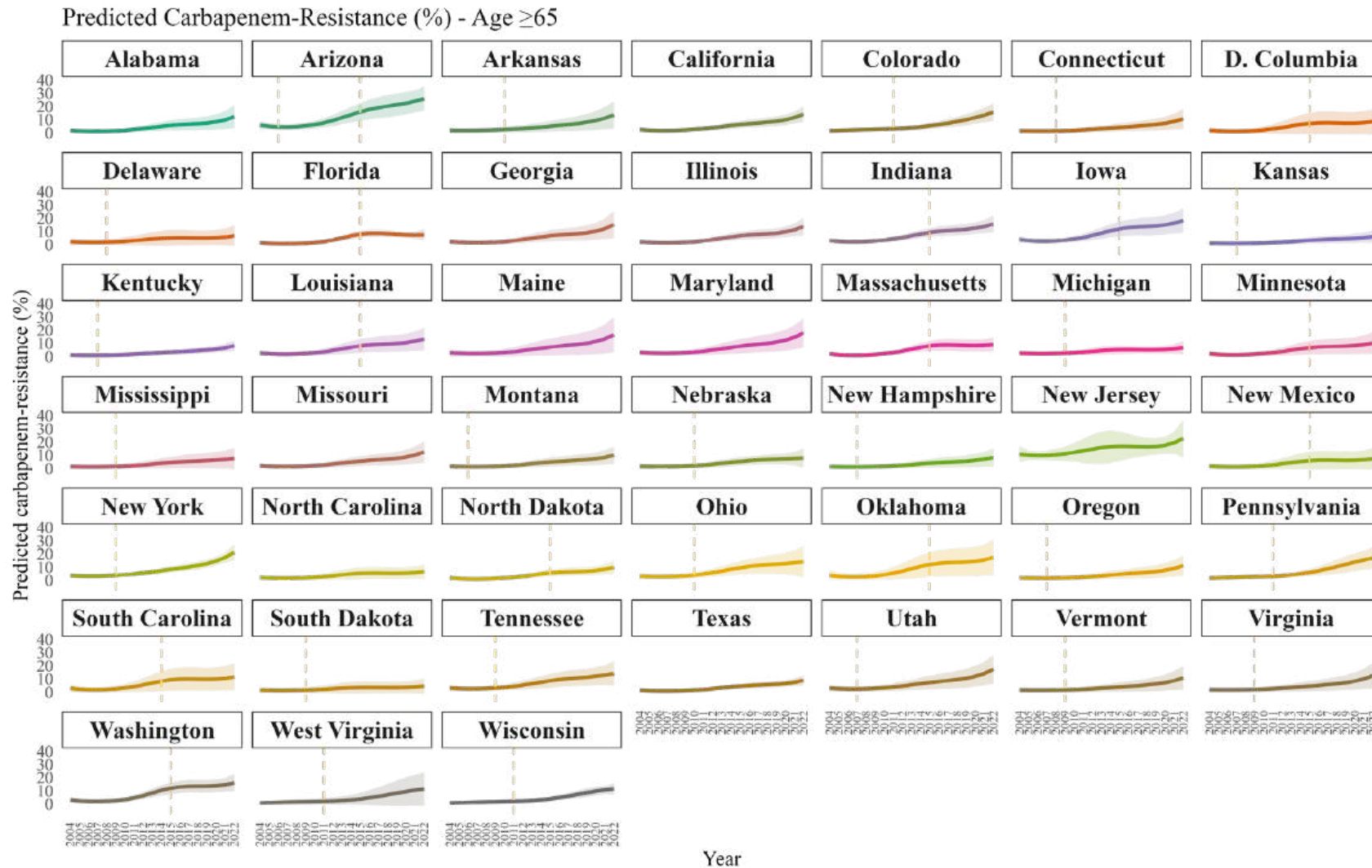
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa).

Figure S59. Growth rate of carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across US-states, age 19≤ and ≤64



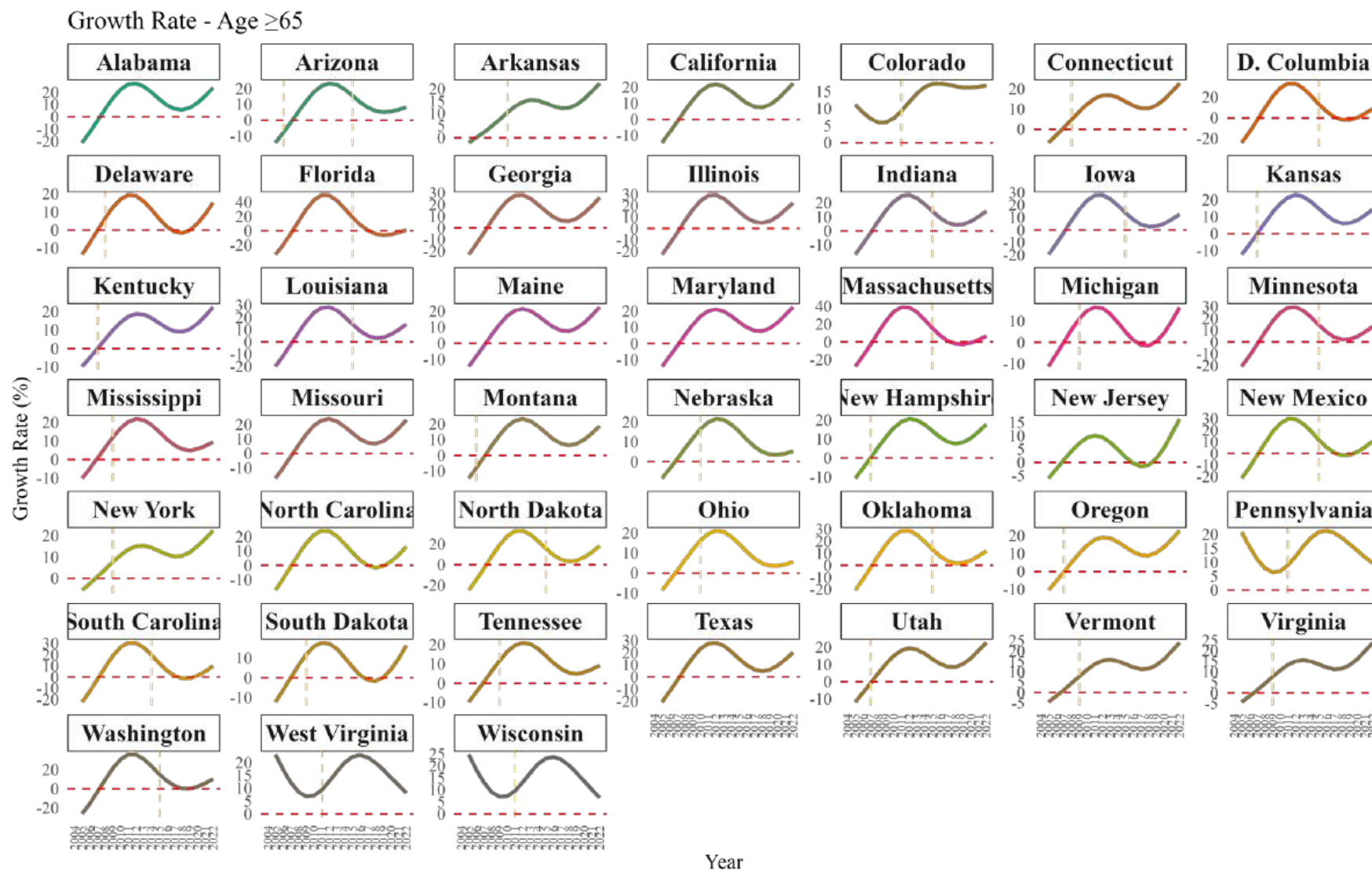
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

Figure S60. Estimated carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across US states, age ≥ 65



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa).

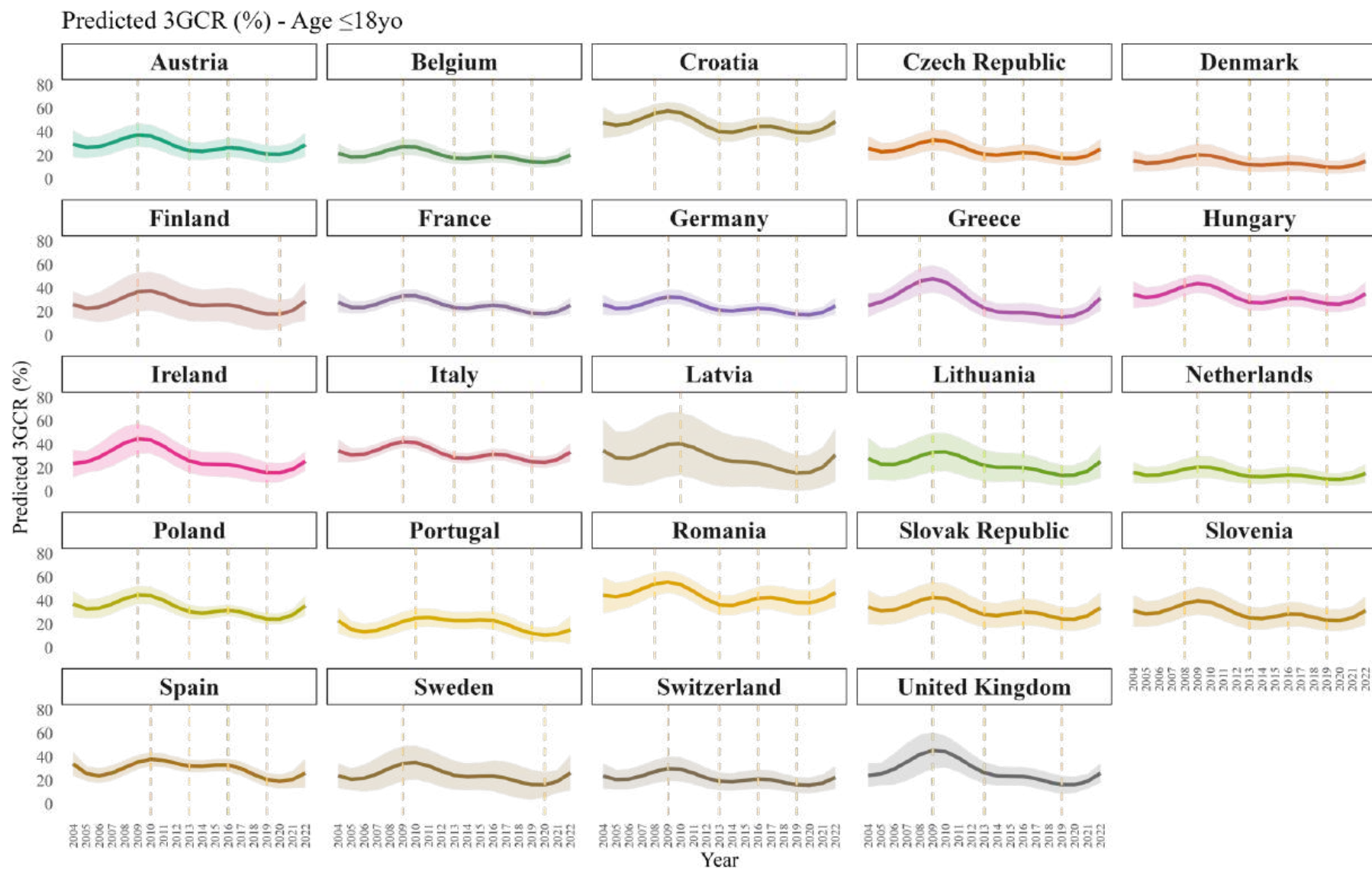
Figure S61. Growth rate of carbapenem-resistance in Enterobacterales using a spatiotemporal GAM across US-states, age ≥ 65



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

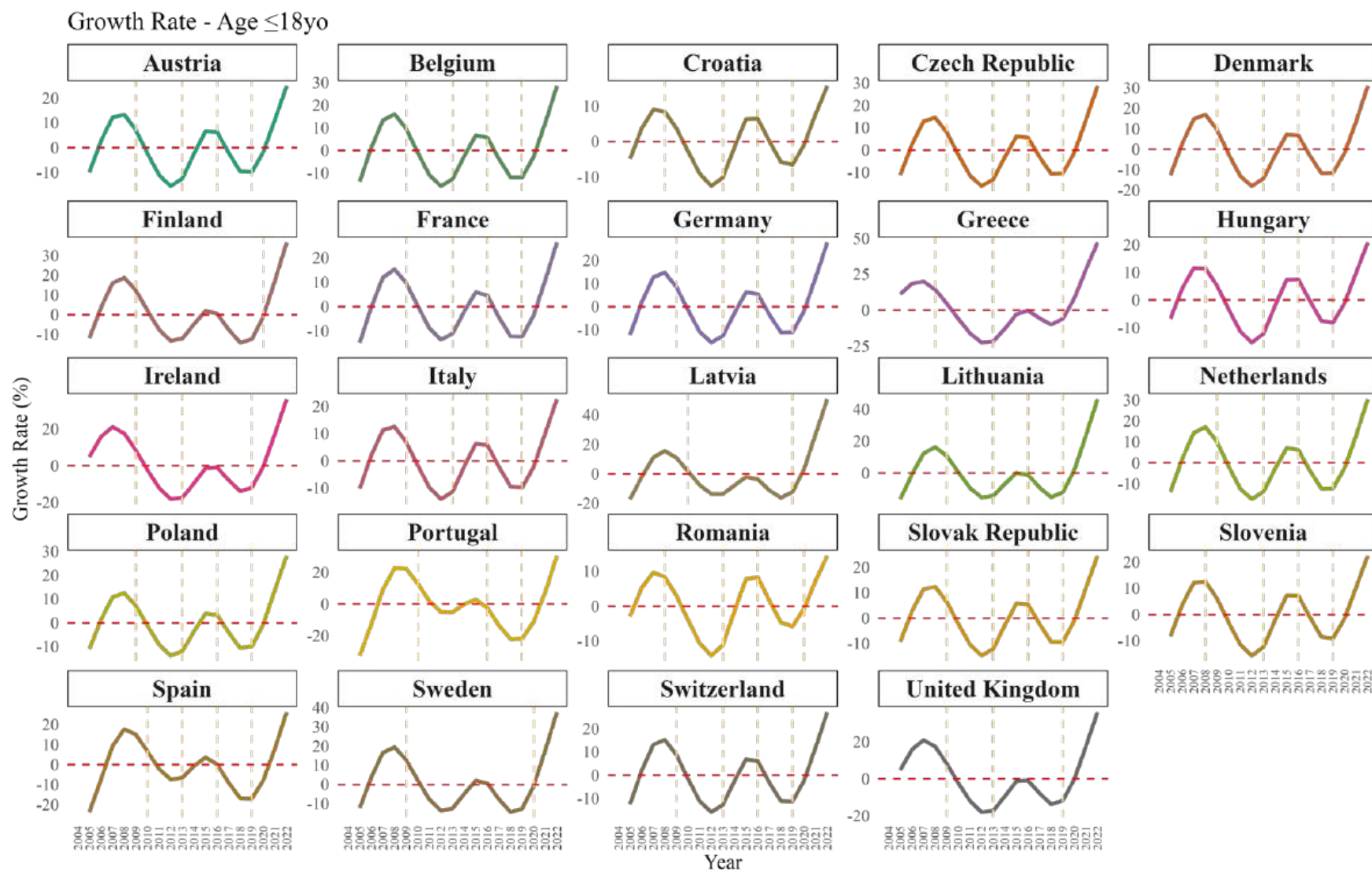
VIII. Subgroup analysis by age-groups for 3rd-generation cephalosporin-resistance, **Europe and the US**

Figure S62. Estimated 3GCR in Enterobacterales using a spatiotemporal GAM across **European countries, age ≤18**



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). 3GCR= Third generation cephalosporin resistance.

Figure S63. Growth rate of third generation cephalosporin-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, age ≤ 18**



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

Figure S64. Estimated 3GCR in Enterobacterales using a spatiotemporal GAM across **European countries, age 19≤ and ≤64**

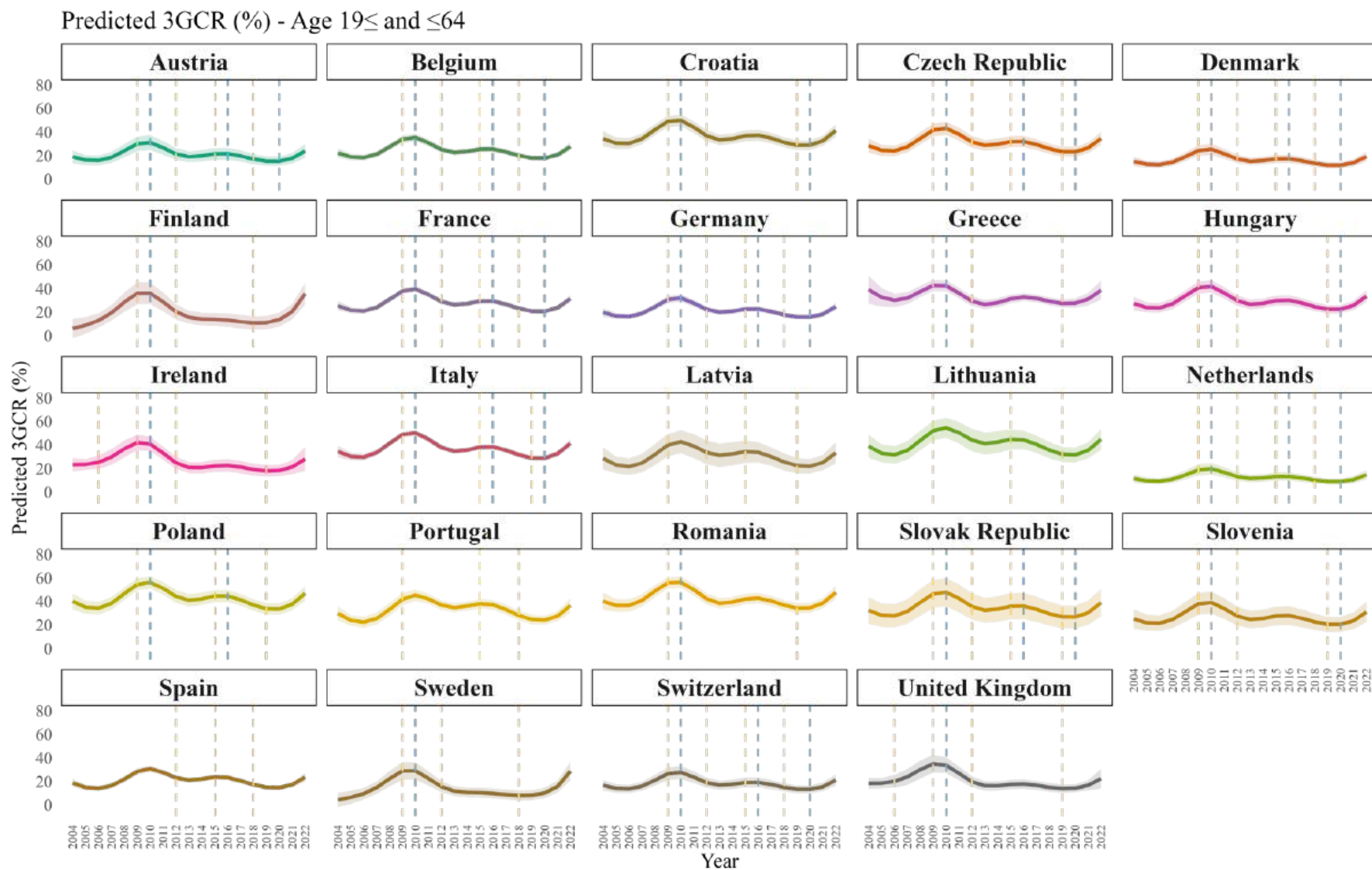


Figure S65. Growth rate of third generation cephalosporin-resistance in Enterobacterales using a spatiotemporal GAM across **European countries, age 19≤ and ≤64**

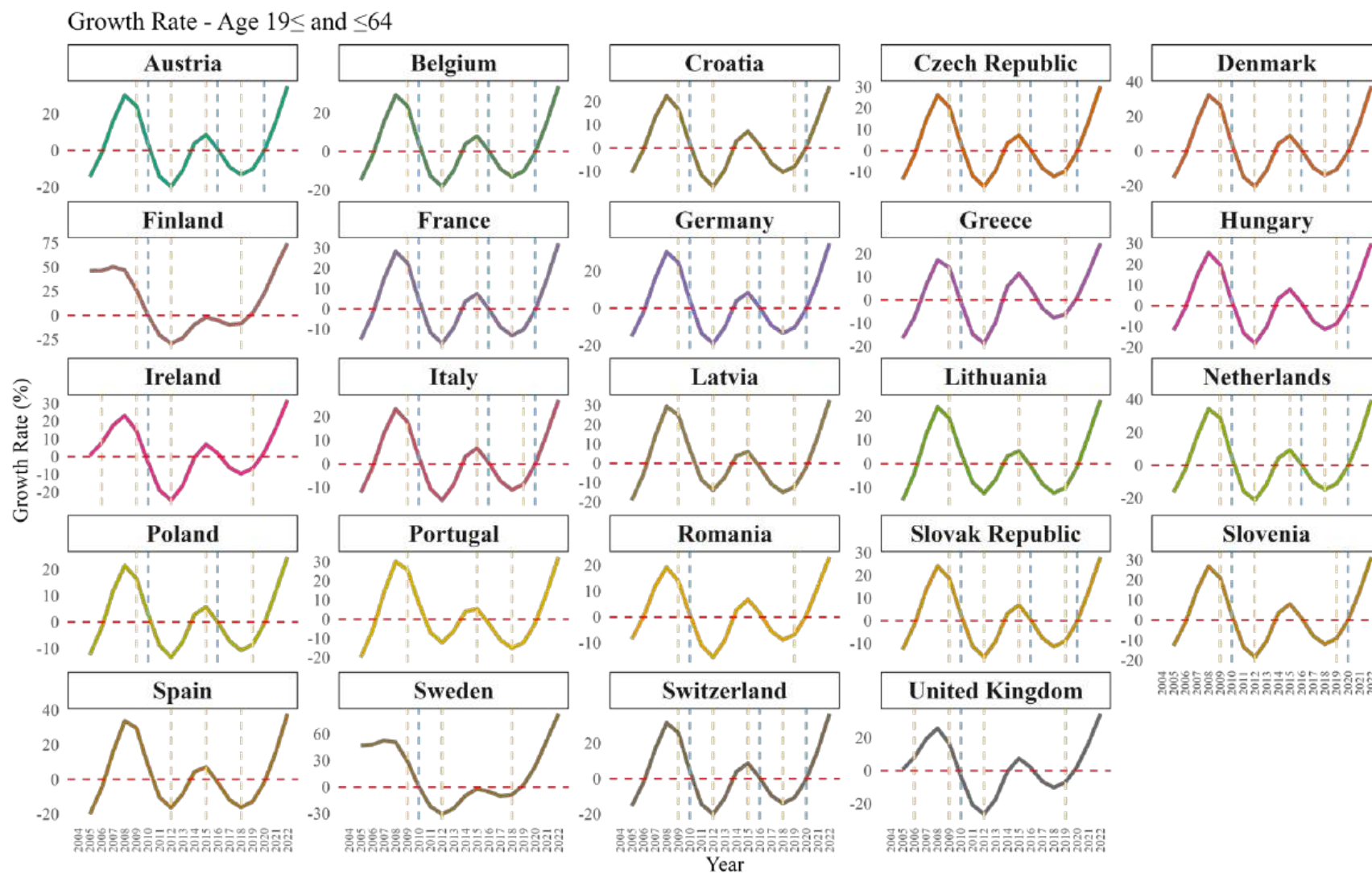


Figure S66. Estimated 3GCR in Enterobacteriales using a spatiotemporal GAM across **European countries, age ≥ 65**

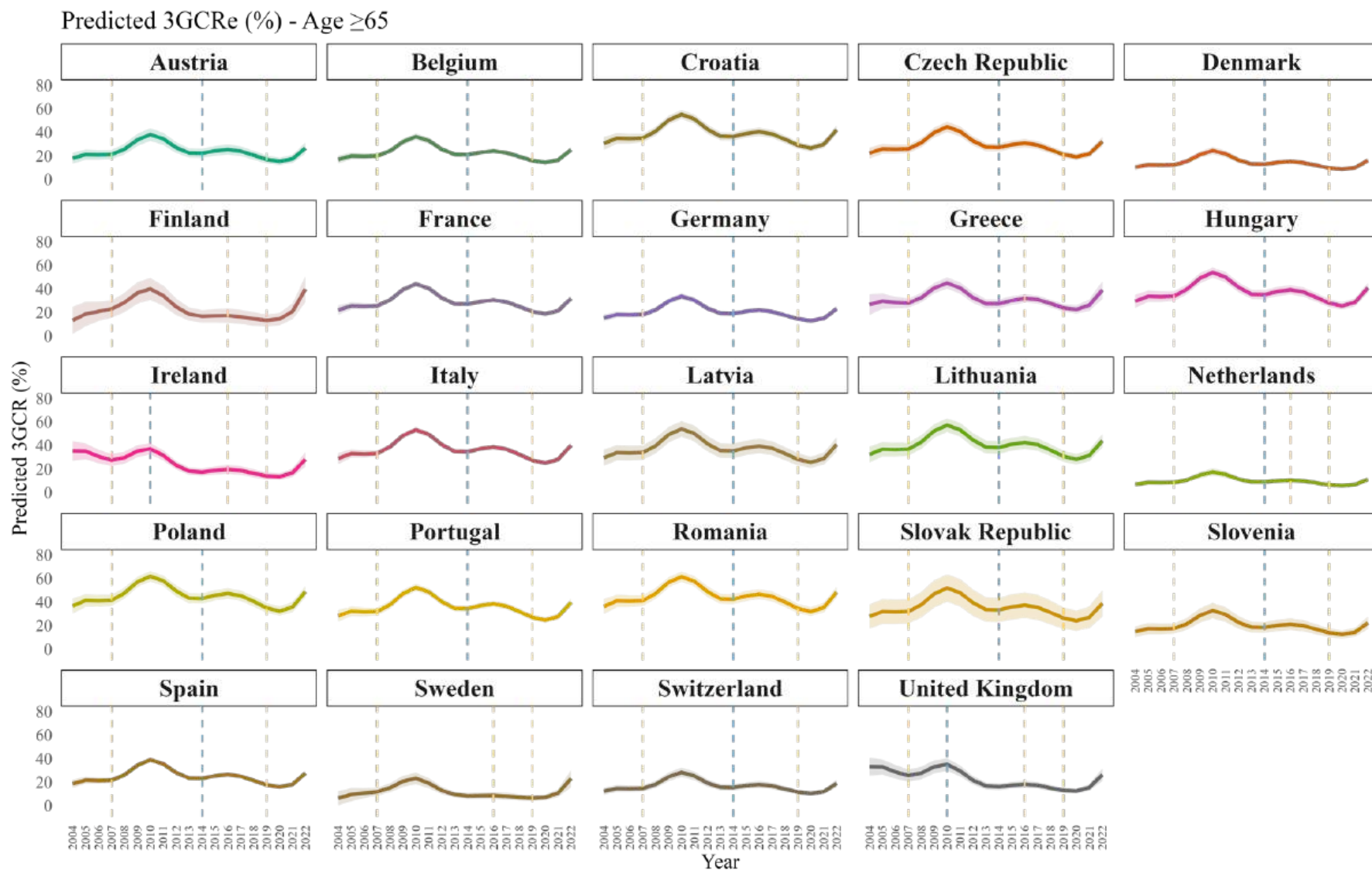


Figure S67. Growth rate of 3GCR in Enterobacterales using a spatiotemporal GAM across **European countries, age ≥ 65**

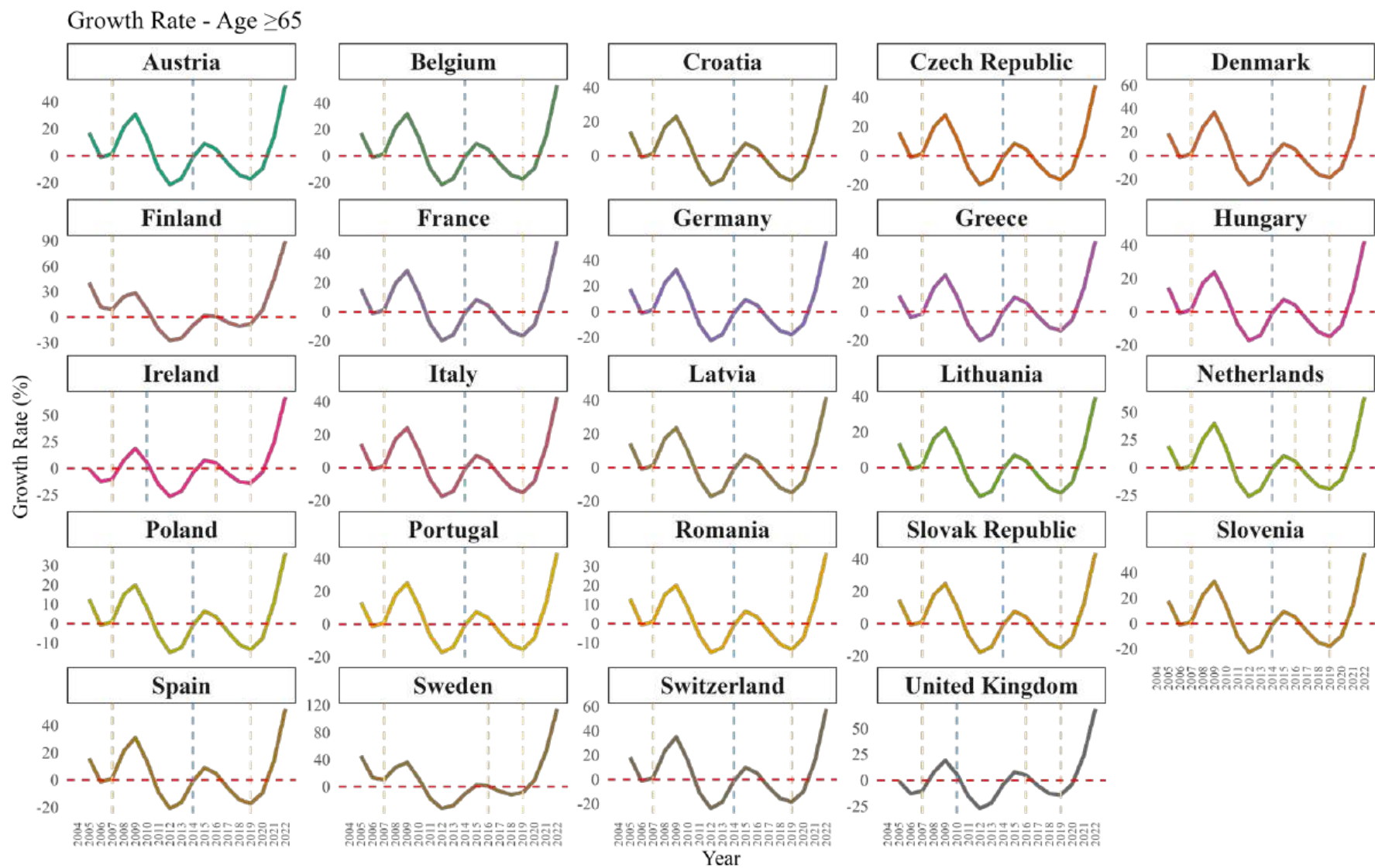
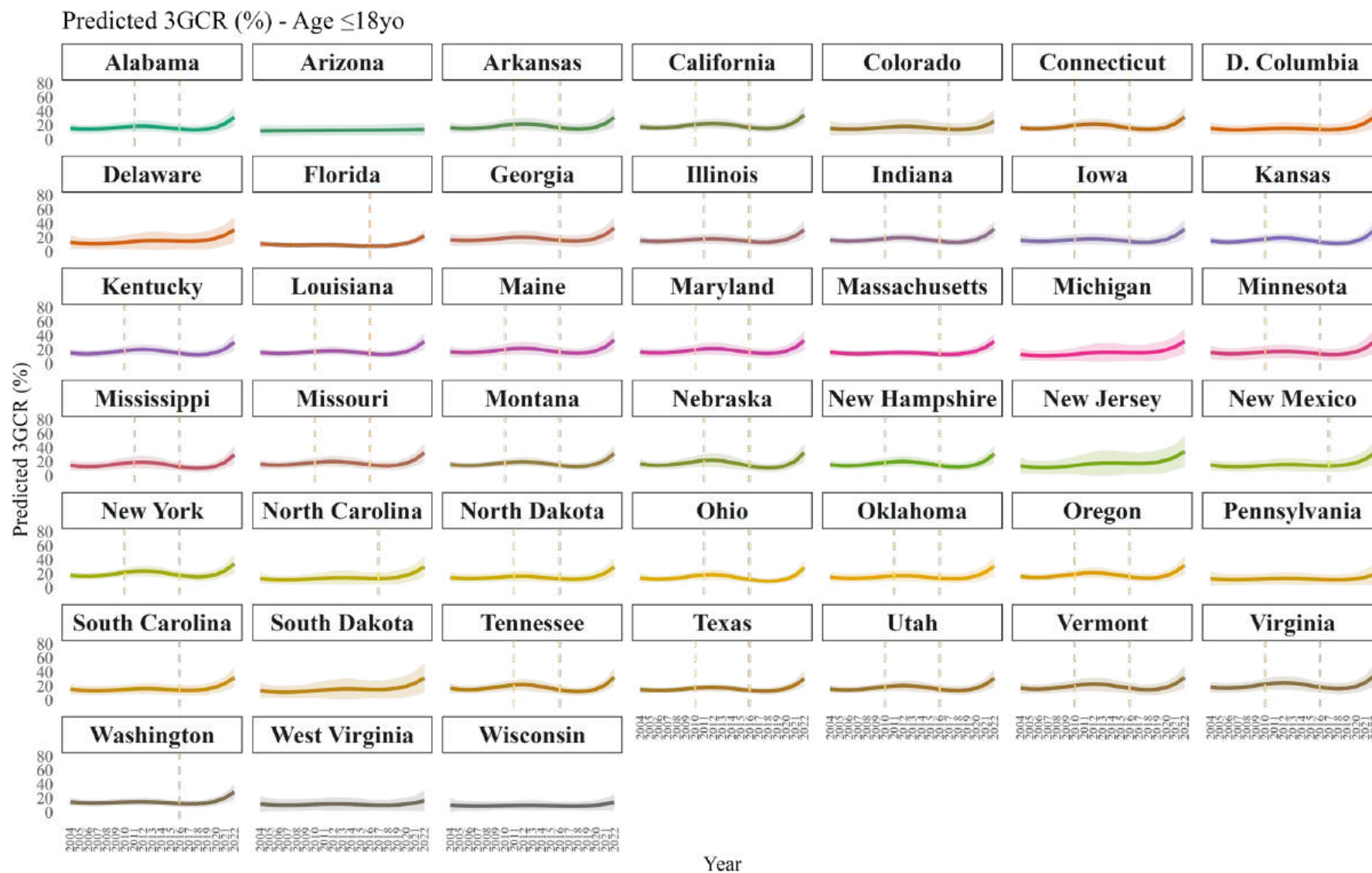
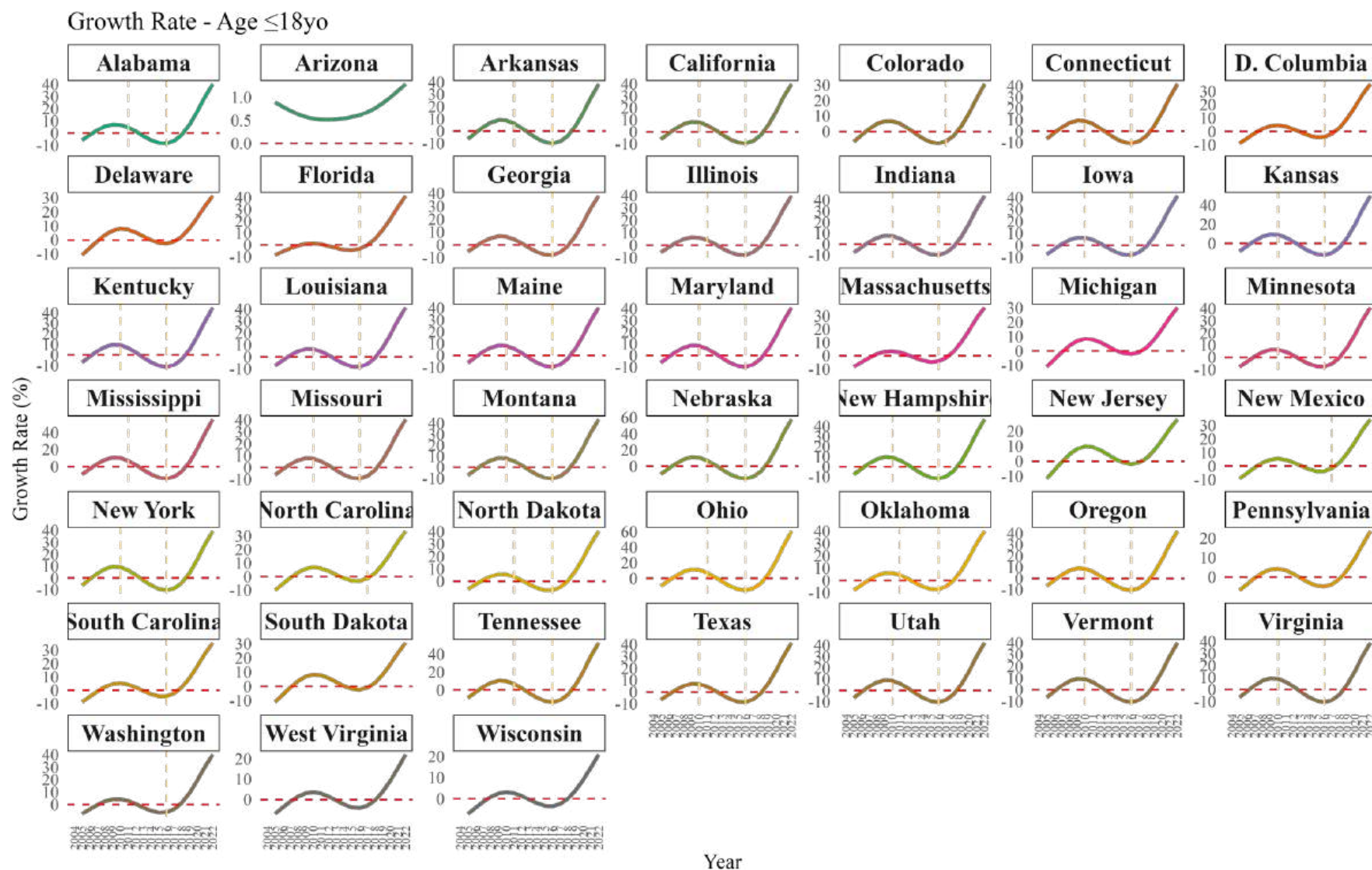


Figure S68. Estimated 3GCR in Enterobacteriales using a spatiotemporal GAM across **US states, age ≤18**



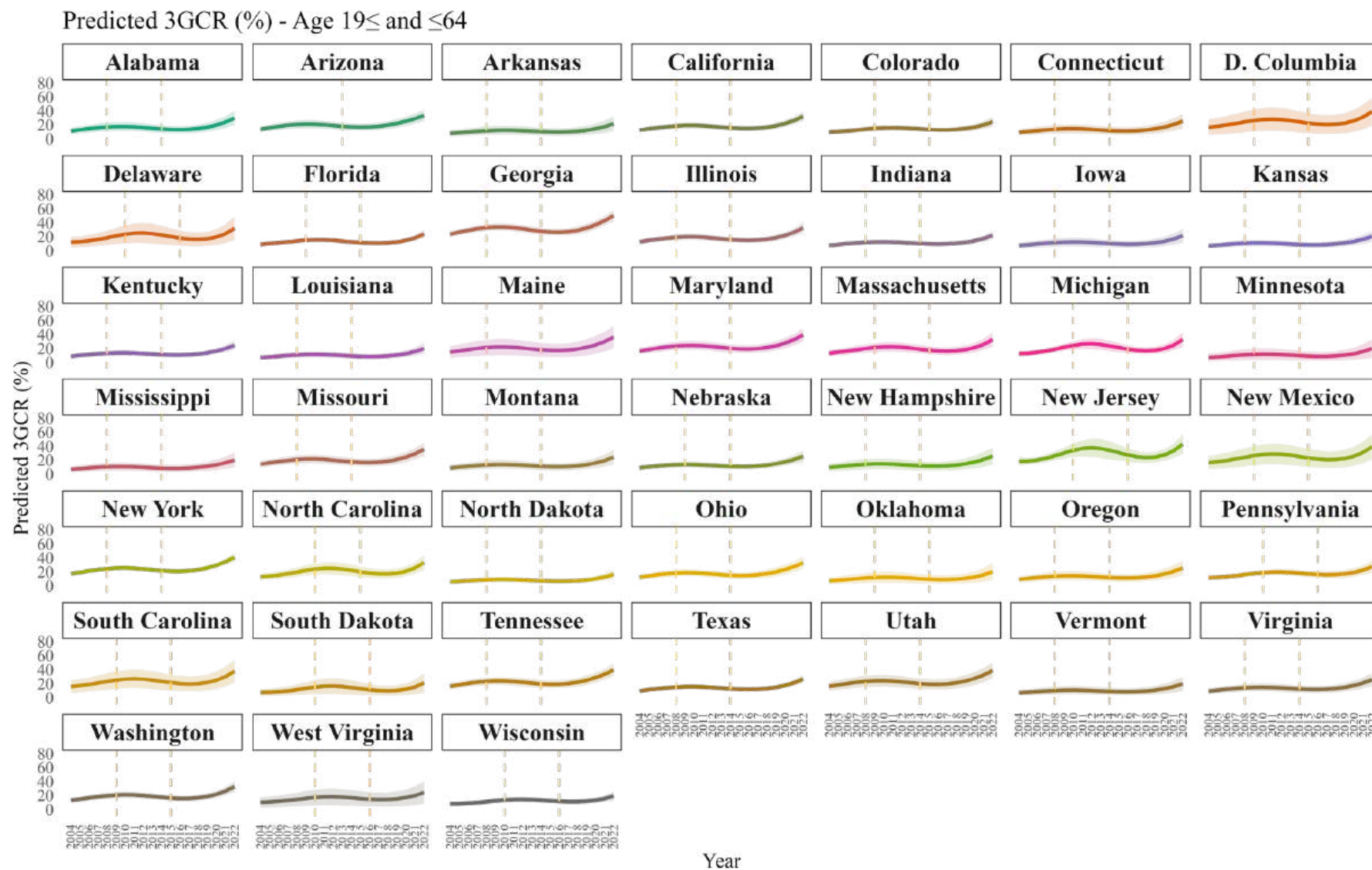
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). 3GCR= Third generation cephalosporin resistance.

Figure S69. Growth rate of third-generation cephalosporin-resistance in Enterobacterales using a spatiotemporal GAM across **US-states, age ≤ 18**



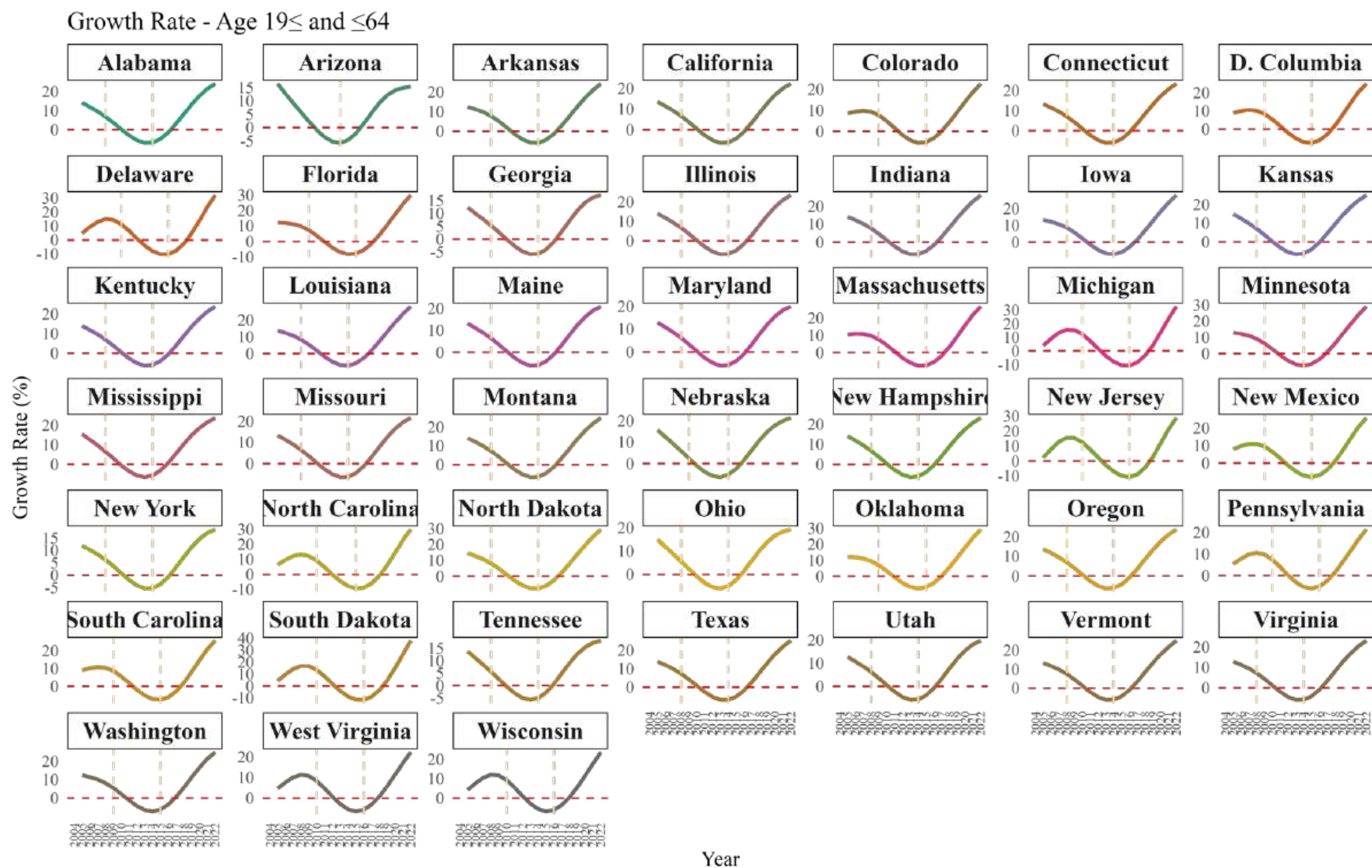
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

Figure S70. Estimated 3GCR in Enterobacteriales using a spatiotemporal GAM across US states, age 19≤ and ≤64



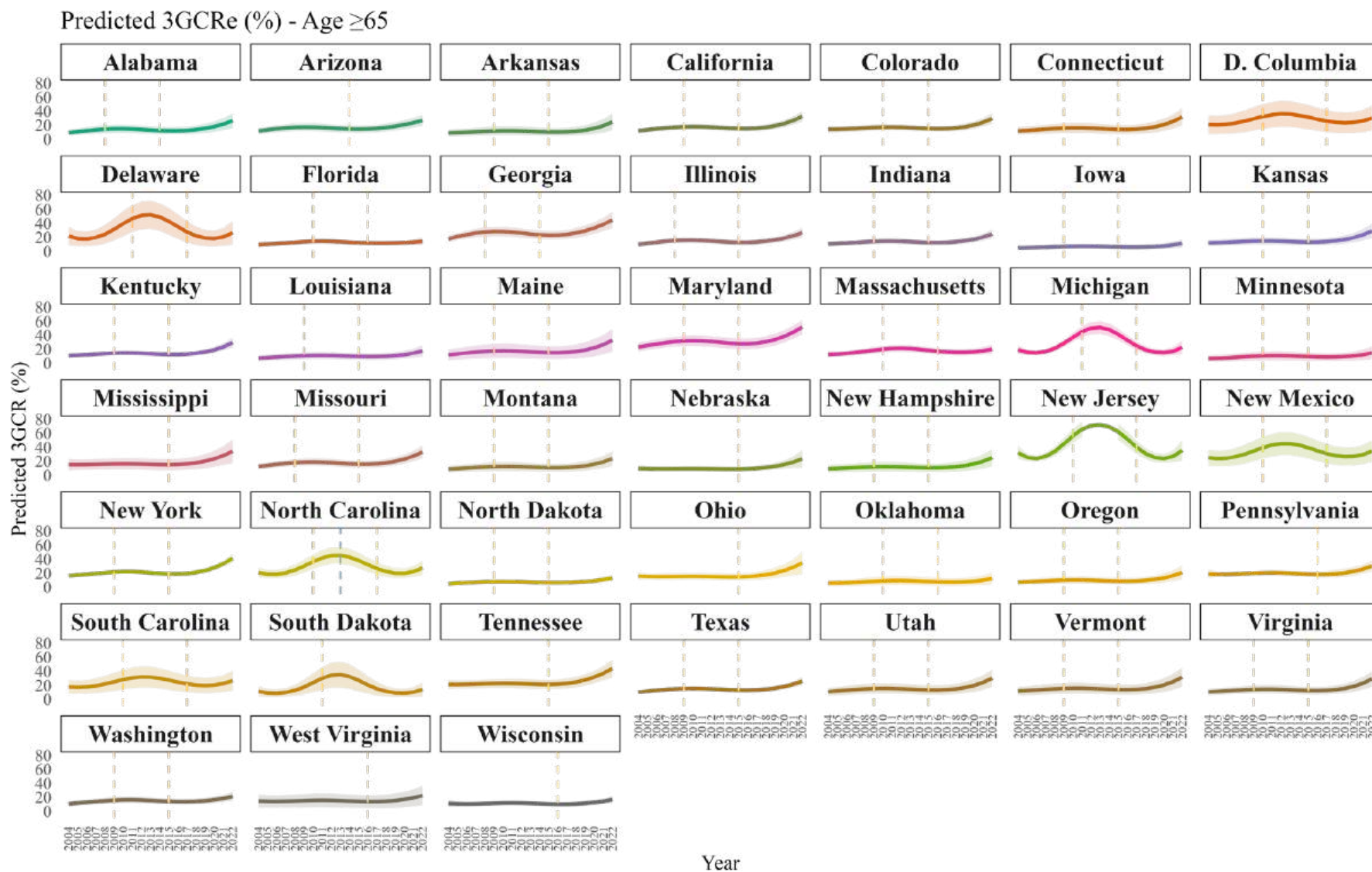
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). 3GCR= Third generation cephalosporin resistance.

Figure S71. Growth rate of third-generation cephalosporin-resistance in Enterobacterales using a spatiotemporal GAM across **US-states, age 19≤ and ≤64**



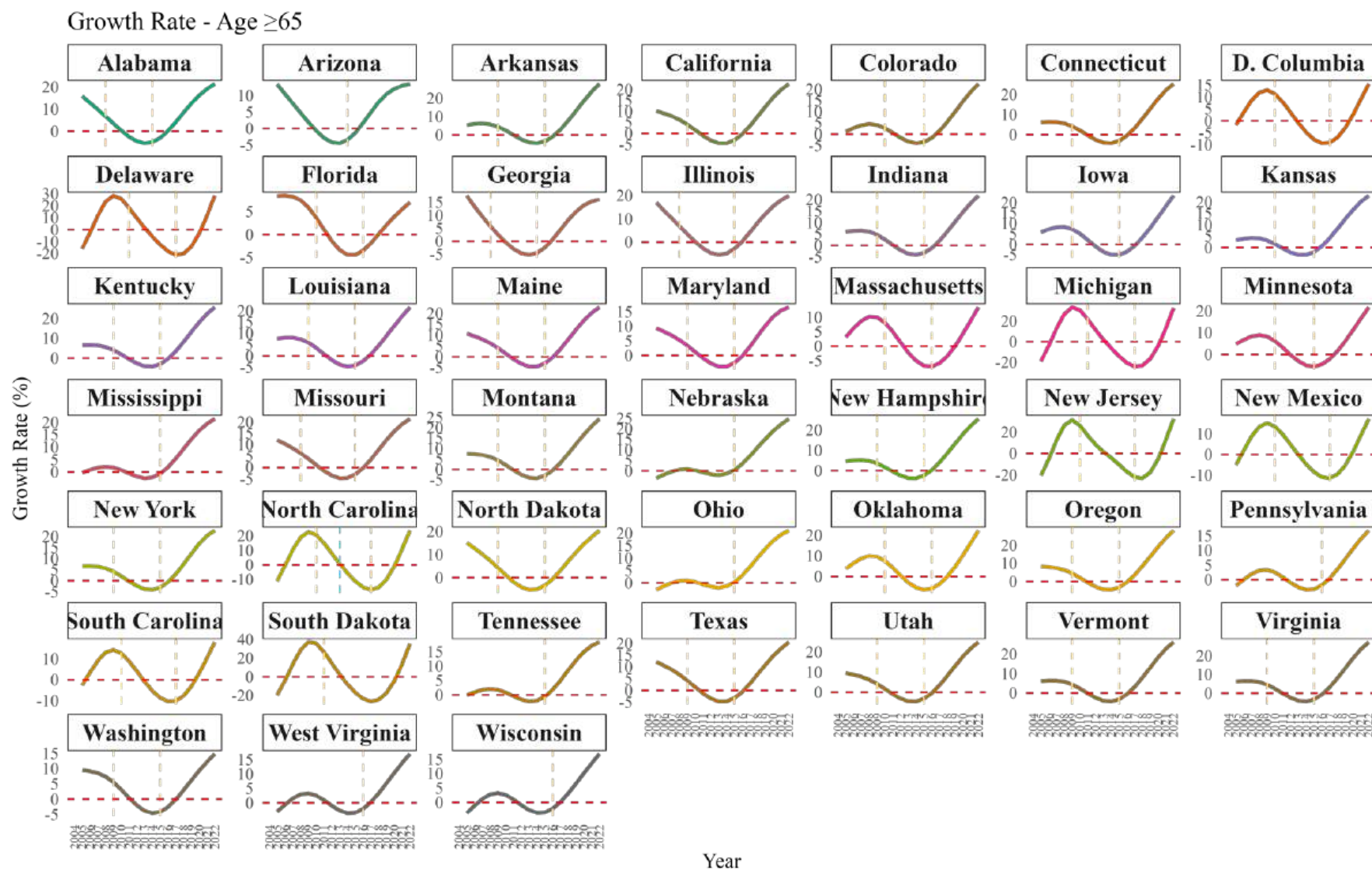
Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

Figure S72. Estimated 3GCR in Enterobacteriales using a spatiotemporal GAM across **US states, age ≥ 65**



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). 3GCR= Third generation cephalosporin resistance.

Figure S73. Growth rate of third-generation cephalosporin-resistance in Enterobacterales using a spatiotemporal GAM across **US-states, age ≥ 65**



Notes: GAM= Generalised additive model. Yellow dashed lines indicate significant break points for the second derivative (significant change respect to previous point; decreasing/increasing shift). Blue lines indicate significant shift in first derivative (from positive to negative or vice versa). The red horizontal dashed line means zero growth.

Table S2. STROBE statement checklist

STROBE Statement—checklist of items that should be included in reports of observational studies.²

	Item No	Recommendation
Title and abstract	1	(a) Indicate the study's design with a commonly used term in the title or the abstract (b) Provide in the abstract an informative and balanced summary of what was done and what was found
Introduction		
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported
Objectives	3	State specific objectives, including any prespecified hypotheses
Methods		
Study design	4	Present key elements of study design early in the paper
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection
Participants	6	(a) <i>Cohort study</i> —Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up <i>Case-control study</i> —Give the eligibility criteria, and the sources and methods of case ascertainment and control selection. Give the rationale for the choice of cases and controls <i>Cross-sectional study</i> —Give the eligibility criteria, and the sources and methods of selection of participants (b) <i>Cohort study</i> —For matched studies, give matching criteria and number of exposed and unexposed <i>Case-control study</i> —For matched studies, give matching criteria and the number of controls per case
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable
Data sources/measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group
Bias	9	Describe any efforts to address potential sources of bias
Study size	10	Explain how the study size was arrived at
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why
Statistical methods	12	(a) Describe all statistical methods, including those used to control for confounding (b) Describe any methods used to examine subgroups and interactions (c) Explain how missing data were addressed (d) <i>Cohort study</i> —If applicable, explain how loss to follow-up was addressed <i>Case-control study</i> —If applicable, explain how matching of cases and controls was addressed <i>Cross-sectional study</i> —If applicable, describe analytical methods taking account of sampling strategy (e) Describe any sensitivity analyses

Continued on next page

Results		
Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed (b) Give reasons for non-participation at each stage (c) Consider use of a flow diagram
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders (b) Indicate number of participants with missing data for each variable of interest (c) <i>Cohort study</i> —Summarise follow-up time (eg, average and total amount)
Outcome data	15*	<i>Cohort study</i> —Report numbers of outcome events or summary measures over time <i>Case-control study</i> —Report numbers in each exposure category, or summary measures of exposure <i>Cross-sectional study</i> —Report numbers of outcome events or summary measures
Main results	16	(a) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included (b) Report category boundaries when continuous variables were categorized (c) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses
Discussion		
Key results	18	Summarise key results with reference to study objectives
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence
Generalisability	21	Discuss the generalisability (external validity) of the study results
Other information		
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based

*Give information separately for cases and controls in case-control studies and, if applicable, for exposed and unexposed groups in cohort and cross-sectional studies.

Note: An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at <http://www.plosmedicine.org/>, Annals of Internal Medicine at <http://www.annals.org/>, and Epidemiology at <http://www.epidem.com/>). Information on the STROBE Initiative is available at www.strobe-statement.org.

References

1. Browne AJ, Chipeta MG, Haines-Woodhouse G, et al. Global antibiotic consumption and usage in humans, 2000–18: a spatial modelling study. *The Lancet Planetary Health* 2021; **5**(12): e893-e904.
2. Von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *The lancet* 2007; **370**(9596): 1453-7.